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**Robustness and Reliability in Sequential Analysis:
A Comparative Evaluation of E-Values and
Standard Significance Testing**

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Abstract

Modern research often involves interim monitoring, data-dependent stopping, follow-up studies, and multiple testing. E-values provide a framework for quantifying evidence while retaining rigorous error guarantees when the timing or continuation of an analysis depends on the observed data. This thesis presents a practical guide to e-values and safe testing, connecting their mathematical foundations to implementation and applied analysis.

Starting from the defining expectation bound, the thesis explains the likelihood-ratio and betting interpretations of e-values, their connection to hypothesis testing through Markov's inequality, and their sequential extension through conditional e-variables, test supermartingales, and Ville's inequality. Confidence sequences provide the corresponding framework for anytime-valid estimation, while the e-Benjamini–Hochberg procedure (eBH) and the e-Partitioning Principle extend the approach to multiple testing.

The methods are illustrated in R, primarily through the `safestats` package, using safe versions of the z-test, t-test, two-proportions test, and logrank test alongside classical procedures. A case study based on ACTG 175 demonstrates fixed-horizon analysis, sequential monitoring, and multiple treatment comparisons. An analysis of the BCG vaccine studies illustrates the accumulation of evidence across separate studies, and simulations examine type-I error, power, and stopping behaviour. Throughout, the safe procedures are compared with their classical counterparts on the same data and designs.

By combining theory, software, simulation, and real-data applications, the thesis provides an accessible route into e-value methods and shows how they can support flexible statistical analyses while maintaining transparent error guarantees under the assumptions of the chosen construction.

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Chapter 1

Introduction

Research rarely unfolds as a single fixed analysis. Researchers inspect data as they accumulate, reconsider recruitment in the light of interim results, stop early when an effect looks convincing, and initiate follow-up studies because earlier findings looked promising. Conventional fixed-horizon testing, by contrast, asks the researcher to specify the sample size and analysis plan before the data are examined, and its type-I error guarantee holds when that procedure is followed. Classical statistics also offers group-sequential and alpha-spending methods for planned interim monitoring (O’Brien and Fleming 1979; Lan and DeMets 1983). What fails is the unadjusted reuse of a fixed-horizon rule. Repeatedly recomputing the same fixed-horizon p-value and stopping at the first favourable result offers more chances to cross a fixed threshold and can drive the false-positive probability far above its nominal level (Ly et al. 2025; Ramdas et al. 2023). A fresh follow-up study can still be analysed validly on its own. The difficulty arises at the level of a study series when decisions to launch, continue, publish or combine studies depend on earlier results, which can create accumulation bias (ter Schure and Grünwald 2019). More broadly, misuse of p-values, selective analysis and incomplete reporting have been identified as significant contributors to concerns about reproducibility (Wasserstein and Lazar 2016). One response proposed lowering the conventional significance threshold for claims of new discoveries (Benjamin et al. 2018).

E-values and safe testing address these problems directly. Let $\mathcal{P}$ denote the null hypothesis. Following the terminology of Vovk and Wang (2021) and the formal treatment of Ramdas and Wang (2025), an e-variable for $\mathcal{P}$ is a non-negative random variable satisfying

$$\mathbb{E}_P[E] \leq 1 \quad \text{for every } P \in \mathcal{P}.$$

Its realised value after observing the data is called an e-value, and large values count as evidence against the null. Markov’s inequality turns the rule “reject when $E \geq 1/\alpha$ ” into a level- α test, so at the five per cent level an e-value of twenty warrants rejection.

As data accumulate, conditionally valid one-step e-variables, each evaluated on fresh information given the past, can be multiplied to form a test supermartingale. Every test supermartingale is an e-process, so its value at any stopping time remains a valid e-variable. Ville’s inequality bounds the probability that a non-negative supermartingale with initial expectation at most one ever crosses a given level by the reciprocal of that level (Ville 1939; Ramdas et al. 2023). This is a statement about the whole path rather than one

pre-chosen sample size, and it yields anytime-validity, meaning that the error guarantee holds uniformly over stopping times.

An analyst may therefore monitor continuously and stop when the evidence is convincing. Evidence can also be continued across later stages or studies without spending the error budget twice, provided that each new stage contributes a conditionally valid e-variable for the same null, evaluated on fresh data (Grünwald et al. 2024; Ramdas and Wang 2025). Optional stopping preserves validity at the chosen stopping time, while optional continuation preserves validity through the product of these conditionally valid contributions. These are the forms of flexibility that defeat an unadjusted fixed-horizon analysis.

Every e-value has a frequentist interpretation through its error bound and a betting interpretation as realised wealth against the null. For e-values constructed as likelihood ratios or Bayes factors, the same number also has a model-comparison interpretation. The betting view is especially useful. An e-value records how much a pre-specified stake against the null has multiplied, and the expectation bound says that the bettor cannot make a systematic profit when the null holds. These ideas are developed in the safe-testing framework of Grünwald et al. (2024), with a broader rethinking of the goals of testing in Grünwald (2024) and a systematic treatment in Ramdas and Wang (2025). They build on the martingale foundations going back to Ville and the game-theoretic view of probability associated with Shafer and Vovk (Shafer 2021; Ramdas et al. 2023).

Even so, e-values have not yet become a standard tool in applied statistics, and practitioner-oriented material remains less abundant than theoretical work. The framework is recent, its conventions reverse familiar p-value habits, and complete worked analyses in the e-value idiom remain relatively scarce. Two recent works address this from opposite directions. Ramdas and Wang (2025) give a systematic treatment at research level, and Ly et al. (2025) give a tutorial for practitioners, organised around sample-size planning and worked largely in the t-test setting, which prioritises readability for an applied audience over rigorous mathematical presentation and defers the latter to specialised works. This thesis sits between them: it keeps the derivations that a tutorial defers, at a level a master’s student can follow, and covers material that practitioner-oriented treatments have so far left aside, including safe counterparts of four standard tests, the e-process machinery behind optional stopping and continuation, and false discovery rate control with e-values.

It is intended as a practical guide to e-values for statistics students and for the practitioners they go on to become. The contribution is one of accessibility rather than novelty, and no new theorems are claimed. The guide offers a route into the subject from first principles and explains how the methods are realised in software through the **safestats** R package (Turner et al. 2022). All code underlying the analyses and simulations in this thesis is available at <https://github.com/KilianG01/evalues-thesis>. One qualification matters throughout: the word “safe” names a precise technical guarantee, not an unconditional assurance. The guarantee is relative to the stated null model and to the assumptions of the construction being used, including correct model specification, a limitation emphasised in the published discussion of the safe-testing framework (Pawel and Held 2024).

The rest of the thesis follows the natural development of the subject. It opens with the foundations of e-values, setting out the distinction between e-variables and their realised e-values, the sense in which they are valid, and the complementary readings of evidence

through likelihood ratios and betting. It then specialises the idea to the standard testing problems an applied statistician meets, building safe counterparts of the z-test, the t-test, the test for two proportions and the logrank test through the **safestats** package. From there it turns to the sequential core of the theory, taking in test supermartingales, e-processes, optional stopping and Ville's inequality, together with the confidence sequences that accompany anytime-valid tests and the product construction for combining conditionally valid contributions across stages. A later chapter takes up multiple testing with e-values, including the eBH procedure (Wang and Ramdas 2022) and the e-Partitioning Principle for false discovery rate control (Goeman et al. 2025). The main body closes with a case study applying these methods to a real randomised clinical trial, the ACTG 175 study of antiretroviral therapies (Hammer et al. 1996), alongside an analysis of the BCG vaccination trials that illustrates optional continuation across separate studies. A final discussion chapter draws together the main conclusions, limitations and directions for future work. In keeping with the character of a guide, the review of prior work is woven in rather than set apart. There is no separate survey chapter, and each chapter opens by citing the specific work it builds on, so that methods and sources are introduced together at the point where they are needed.

Chapter 2

Foundations of E-Values

In classical statistics, the p-value remains the default testing summary for evidence against a null hypothesis. Yet its interpretation is notoriously subtle. An unadjusted analysis under optional stopping does not retain its nominal guarantee, and combining dependent p-values requires dedicated procedures that may be conservative. E-values offer a complementary framework for quantifying evidence. This chapter introduces the core concepts, building from definitions to intuition and first comparisons with p-values. Each concept is accompanied by R code to make the ideas concrete.

2.1 Setup and notation

We work on a sample space Ω equipped with a σ -algebra $\mathcal{F}$. Let $\mathcal{M}_1$ denote the set of all probability measures on $(\Omega, \mathcal{F})$. Some unknown distribution $P^\dagger \in \mathcal{M}_1$ governs our data X , which may be a single observation or a vector $(X_1, \dots, X_n)$. When this distribution belongs to a parametric family, we write $\theta^\dagger$ for the true parameter value.

Definition 2.1 (Hypothesis (adapted from Ramdas and Wang 2025, Definition 1.1)). A **hypothesis** is a set of probability measures in $\mathcal{M}_1$. A hypothesis is **simple** if it is a singleton $\{P\}$. Otherwise, it is **composite**.

Throughout, we write $\mathcal{P}$ for the null hypothesis and $\mathcal{Q}$ for the alternative hypothesis, and use P, Q for individual distributions within these sets. The null hypothesis H_0 is the claim that $P^\dagger \in \mathcal{P}$, and the alternative H_1 is the claim that $P^\dagger \in \mathcal{Q}$.

Example: Suppose our data X is normally distributed with an unknown mean μ and a known variance of 1. If we test the claim that the mean is exactly zero, this forms a **simple** null hypothesis because it specifies a single, exact distribution: $\mathcal{P} = \{\mathcal{N}(0, 1)\}$. If we test the claim that the mean is strictly positive, this forms a **composite** alternative hypothesis because it encompasses an infinite set of possible distributions: $\mathcal{Q} = \{\mathcal{N}(\mu, 1) \mid \mu > 0\}$.

2.2 What are e-variables and e-values?

The central definition of this thesis is remarkably simple.

Definition 2.2 (E-variables, p-variables, and tests (adapted from Ramdas and Wang 2025, Definition 1.2)). Let $\mathcal{P} \subseteq \mathcal{M}_1$ be the null hypothesis.

- (i) An **e-variable** E for $\mathcal{P}$ is a $[0, \infty]$ -valued random variable satisfying

$$\mathbb{E}_P[E] \leq 1 \quad \text{for all } P \in \mathcal{P}.$$

An e-variable is **exact** if $\mathbb{E}_P[E] = 1$ for all $P \in \mathcal{P}$.

- (ii) A **p-variable** U for $\mathcal{P}$ is a $[0, \infty)$ -valued random variable satisfying

$$P(U \leq \alpha) \leq \alpha \quad \text{for all } \alpha \in (0, 1) \text{ and all } P \in \mathcal{P}.$$

The $[0, \infty)$ convention follows Ramdas and Wang (2025); classical expositions more commonly restrict attention to $[0, 1]$ -valued p-variables, but the distinction is immaterial here since any $[0, \infty)$ -valued p-variable can be truncated to $[0, 1]$ without affecting the defining inequality.

- (iii) A **test** φ is a $[0, 1]$ -valued random variable. It has **level** α for $\mathcal{P}$ if $\mathbb{E}_P[\varphi] \leq \alpha$ for every $P \in \mathcal{P}$.

The value taken by the e-variable upon observing the data is called an **e-value**. The “e” stands for both *expectation* (the defining property) and *evidence* (the interpretation).

The literature sometimes uses “e-value” for both the random statistic and its realised value. This thesis retains the distinction in formal definitions and mathematical statements, while occasionally using the shorter terminology in applied discussion when no ambiguity can arise.

The constraint on e-variables is an **expectation** constraint: $\mathbb{E}_P[E] \leq 1$. The constraint on p-variables is a **probability** constraint: $P(U \leq \alpha) \leq \alpha$. This seemingly small difference has profound consequences for how the two quantities can be used, combined, and interpreted.

Several points deserve emphasis:

- The **trivial e-variable** $E \equiv 1$ is always valid for any $\mathcal{P}$. It provides no evidence against the null, but it establishes that at least one e-variable always exists.
- An e-variable is allowed to take the value ∞ , but only on events that have probability zero under every $P \in \mathcal{P}$.
- Unlike p-values, **larger e-values indicate stronger evidence** against the null. For simple-vs-simple likelihood-ratio e-values, an e-value of 20 means that the observed data are 20 times as likely under the alternative as under the null. For general e-values, the interpretation is that Markov's inequality guarantees a p-value of at most $1/20 = 0.05$.
- E-values and p-values are *not* on the same scale. We should not simply compare E with $1/p$ for a p-value p ; as we will see when discussing calibrators, the conversion between the two inherently loses information.

2.3 From e-values to tests: Markov's inequality

The most immediate use of an e-value is as a test statistic. The bridge is Markov's inequality, which gives us a level- α test for free.

Proposition 2.1 (Markov's inequality for e-values (Ramdas and Wang (2025), Proposition 2.1)). *Let E be an e-variable for $\mathcal{P}$. Then for all $P \in \mathcal{P}$ and $\alpha \in (0, 1]$,*

$$P(E \geq 1/\alpha) \leq \alpha.$$

Hence:

- $1/E$ is a p -variable for $\mathcal{P}$,
- $\mathbf{1}\{E \geq 1/\alpha\}$ is a level- α binary test.

Proof. Since $\mathbf{1}\{x \geq 1\} \leq x$ for all $x \geq 0$ (the indicator of exceeding 1 is bounded by the value itself), we have $P(E \geq 1/\alpha) = \mathbb{E}_P[\mathbf{1}\{\alpha E \geq 1\}] \leq \mathbb{E}_P[\alpha E] = \alpha \mathbb{E}_P[E] \leq \alpha$. $\square$

In words: **reject the null whenever the e-value exceeds $1/\alpha$** . At $\alpha = 0.05$, we reject when $E \geq 20$. At $\alpha = 0.01$, we reject when $E \geq 100$.

The Markov rule is universally valid, but it is often **conservative**, because Markov's inequality is loose for many e-variables. In such cases the rejection probability under the null falls below α , and fixed-sample power can be lower than for a test calibrated to the full null distribution. How much is lost depends on the choice of e-variable, not on the formalism itself, as Section 2.11 explains. If the distribution of E under the null is known, one can use a tighter threshold (Ramdas and Wang 2025, sec. 2.2); otherwise $1/\alpha$ remains the universally valid default.

2.4 The betting interpretation

E-values have a natural interpretation as **bets against the null hypothesis**. This perspective, rooted in game-theoretic probability, provides the deepest intuition for why e-values behave as they do (see Section 1.5 in Ramdas and Wang 2025).

Consider two parties:

- **Forecaster** believes the null hypothesis P is true.
- **Skeptic** believes the alternative Q is more accurate.

Forecaster offers a contract: Skeptic pays \$1 upfront, and after observing data ω , Forecaster pays Skeptic $E(\omega)$ dollars, where E is some agreed-upon nonnegative function of the data. When would Forecaster agree to this contract? Precisely when $\mathbb{E}_P[E] \leq 1$, because then Forecaster expects not to lose money. In other words: E must be an e-variable.

Meanwhile, Skeptic will only take the bet if they expect to profit under Q , meaning $\mathbb{E}_Q[E] > 1$. The realised e-value $E(\omega)$ is the **betting score**: it tells us how much Skeptic's wealth has multiplied.

P-values also have a betting interpretation, but a flawed one. For a p -variable U and any threshold $r \in (0, 1]$, the quantity $\mathbf{1}\{U \leq r\}/r$ is an e-variable. But the realised p -value equals the *smallest* r at which Skeptic would have profited, the best bet *in hindsight*. An e-value, by contrast, is a single pre-committed bet. This is why $1/p$ for a p -value p overstates the evidence: it involves “double-dipping” by choosing the best bet after seeing the data.

2.5 First example: the Bernoulli likelihood ratio

The most natural source of e-values is the **likelihood ratio**. Suppose we test a simple null P against a simple alternative Q , with $Q \ll P$. Then

$$E = \frac{dQ}{dP}(X)$$

is an e-variable for P , because

$$\mathbb{E}_P[E] = \mathbb{E}_P \left[\frac{dQ}{dP}(X) \right] = \int \frac{q(x)}{p(x)} p(x) dx = 1.$$

In fact, for a simple null, exactness is equivalent to admissibility, and every exact e-variable equals a likelihood ratio dS/dP , P -almost surely, for some distribution $S \ll P$ (Ramdas and Wang 2025, Proposition 2.12).

To make this concrete, consider testing $P = \text{Bernoulli}(p_0)$ against $Q = \text{Bernoulli}(q)$ with $q > p_0$, based on n iid observations. Write $S_n = \sum_{i=1}^n X_i$ for the number of successes. The likelihood ratio e-variable is (Ramdas and Wang 2025, Equation 1.5)

$$E_n = \left(\frac{q}{p_0} \right)^{S_n} \left(\frac{1-q}{1-p_0} \right)^{n-S_n}.$$

We extend Example 1.4 from Ramdas and Wang (2025). A scientist observes the dataset

(1, 1, 0, 1, 1, 1, 1)

and tests $H_0 : p = 1/2$. We calculate the e-value for several choices of the alternative q , including the extreme case $q = 1$.

```
x <- c(1, 1, 0, 1, 1, 1, 1)
n <- length(x)
s <- sum(x)
p0 <- 0.5

q_values <- c(0.6, 0.7, 0.8, 0.9, 1.0)
e_values <- sapply(q_values, function(q) {
  (q / p0)^s * ((1 - q) / (1 - p0))^(n - s)
})

q_mle <- s / n
e_mle <- (q_mle / p0)^s *
  ((1 - q_mle) / (1 - p0))^(n - s)

data.frame(
  q = q_values,
  e_value = round(e_values, 2)
)
```

	q	e_value
1	0.6	2.87
2	0.7	6.32
3	0.8	10.74
4	0.9	12.24
5	1.0	0.00

The e-value depends on the choice of the alternative q , but it is a valid e-variable regardless.

Notice two important behaviours here. First, if we choose $q = 1$, the e-value drops to 0. This makes intuitive sense: an alternative with $q = 1$ asserts that a failure is strictly

impossible. Since the observed dataset contains a 0, that alternative is falsified, and the e-value reflects this: Skeptic loses their entire \$1 stake.

Second, even the best pre-specified alternative in our grid does not produce enough evidence to reject at $\alpha = 0.05$. Among the grid values, $q = 0.9$ gives the largest e-value, about 12.24. Treating q as a free parameter, the function $q \mapsto E(q; \mathbf{x})$ is maximised at the maximum-likelihood value $q = 0.875$, where it takes the value 12.57. This is the largest e-value any pre-specified alternative in $(0, 1)$ could have produced for this dataset, which remains below the Markov threshold $1/\alpha = 20$. An important caveat: a practitioner who *chooses* q after seeing the data would be using the data twice, and the resulting statistic would not be a valid e-variable. The proper sequential analogue, which uses only *past* data to choose the next bet, is the plug-in strategy introduced in Section 2.10.

To see how e-values accumulate evidence multiplicatively, consider a slightly larger dataset with the same qualitative pattern: 13 successes in 16 trials.

```
n_large <- 16
s_large <- 13
p0 <- 0.5

q_values_large <- c(0.6, 0.7, 0.8, 0.9)
e_values_large <- sapply(q_values_large, function(q) {
  (q / p0)^s_large * ((1 - q) / (1 - p0))^(n_large - s_large)
})

data.frame(
  q = q_values_large,
  e_value = round(e_values_large, 2)
)
```

	q	e_value
1	0.6	5.48
2	0.7	17.14
3	0.8	28.82
4	0.9	16.66

With this larger sample, the evidence against the null compounds multiplicatively. If the scientist had pre-specified an alternative of $q = 0.8$, the resulting e-value is 28.82. Because this exceeds 20, we can now reject the null hypothesis $H_0 : p = 1/2$ at level $\alpha = 0.05$.

Unknown sampling schemes and stopping rules. A useful property of the pre-specified likelihood-ratio process in this example is that its validity is preserved under a broad range of stopping rules, even though the appropriate classical p-value depends on the sampling scheme.

To see why this matters, consider the dataset from the previous section: 7 successes and 1 failure in 8 coin tosses. Suppose we do not know whether these data were collected via design (a) “collect exactly 8 observations” or design (b) “collect until 5 consecutive heads.” Notice that our specific dataset, $(1, 1, 0, 1, 1, 1, 1, 1)$, triggers the stopping condition for both designs exactly on the 8th toss. The physical data sitting on the desk is identical in both scenarios.

Under design (b), the sample size is *random*: the experimenter keeps flipping until a run

of 5 consecutive heads appears, which could happen at flip 5 (if the first five are all heads) or much later. The p-value under this design is therefore $P_0(\tau \leq 8)$, the probability that the stopping time τ , the first time a run of 5 consecutive heads is completed, occurs at or before the 8th flip. Because there are only $2^8 = 256$ possible sequences of eight tosses, this probability can be computed exactly.

```
# P-value under design (a): fixed sample size n = 8
p_fixed <- pbinom(s - 1, size = n, prob = p0, lower.tail = FALSE)

# P-value under design (b): exact enumeration of all 2^8 sequences.
sequences <- as.matrix(
  do.call(expand.grid, rep(list(c(0L, 1L)), 8L))
)

has_run_of_five <- apply(sequences, 1L, function(z) {
  runs <- rle(z)
  any(runs$values == 1L & runs$lengths >= 5L)
})

stopifnot(sum(has_run_of_five) == 20L)
p_sequential <- mean(has_run_of_five)

# E-value with q = 0.7 (identical for both designs)
q <- 0.7
e_val <- (q / p0)^s * ((1 - q) / (1 - p0))^(n - s)

cat("Design (a) fixed n=8:           p-value =", round(p_fixed, 4), "\n")
```

Design (a) fixed n=8: p-value = 0.0352

```
cat("Design (b) stop at 5 in a row: p-value =", round(p_sequential, 4), "\n")
```

Design (b) stop at 5 in a row: p-value = 0.0781

```
cat("E-value (q=0.7):", round(e_val, 4),
    "(same for both)\n")
```

E-value (q=0.7): 6.3248 (same for both)

Under the fixed-sample design, the appropriate p-value is below 0.05, whereas under the stopping-rule design it is above 0.05. A scientist who receives only the observations cannot determine which of these design-specific p-values applies. By contrast, the pre-specified likelihood-ratio process yields the same realised e-value for the same observed sequence and remains valid under either stopping description because it is an e-process, as developed in Chapter 4. This conclusion assumes that the observations are retained from the beginning. It does not cover cherry-picking, deleting early unfavourable observations, or reporting the best of several batches.

2.6 Second example: the Gaussian likelihood ratio

Following the example in Section 1.6 of Ramdas and Wang (2025), test $P = N(0, 1)$ against $Q = N(\mu, 1)$ with $\mu > 0$, based on n iid observations $X_1, \dots, X_n$. Let $S_n = \sum_{i=1}^n X_i$. The likelihood ratio for a single observation X_i is

$$\frac{q(X_i)}{p(X_i)} = \frac{\phi(X_i - \mu)}{\phi(X_i)} = \frac{\exp\left(-\frac{1}{2}(X_i - \mu)^2\right)}{\exp\left(-\frac{1}{2}X_i^2\right)} = \exp\left(\mu X_i - \frac{\mu^2}{2}\right),$$

where ϕ denotes the standard normal density. Taking the product over all n observations gives the likelihood-ratio e-variable (Ramdas and Wang 2025, Equation 1.1)

$$E_n = \prod_{i=1}^n \exp\left(\mu X_i - \frac{\mu^2}{2}\right) = \exp\left(\mu S_n - \frac{n\mu^2}{2}\right).$$

Setting $Z = S_n/\sqrt{n}$ and $\delta = \mu\sqrt{n}$ gives the equivalent form

$$E_n = \exp(\delta Z - \delta^2/2)$$

(Ramdas and Wang 2025, Equation 1.2), where $Z \sim N(0, 1)$ under the null. The classical one-sided p-value is $P_n = 1 - \Phi(Z)$.

Both are functions of the same sufficient statistic Z , but they quantify evidence differently. We plot them side by side.

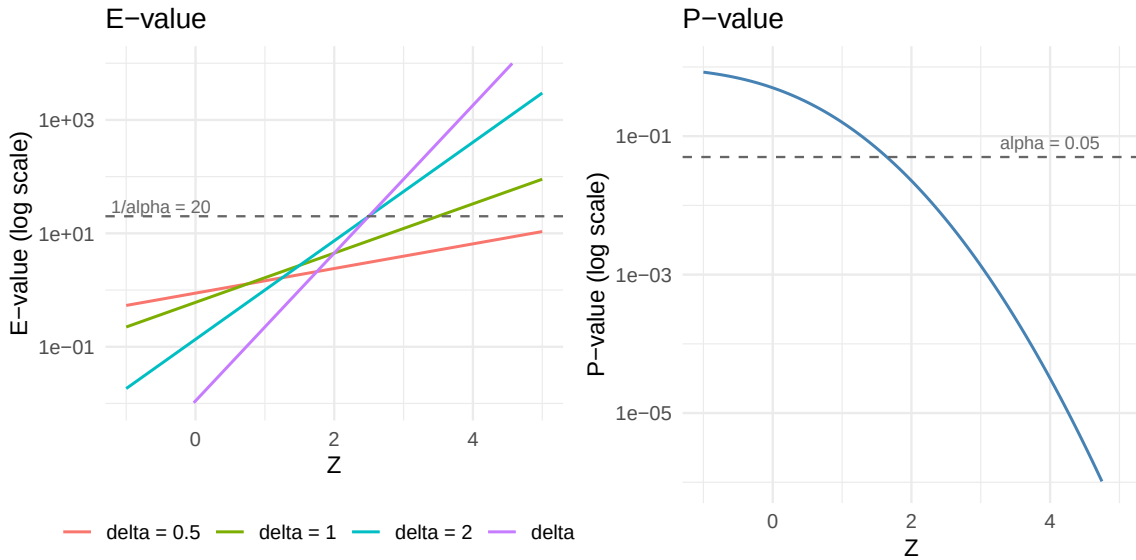


Figure 2.1: E-values and p-values as functions of the test statistic Z for the one-sided Gaussian test. Dashed lines show the rejection thresholds at $\alpha = 0.05$.

Figure 2.1 shows the two evidence scales as functions of the same statistic. Two things stand out. First, the p-value is a single curve and does not depend on the alternative, while the e-value depends on our choice of δ . Second, the e-value threshold for rejection ($E \geq 20$ at $\alpha = 0.05$) requires a larger Z than the p-value threshold ($Z \geq 1.645$). This is the conservatism of Markov's inequality discussed in Proposition 2.1.

2.7 Combining e-values

One of the most practically important properties of e-values is how cleanly they combine. The following facts follow immediately from the definition.

Proposition 2.2 (Combination rules (parts (iii) and (iv) of Fact 1.3 in Ramdas and Wang (2025))). *Let $E_1, \dots, E_K$ be e-variables for $\mathcal{P}$. Parts (i) and (ii) of Fact 1.3 construct p-variables from cumulative distribution functions, while parts (v) and (vi) give monotonicity properties. We state here only the two e-variable combination rules in parts (iii) and (iv).*

- (iii) Their **arithmetic mean** $\bar{E} = \frac{1}{K} \sum_{k=1}^K E_k$ is an e-variable for $\mathcal{P}$ (under arbitrary dependence).
- (iv) If $E_1, \dots, E_K$ are **independent** under each $P \in \mathcal{P}$, then their **product** $\prod_{k=1}^K E_k$ is an e-variable for $\mathcal{P}$.

Both facts follow from linearity of expectation (for the average) and the factorisation of expectations for independent random variables (for the product).

These e-variable combination rules are unusually direct. Dependent p-values can also be merged, but not by taking their unadjusted arithmetic mean in general. Valid p-merging requires a dedicated construction and may involve conservatism (Ramdas and Wang 2025, chap. 12). For e-variables, the arithmetic mean itself is valid under arbitrary dependence.

Combining e-values in R. We demonstrate both rules. Suppose $K = 5$ independent labs each collect $n = 20$ observations from $N(\theta^\dagger, 1)$ and test $H_0 : \theta = 0$ vs $H_1 : \theta = 0.3$ using the likelihood ratio.

```
mu <- 0.3          # alternative used in e-value construction
theta_true <- 0.3  # true data-generating parameter
n <- 20
K <- 5

# Simulate K independent experiments
set.seed(123)
e_vals <- sapply(seq_len(K), function(k) {
  x_k <- rnorm(n, mean = theta_true, sd = 1)
  exp(mu * sum(x_k) - n * mu^2 / 2)
})

cat("Individual e-values:", paste(sprintf("%.2f", e_vals), collapse = ", "),
    "\n")
```

Individual e-values: 5.75, 1.81, 4.66, 1.20, 23.35

```
cat("Arithmetic mean:      ", sprintf("%.2f", mean(e_vals)), "\n")
```

Arithmetic mean: 7.35

```
cat("Product:              ", sprintf("%.2f", prod(e_vals)), "\n")
```


Product: 1355.84

The individual e-values range from 1.20 to 23.35.

- The **average** gives a combined e-value of 7.4, and requires no independence assumption: it is a valid e-variable whenever each marginal one is.
- Because the experiments are independent, their e-values may also be multiplied. The product above crosses the level-0.05 Markov threshold of 20.

To illustrate the behaviour of these combination rules under the null, we simulate 10,000 repetitions in which H_0 is true ($\theta^\dagger = 0$).

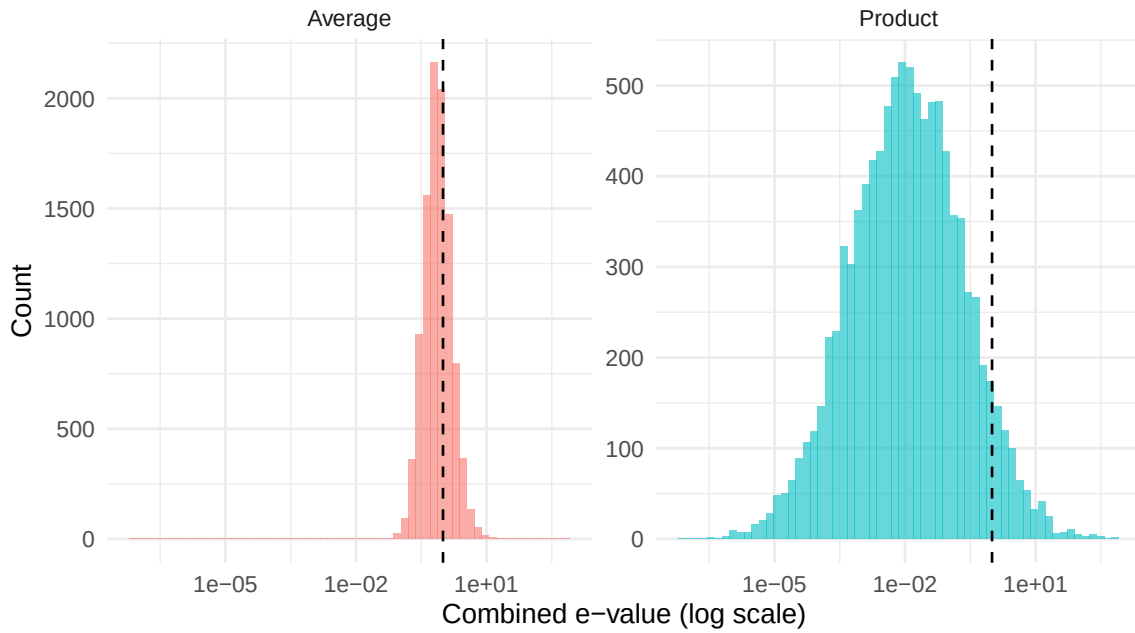


Figure 2.2: Null distributions of averaged and multiplied e-values across 10,000 repetitions of 5 independent experiments. Both have expectation one, but the product is far more right-skewed.

Figure 2.2 illustrates two very different null distributions. In the betting language of Section 2.4, no valid strategy can expect to profit when the null holds, and both combined e-variables here have exact expectation one. The simulated means are 0.995 for the average and 0.872 for the product. The corresponding proportions exceeding the threshold 20 are 0.0002 and 0.0053, respectively.

That guarantee constrains the expectation alone: it does not require the realised values, or even their Monte Carlo sample means, to be concentrated near one. The arithmetic mean is comparatively stable. The product is highly right-skewed: in this setting its theoretical median is 0.011 and its standard deviation is approximately 90, even though its expectation is one. The Monte Carlo standard error of the estimated product mean across 10,000 repetitions is therefore approximately 0.90, compared with 0.010 for the averaged e-variable.

The simulation illustrates the theoretical expectation and threshold-crossing guarantees but does not verify them. Those guarantees follow from Proposition 2.2 and Proposition 2.1. The empirical crossing proportions are Monte Carlo summaries to be read

against the theoretical upper bound of 0.05, and both fall far below it, a concrete instance of the looseness of the Markov threshold discussed in Section 2.3.

2.8 Calibrators: converting between p-values and e-values

Since both e-values and p-values are commonly used in practice, it is important to understand how to convert between them.

2.8.1 E-to-p calibration

By Proposition 2.1, $g(e) = \min(1/e, 1)$ converts any e-value into a p-value. This is the **unique admissible deterministic e-to-p calibrator** (Ramdas and Wang 2025, Proposition 2.4). In other words, there is essentially only one sensible way to go from e-values to p-values: inversion.

2.8.2 P-to-e calibration

Going the other direction is richer. A **p-to-e calibrator** is a decreasing function $f : [0, \infty) \rightarrow [0, \infty]$ with $f = 0$ on $(1, \infty)$ such that $f(U)$ is an e-variable whenever U is a p-variable. The characterisation is elegant: f is a p-to-e calibrator if and only if $\int_0^1 f(p) dp \leq 1$ (Ramdas and Wang 2025, Proposition 2.5).

Some useful admissible calibrators include:

- **Power calibrator:** $f(p) = \kappa p^{\kappa-1}$ for $\kappa \in (0, 1)$. For small κ , its dependence on p approaches the p^{-1} shape, while the prefactor κ keeps the integral equal to one (Ramdas and Wang 2025, Equation 2.1).
- **Negative log:** $f(p) = -\log p$. Simple and useful, but penalises small p-values less aggressively. (Ramdas and Wang 2025, Equation 2.5)
- **Mixture calibrator:** $f(p) = \frac{1-p+p \log p}{p(\log p)^2}$. This is the mixture of all power calibrators (Ramdas and Wang 2025, Equation 2.2).

An important lesson is that **round-tripping can lose evidence**. A p-to-e-to-p round trip cannot improve the p-value and may make it substantially larger. We illustrate this with a concrete computation.

2.8.3 Calibrators in R

We start from z-test p-values and convert them to e-values via several calibrators, then convert back.

```
# Define calibrators on [0, infinity), with value zero above one.
f_power <- function(p, kappa = 0.5) {
  stopifnot(all(p >= 0), kappa > 0, kappa < 1)
  ifelse(p <= 1, kappa * p^(kappa - 1), 0)
}

f_log <- function(p) {
  stopifnot(all(p >= 0))
  ifelse(p <= 1, -log(p), 0)
}
```

```

f_mixture <- function(p) {
  stopifnot(all(p >= 0))

  out <- numeric(length(p))
  out[p == 0] <- Inf

  inside <- p > 0 & p < 1
  out[inside] <-
    (1 - p[inside] + p[inside] * log(p[inside])) /
    (p[inside] * log(p[inside])^2)

  out[p == 1] <- 1 / 2
  out[p > 1] <- 0
  out
}

p_grid <- seq(0.001, 0.999, length.out = 500)

df_cal <- data.frame(
  p = rep(p_grid, 4),
  e = c(
    f_power(p_grid, 0.5),
    f_log(p_grid),
    f_mixture(p_grid),
    1 / p_grid
  ),
  calibrator = rep(
    c(
      "Power (kappa=0.5)",
      "Negative log",
      "Mixture",
      "1/p (not a calibrator)"
    ),
    each = length(p_grid)
  )
)

ggplot(df_cal, aes(x = p, y = e, color = calibrator, linetype = calibrator))
  ↪ +
  geom_line(linewidth = 0.7) +
  scale_y_log10(limits = c(0.1, 1000)) +
  scale_linetype_manual(
    values = c(
      "Power (kappa=0.5)" = "solid",
      "Negative log" = "solid",
      "Mixture" = "solid",
      "1/p (not a calibrator)" = "dashed"
    )
  ) +
  labs(x = "p-value", y = "e-value (log scale)",
       color = NULL, linetype = NULL) +
  theme(legend.position = "bottom")

```

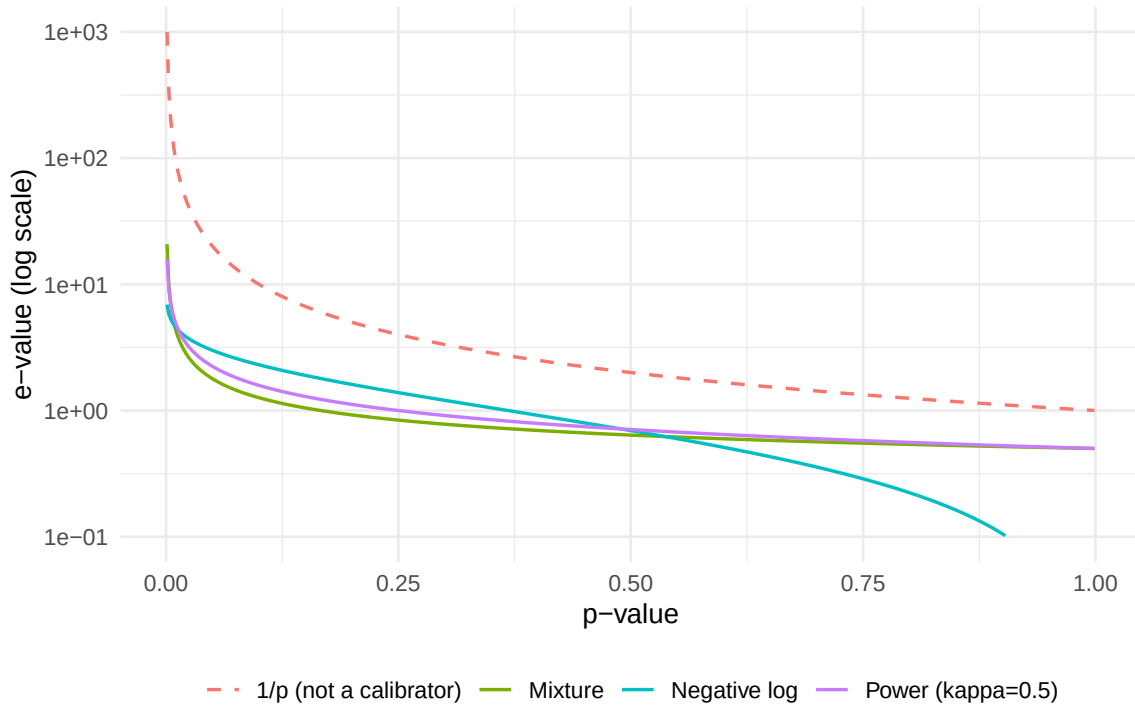


Figure 2.3: Three p-to-e calibrators applied to p-values from a standard z-test, with the reciprocal $1/p$ shown for comparison.

Every valid calibrator satisfies $f(p) \leq 1/p$ (Ramdas and Wang 2025, sec. 2.3). The argument is one line: if $f(p_1) > 1/p_1$ for some $p_1 \in (0, 1]$, then, because f is decreasing,

$$\int_0^1 f(p) dp \geq \int_0^{p_1} f(p) dp \geq p_1 f(p_1) > 1,$$

contradicting the integrability condition. The admissible calibrators displayed in Figure 2.3 lie strictly below this bound. The function $1/p$ itself is not a p-to-e calibrator: its integral diverges,

$$\int_0^1 p^{-1} dp = \infty,$$

so it fails the defining integrability condition. Admissibility is the sharper optimality notion within the class of valid calibrators. Proposition 2.5 of Ramdas and Wang (2025) states that a calibrator is admissible if and only if it is upper semicontinuous, satisfies $f(0) = \infty$, and has

$$\int_0^1 f(p) dp = 1.$$

Integral equality alone is therefore not sufficient.

```
# Round-trip loss: p -> e -> p'
p_start <- 0.01

cat("Starting p-value:", p_start, "\n\n")
```

Starting p-value: 0.01

```

for (cal_name in c("Power (kappa=0.5)", "Negative log", "Mixture")) {
  e <- switch(
    cal_name,
    "Power (kappa=0.5)" = f_power(p_start, 0.5),
    "Negative log" = f_log(p_start),
    "Mixture" = f_mixture(p_start)
  )
  p_back <- min(1 / e, 1) # e-to-p calibration

  # Print with consistent formatting
  cat(sprintf("%-18s : p -> %6.2f -> p' = %.4f\n", cal_name, e, p_back))
}

```

```

Power (kappa=0.5) : p -> 5.00 -> p' = 0.2000
Negative log      : p -> 4.61 -> p' = 0.2171
Mixture          : p -> 4.45 -> p' = 0.2247

```

The round-trip loss is substantial. For each of the smooth calibrators used here, the recovered p-value p' is strictly larger than the starting value. More generally, a p-to-e-to-p round trip cannot improve the p-value and may lose substantial information.

2.9 E-power and log-optimality

While Section 2.3 tells us how to *use* e-values for testing, it says nothing about which e-variable to choose. Ramdas and Wang (2025) develop the theory of efficiency for e-values (Definition 3.11), centred on a notion called **e-power**.

Definition 2.3 (E-power (adapted from Ramdas and Wang 2025, Definition 3.11)). The **e-power** of an e-variable E against an alternative Q is

$$\Lambda_Q(E) = \mathbb{E}_Q[\log E],$$

provided that the expectation is well-defined. An e-variable has **positive e-power** against Q if $\mathbb{E}_Q[\log E] > 0$ for each $Q \in \mathcal{Q}$.

2.9.1 Why the logarithm?

The choice of the logarithm is not arbitrary. It is justified by three converging arguments:

1. **Product structure.** In the simplest setting, suppose we observe iid data and use the same construction at each step, producing iid e-variables $E_1, E_2, \dots$ under the alternative Q . Their product $M_n = \prod_{i=1}^n E_i$ accumulates evidence multiplicatively. Provided that $\mathbb{E}_Q[|\log E_1|] < \infty$, the strong law of large numbers gives $\frac{1}{n} \log M_n \rightarrow \mathbb{E}_Q[\log E_1]$ almost surely. The e-power is therefore the asymptotic growth rate of the accumulated evidence. This convergence extends to adaptive, non-iid settings under weaker conditions via e-processes; see Chapter 7 of Ramdas and Wang (2025).
2. **Kelly criterion.** In the betting interpretation, a gambler who repeatedly bets at fair odds maximises long-run wealth growth by maximising expected log wealth (Kelly 1956). The e-power is the expected log wealth from a single round.

3. **Axiomatic derivation.** On the class $\mathcal{X}$ of nonnegative random variables that are bounded above and away from zero, and for an atomless alternative distribution Q , Ramdas and Wang (2025) show (Theorem 3.28) that any ranking satisfying two natural axioms, *homogeneity* (scaling the e-variable by a constant preserves the ranking) and *asymptotic consistency* (if a product process achieves asymptotic power 1, so does any process ranked higher), must be a strictly increasing transform of $\mathbb{E}_Q[\log E]$.

2.9.2 The log-optimal e-variable

For testing a simple null $\{P\}$ against a simple alternative $\{Q\}$ with $Q \ll P$, which e-variable should we choose? The answer is given by the following result (Theorem 3.20 in Ramdas and Wang (2025)):

Theorem 2.1 (Log-optimality of the likelihood ratio). *For testing P against Q with $Q \ll P$, the P -almost surely unique log-optimal e-variable is $E^* = dQ/dP$. Its e-power equals the Kullback–Leibler divergence, which may be infinite:*

$$\mathbb{E}_Q[\log E^*] = \text{KL}(Q\|P) = \int q(x) \log \frac{q(x)}{p(x)} dx.$$

This result is the **e-value analogue of the Neyman–Pearson lemma**. Where Neyman–Pearson tells us that the most powerful level- α test thresholds the likelihood ratio, Theorem 2.1 tells us that the likelihood ratio itself is the optimal e-variable in the sense of maximising e-power.

2.9.3 E-power in R

For the Gaussian example, the e-power of $E_n = \exp(\mu S_n - n\mu^2/2)$ under the true alternative $\theta^\dagger$ is $n(\mu\theta^\dagger - \mu^2/2)$, which is maximised at $\mu = \theta^\dagger$ with optimal value $n\theta^{\dagger 2}/2 = \text{KL}(Q\|P)$ (Ramdas and Wang 2025, Example 3.13). We illustrate this by simulation and compare the empirical curve with the theoretical expression.

```
set.seed(2026)

theta_true <- 0.3
n <- 50
mu_grid <- seq(0.05, 0.8, by = 0.025)
n_rep_epower <- 5000

# For each mu, simulate the e-power
epower <- sapply(mu_grid, function(mu) {
  log_e <- replicate(n_rep_epower, {
    x <- rnorm(n, mean = theta_true, sd = 1)
    mu * sum(x) - n * mu^2 / 2
  })
  mean(log_e)
})

# Theoretical curve: n * (mu * theta_true - mu^2 / 2)
epower_theory <- n * (mu_grid * theta_true - mu_grid^2 / 2)
```

```
df_ep <- data.frame(
  mu = rep(mu_grid, 2),
  epower = c(epower, epower_theory),
  type = rep(c("Simulated", "Theoretical"), each = length(mu_grid))
)

ggplot(df_ep, aes(x = mu, y = epower, color = type, linetype = type)) +
  geom_line(linewidth = 0.8) +
  geom_vline(xintercept = theta_true, linetype = "dashed", color = "grey40")
  ↪ +
  annotate("text", x = theta_true + 0.05, y = max(epower_theory) * 0.9,
    label = expression(mu == theta[true] ~ "= 0.3"),
    hjust = 0, size = 3) +
  labs(x = expression(mu ~ "(alternative used in e-value)"),
    y = "E-power (mean of log E)",
    color = NULL, linetype = NULL) +
  theme(legend.position = "bottom")
```

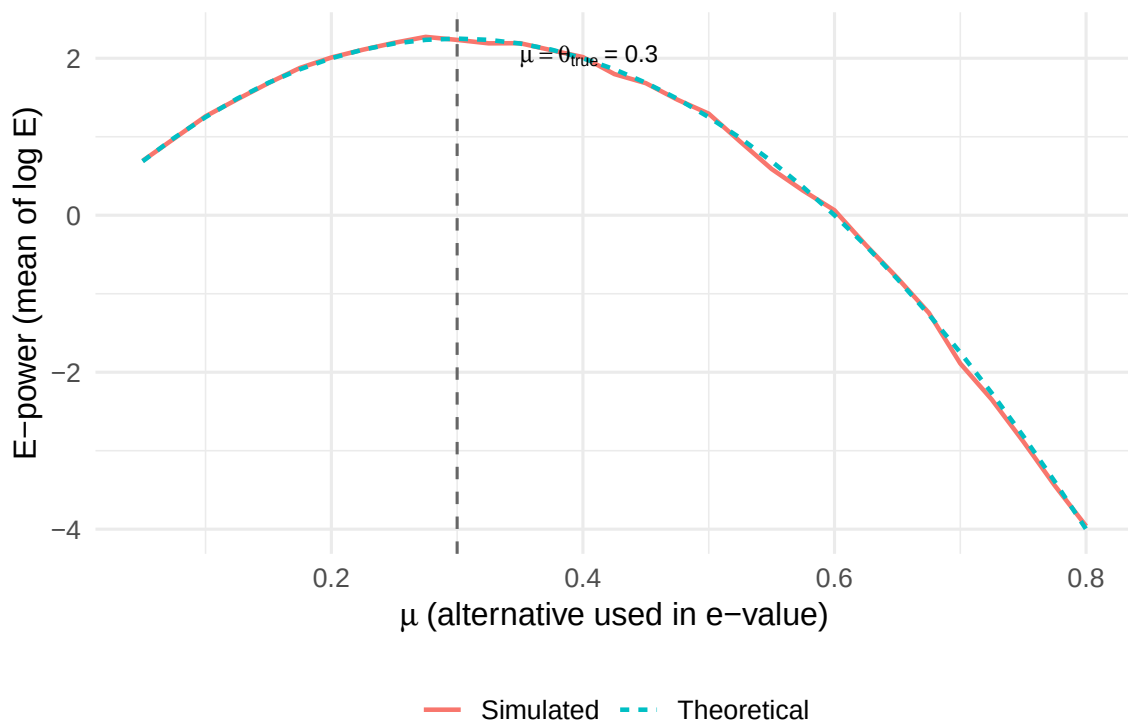


Figure 2.4: Empirical e-power (mean of $\log E$) for different choices of the alternative μ used to construct the e-value. The true parameter is $\theta^\dagger = 0.3$. The dashed vertical line marks the optimum.

Figure 2.4 shows close agreement between the empirical and theoretical curves, with both maximised at $\mu = \theta^\dagger = 0.3$, as predicted by Theorem 2.1. Using a moderately misspecified alternative, such as $\mu = 0.5$, still yields a valid e-variable with positive but lower e-power, so evidence accumulates more slowly. More severe misspecification is worse: at $\mu = 2\theta^\dagger = 0.6$ the e-power is exactly zero, and beyond that it turns negative, meaning the e-variable is expected to shrink under the alternative. A negative e-power does not invalidate the e-variable, which remains valid under the null, but it means that the bet loses wealth on

average even when the alternative is true.

2.10 Composite hypotheses and adaptive strategies

In practice, simple-vs-simple testing is rare. The alternative is usually composite: we know the null but not which specific alternative generated the data. In the betting language of Section 2.4, the Skeptic believes the coin is biased but does not know *how much*. Which bet should they place?

Two general strategies address this: **hedging your bets upfront** (method of mixtures) or **learning as you go** (plug-in method). Both produce valid e-variables regardless of which specific alternative is true (Ramdas and Wang 2025, sec. 3.7).

2.10.1 Method of mixtures: hedging your bets

Suppose we test a fair coin ($H_0 : p = 0.5$) and believe the true bias is somewhere in the range $q \in \{0.6, 0.7, 0.8\}$, but we do not know which. Rather than committing to a single alternative, we can **average** the likelihood ratio e-values across all candidates. If we assign equal weight to each, we compute

$$E_n = \frac{1}{3} \left[E_n^{(0.6)} + E_n^{(0.7)} + E_n^{(0.8)} \right]$$

where $E_n^{(q)}$ is the likelihood ratio e-value under alternative q . Because each $E_n^{(q)}$ has expectation at most 1 under the null, so does their average. The mixture is a valid e-variable no matter which q is true.

More generally, we can replace the discrete average with a continuous mixture using a probability measure ν over the set of alternatives $\mathcal{Q}$ (Ramdas and Wang 2025, Equation 3.7):

$$E_n = \int \prod_{i=1}^n \frac{q(X_i)}{p(X_i)} \nu(dq).$$

This is always a valid e-variable. Under regularity and support conditions on ν , it achieves asymptotic log-optimality: $\frac{1}{n} \mathbb{E}_{Q^n} [\log E_n] \rightarrow \text{KL}(Q \| P)$ for any $Q \in \mathcal{Q}$ (Ramdas and Wang 2025, Equation 3.6). The mixing distribution ν plays a role similar to a Bayesian prior, but validity does not depend on ν being “correct”. Only asymptotic efficiency does.

2.10.2 Plug-in method: learning as you go

Instead of hedging upfront, we can **adapt** our bet as data accumulates. The idea is simple: use the first $i-1$ observations to estimate the alternative, then place our bet on observation i using that estimate.

Concretely, suppose we are testing a fair coin and have already seen 9 tosses with 7 heads. Our current best guess for the bias is $\hat{q}_9 = 7/9 \approx 0.78$. We use this as the alternative to construct the likelihood ratio for the 10th toss. Crucially, the estimate $\hat{q}_9$ does not use the 10th observation. It looks only backward, never forward.

In general, let $\hat{Q}_{i-1}$ be an estimate of Q based on $(X_1, \dots, X_{i-1})$, with density $\hat{q}_{i-1}$. The plug-in e-variable is (Ramdas and Wang 2025, Equation 3.8)

$$E_n = \prod_{i=1}^n \frac{\hat{q}_{i-1}(X_i)}{p(X_i)}.$$

Because each factor uses only past data to form its bet, the running product is a non-negative martingale under P : at every step, the expected value of the next factor, given everything seen so far, equals 1 whenever $\hat{q}_{i-1}$ is a proper probability density, since

$$\mathbb{E}_P \left[\frac{\hat{q}_{i-1}(X_i)}{p(X_i)} \mid X_1, \dots, X_{i-1} \right] = \int \frac{\hat{q}_{i-1}(x)}{p(x)} p(x) dx = \int \hat{q}_{i-1}(x) dx = 1.$$

If $\hat{q}_{i-1}$ is instead a sub-probability density, the conditional expectation is at most 1 and the running product is a supermartingale; the validity argument goes through in both cases (see the discussion preceding Example 1.5 in Ramdas and Wang 2025). Common choices for $\hat{Q}_{i-1}$ include the MLE or posterior mean from the first $i-1$ observations (where the initial $\hat{Q}_0$ is simply set to the null distribution or a neutral prior).

2.10.3 Adaptive strategies in R

We implement several strategies from Example 1.5 in Ramdas and Wang (2025) for testing $H_0 : \theta = 0$ vs $H_1 : \theta > 0$ with $N(\theta, 1)$ data, where the true parameter is $\theta^\dagger = 0.3$.

```
theta_true <- 0.3
n <- 500
set.seed(42)
x <- rnorm(n, mean = theta_true, sd = 1)

# (i) Oracle: knows the true theta
log_M_oracle <- cumsum(theta_true * x - theta_true^2 / 2)

# (ii) Misspecified: uses theta = 0.1
theta_mis <- 0.1
log_M_mis <- cumsum(theta_mis * x - theta_mis^2 / 2)

# (iii) Clipped MLE plug-in: theta_i is estimated from X_1, ..., X_{i-1}
# For the first observation we have no data, so we use theta = 0.1 as a
# starting guess (following Ramdas & Wang). This means we still extract
# some evidence from the very first data point, unlike initialising at 0.
log_M_mle <- numeric(n)
theta_init <- 0.1
log_M_mle[1] <- theta_init * x[1] - theta_init^2 / 2
for (i in 2:n) {
  theta_hat <- max(mean(x[1:(i-1)]), 0.001) # MLE, clipped to be positive
  log_M_mle[i] <- log_M_mle[i-1] + theta_hat * x[i] - theta_hat^2 / 2
}

# (iv) Mixture over a grid of theta values
theta_grid <- seq(0.01, 0.6, length.out = 30)
log_LR_grid <- sapply(theta_grid, function(th) {
  cumsum(th * x - th^2 / 2)
})
# Mixture: average the likelihood ratios on the original scale, then take
# ↪ log.
# We use the log-sum-exp trick for numerical stability (see note below).
log_M_mix <- apply(log_LR_grid, 1, function(row) {
  max_val <- max(row)
  max_val + log(mean(exp(row - max_val)))
})
```

```
df_strat <- data.frame(
  t = rep(1:n, 4),
  log_M = c(log_M_oracle, log_M_mis, log_M_mle, log_M_mix),
  strategy = rep(c("(i) Oracle (theta=0.3)",
                    "(ii) Misspecified (theta=0.1)",
                    "(iii) Clipped MLE plug-in",
                    "(iv) Mixture"), each = n)
)

ggplot(df_strat, aes(x = t, y = log_M, color = strategy)) +
  geom_line(linewidth = 0.6) +
  geom_hline(yintercept = log(20), linetype = "dashed", color = "grey40") +
  annotate("text", x = 490, y = log(20) - 0.7,
          label = "log(1/alpha) at alpha = 0.05",
          hjust = 1, size = 3, color = "grey40") +
  labs(x = "Sample size", y = "log(e-process)",
       color = NULL) +
  theme(legend.position = "bottom")
```

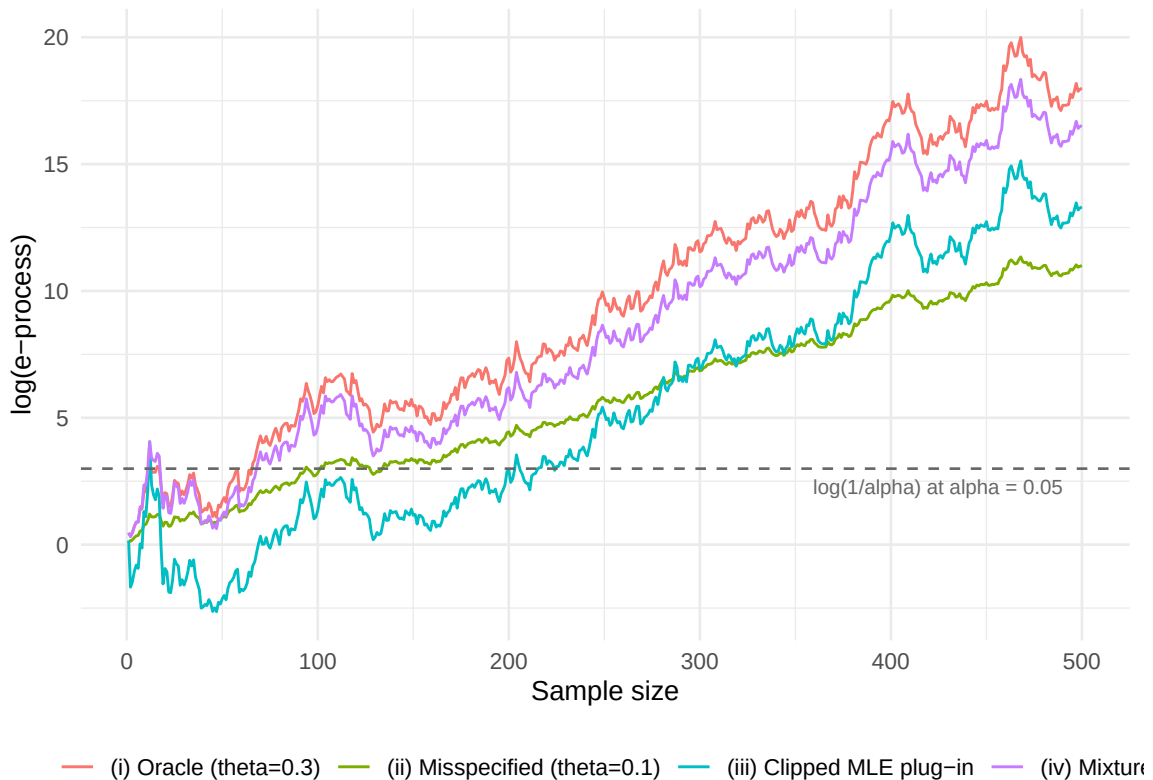


Figure 2.5: Sequential e-processes (log scale) for different strategies of choosing the alternative. The true parameter is $\theta^\dagger = 0.3$.

Figure 2.5 compares the four accumulated-evidence paths. In this run the oracle path, which knows $\theta^\dagger$, grows fastest, consistently with its maximal expected log-growth rate. The misspecified alternative grows more slowly but remains valid. The clipped MLE plug-in and mixture paths move towards the oracle benchmark in this run, consistently with

the asymptotic results discussed in Section 2.10.

A practical implementation note: when computing the mixture e-value, directly evaluating `mean(exp(log_LR))` will cause numerical overflow in R for large sample sizes, since individual log-likelihood ratios can easily exceed 700 (the limit of `exp()` in double precision). The log-sum-exp trick (subtracting the maximum before exponentiating and adding it back afterward) avoids this by keeping all intermediate values in a numerically safe range.

2.11 The cost of the Markov threshold: a fixed-sample power comparison

A natural question: if we build a test from an e-value via Markov's inequality, how does its power compare to the optimal p-value test? (see Ramdas and Wang 2025, sec. 1.2.2 for a detailed discussion of this trade-off). What follows concerns a likelihood-ratio e-variable read through the universal threshold $1/\alpha$, not the e-variable formalism as such: any binary level- α test φ can itself be written as the all-or-nothing e-variable $E = \varphi/\alpha$, which satisfies $\mathbb{E}_P[E] \leq 1$ and rejects exactly when φ does, so it has identical power.

For testing $P = N(0, 1)$ against $Q = N(\mu, 1)$ at level α :

- The **z-test** rejects when $Z > z_\alpha$, where $z_\alpha = \Phi^{-1}(1 - \alpha)$.
- The **Markov test** rejects when $E_n = \exp(\delta Z - \delta^2/2) \geq 1/\alpha$, i.e., when $Z \geq \delta/2 + \log(1/\alpha)/\delta$.

For this comparison, we grant the Markov test an oracle advantage in the growth-rate sense: at each true parameter θ we set $\mu = \theta$, the value that maximises the e-power of Section 2.9. The resulting curve therefore evaluates a different e-variable at each θ , rather than tracing the power function of a single pre-specified procedure; the best single choice of μ is considered after the figure.

```
alpha <- 0.05
n <- 30
theta_grid <- seq(0, 0.8, length.out = 200)

# Z-test power: reject if Z > qnorm(1 - alpha)
z_thresh <- qnorm(1 - alpha)
power_z <- pnorm(z_thresh, mean = theta_grid * sqrt(n), lower.tail = FALSE)

# Markov test power: reject if exp(delta*Z - delta^2/2) >= 1/alpha
# i.e. Z >= delta/2 + log(1/alpha)/delta, with delta = mu*sqrt(n)
# We use the oracle choice mu = theta for each theta
power_markov <- sapply(theta_grid, function(th) {
  delta <- th * sqrt(n)
  if (delta == 0) return(0) # oracle uses mu=0, so E=1 always; never rejects
  e_thresh <- delta / 2 + log(1 / alpha) / delta
  pnorm(e_thresh, mean = th * sqrt(n), lower.tail = FALSE)
})

df_power <- data.frame(
  theta = rep(theta_grid, 2),
  power = c(power_z, power_markov),
  test = rep(c("Z-test (p-value)", "Markov test (oracle-tuned e-value)"),
    each = length(theta_grid))
)
```

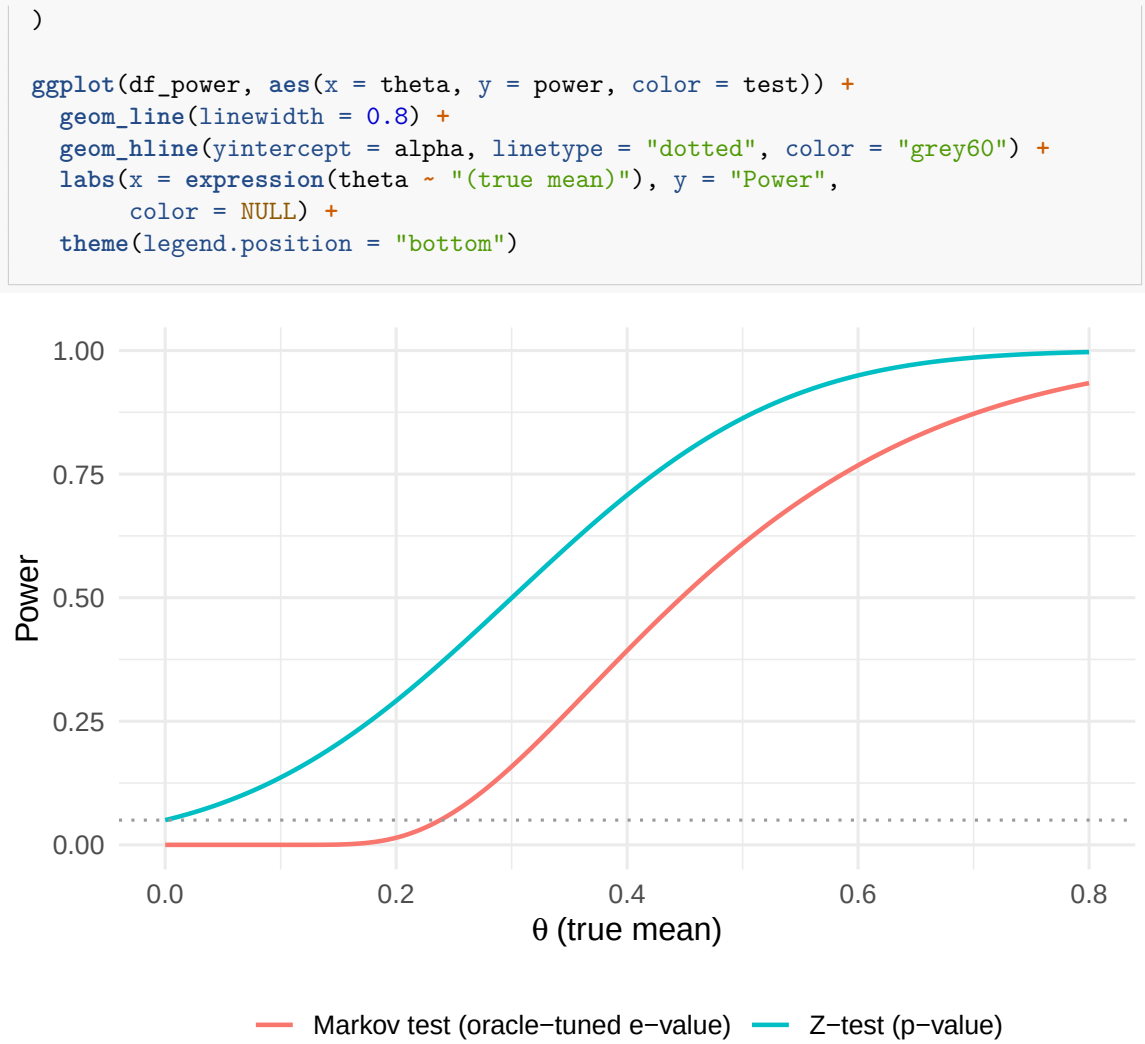


Figure 2.6: Rejection probabilities for the z-test and for the Markov-thresholded likelihood-ratio e-variable tuned to each true θ , at level $\alpha = 0.05$ with $n = 30$ observations.

Figure 2.6 shows that the z-test is uniformly more powerful for fixed-sample testing at $\theta > 0$, reflecting the conservatism of Markov's threshold. The point $\theta = 0$ should be interpreted separately: there the null is true, so the vertical axis represents type-I error rather than power. In this oracle comparison, the Markov test has type-I error 0 because the oracle sets $\mu = 0$, which makes the e-variable equal to 1 identically and hence prevents rejection. This should not be read as a general advantage of e-values: in a practical safe test, μ is fixed in advance, so the e-variable under $\theta = 0$ is a genuine random variable, and its type-I error is at most α rather than exactly zero (as illustrated in Figure 3.3). The substantive trade-off is the power gap for $\theta > 0$, where the z-test rejects more often even when the Markov construction is given oracle knowledge of the true alternative.

The oracle tuning $\mu = \theta$ is optimal for evidence growth, but not for fixed-horizon rejection. For a fixed δ the Markov rule rejects when $Z \geq \delta/2 + \log(1/\alpha)/\delta$, and this boundary is minimised at

$$\delta_{\star} = \sqrt{2 \log(1/\alpha)},$$

where it takes the value $\delta_{\star}$ itself. At $\alpha = 0.05$ the smallest attainable boundary is therefore

2.448, against $z_\alpha = 1.645$ for the classical test, and the rejection probability of that best single choice under the null is 0.0072. Even the best fixed μ therefore rejects less often than the z-test at every $\theta > 0$, so a fixed-sample gap remains that is attributable to the conservative Markov boundary alone. The gap displayed in Figure 2.6 is at least as large, since its pointwise choice $\mu = \theta$ is optimal for e-power rather than for fixed-horizon rejection.

However, this comparison is misleading in isolation. E-variable methods offer forms of flexibility that the oracle fixed-horizon comparison does not capture (see Ramdas and Wang 2025, sec. 1.2.1 and 4.1):

- **Optional stopping.** A valid e-process may be monitored continuously and evaluated at a data-dependent stopping time. Repeatedly recomputing an unadjusted fixed-horizon p-value does not retain the corresponding guarantee.
- **Combination.** E-variables can be averaged under arbitrary dependence and multiplied under independence. P-values can also be combined, but require dedicated merging procedures.
- **Post-hoc error levels.** A single e-variable generates a family of threshold tests that remains valid under a ratio-based post-hoc criterion, even when the reported error level is data-dependent.

The final point is not ordinary type-I error control at a randomly chosen level. Formally, for every data-dependent $\hat{\alpha} \in (0, 1)$,

$$\mathbb{E}_P \left[\frac{\mathbf{1}\{E \geq 1/\hat{\alpha}\}}{\hat{\alpha}} \right] \leq \mathbb{E}_P[E] \leq 1.$$

This is the post-hoc validity criterion of Ramdas and Wang (2025), Definition 4.1 and Proposition 4.3; see also Grünwald (2024). These forms of flexibility are developed in the subsequent chapters.

2.12 Informal evidence grids

Unlike p-values, which have conventional decision thresholds such as 0.05 and 0.01, there are no universally agreed evidential categories for e-values. Jeffreys' scale can nevertheless serve as a rough descriptive guide for e-values that arise naturally as likelihood ratios or as Bayes factors satisfying the e-variable bound (Ramdas and Wang 2025, sec. 2.7, Table 2.19). The categories in Table 2.1 are heuristic descriptions rather than testing thresholds and should not be applied mechanically to arbitrary e-values.

Table 2.1: Informal evidence grid for e-values, adapted from Jeffreys' scale.

E-value range	Level of evidence
1 to $\sqrt{10} \approx 3.16$	Weak
$\sqrt{10}$ to 10	Substantial
10 to $10\sqrt{10} \approx 31.6$	Strong
$10\sqrt{10}$ to 100	Very strong
> 100	Decisive

The categories in Table 2.1 lie on a logarithmic scale, in powers of $\sqrt{10}$, reflecting the multiplicative nature of e-values. The grid applies best to naturally arising evidence

measures such as likelihood ratios, not to all-or-nothing e-variables of the form $\mathbf{1}\{U \leq \alpha\}/\alpha$. Values $E \leq 1$ indicate that the Skeptic did not increase their wealth and therefore provide no evidence against the null.

2.13 Summary of key takeaways

The core messages from this chapter are:

1. **E-variables measure evidence on a multiplicative scale:** an e-variable satisfies $\mathbb{E}_P[E] \leq 1$ under the null, and larger realised e-values indicate more evidence against it.
2. **Markov's inequality gives instant tests:** reject when $E \geq 1/\alpha$. The conversion is universally valid, though often conservative.
3. **E-variables combine cleanly:** arithmetic averages are valid under arbitrary dependence, while products are valid under independence and, more generally, under the conditional-validity construction developed later.
4. **The likelihood ratio is the canonical e-variable:** it is log-optimal for simple hypotheses, with e-power equal to the KL divergence.
5. **Flexibility can cost fixed-sample power:** e-processes support optional stopping, while e-variables also permit direct combination and post-hoc valid threshold families.
6. **Calibrators connect the two worlds:** the only admissible deterministic e-to-p calibrator is $e \mapsto \min(1/e, 1)$, while the class of p-to-e calibrators is rich.
7. **Adaptive strategies handle composite alternatives:** when the true alternative is unknown, methods like continuous mixtures or sequential MLE plug-ins allow us to learn the alternative on the fly while maintaining rigorous validity under the null.

Chapter 3

Safe Tests for Classical Problems

The foundations of Chapter 2 showed that e-values provide a valid and flexible framework for hypothesis testing. In this chapter, we show that several standard tests a statistics student encounters, including the z-test, t-test, tests for contingency tables, and the logrank test, admit principled e-value versions that remain valid under optional stopping and optional continuation while still accumulating evidence efficiently. Following Grünwald et al. (2024), we refer to these as **safe tests**.

Most sections follow a common template. We present the theory and motivation behind the safe version of a classical test, implement it using the **safestats** R package (Turner et al. 2022), compare its behaviour with the classical test on the same data, and demonstrate its validity under optional stopping.

3.1 The safe testing framework

3.1.1 From log-optimality to growth-rate optimality

In Section 2.9 we saw that, for simple-vs-simple testing, the log-optimal e-variable is the likelihood ratio, with e-power equal to the KL divergence. When the alternative is composite, as is typical in practical testing problems, the question becomes which e-variable to use.

Grünwald et al. (2024) propose a principled answer in the **growth-rate optimal (GRO)** e-variable. GRO is an analogue of power optimality for e-values, but in a different sense from the Neyman–Pearson framework. Whereas Neyman–Pearson theory maximises the probability of rejection at a fixed sample size, GRO maximises the expected rate of evidence accumulation. As discussed in Section 2.9, taking the expectation of the logarithm corresponds to maximising the long-run growth rate of a gambler’s wealth, which motivates the term “growth-rate optimality.”

Definition 3.1 (Growth-rate optimality (Grünwald et al. (2024), Section 2)). Let $\mathcal{P}$ be the null hypothesis and let Q be a simple alternative distribution. A **Q -GRO e-variable** is any maximiser, when it exists, of

$$\mathbb{E}_Q[\log E]$$

over all e-variables $E \in \mathcal{E}(\mathcal{P})$, where $\mathcal{E}(\mathcal{P})$ denotes the set of all e-variables for $\mathcal{P}$.

By Theorem 2.1 (the log-optimality theorem from Chapter 2), when $\mathcal{P} = \{P\}$ is a simple null, the GRO e-variable is the likelihood ratio dQ/dP , and the corresponding optimal growth rate is the KL divergence $\text{KL}(Q\|P)$. The main contribution of Grünwald et al. (2024) is to extend this idea to **composite nulls**, the setting of most practical tests. For instance, a t-test has a composite null because the variance σ^2 is unknown, so H_0 specifies a whole family of distributions rather than a single one.

3.1.2 The GROW criterion for composite hypotheses

When both the null and the alternative are composite, for example testing a range $\Theta_0 = \{\theta : \theta \leq \delta^-\}$ against a range $\Theta_1 = \{\theta : \theta \geq \delta^+\}$, we face the problem that the best e-variable depends on the true θ , which we do not know. The **GROW (growth-rate optimal in worst-case)** criterion resolves this by seeking an e-variable that maximises the *worst-case* growth rate over the alternative (Grünwald et al. 2024, sec. 3). More precisely, a GROW e-variable is any maximiser, when it exists, of

$$\inf_{\theta \in \Theta_1} \mathbb{E}_{P_\theta}[\log E]$$

over all e-variables $E \in \mathcal{E}(\mathcal{P})$.

Specifying a minimum relevant effect size δ^+ that separates Θ_0 from Θ_1 is essential here. If we defined the alternative as $\theta > 0$ without any gap, the infimum would be driven by parameters arbitrarily close to the null boundary, and the worst-case growth rate would collapse to zero. In that case, the criterion no longer distinguishes practically useful e-variables. This is one reason why safe testing asks the researcher to state which effect sizes are scientifically relevant. By contrast, Grünwald et al. (2024) (Example 3) show that a standard Bayes-factor construction based on a diffuse Gaussian prior over the whole alternative can become overly conservative in this setting, requiring signals of order $\sqrt{(\log n)/n}$ for rejection rather than the classical $1/\sqrt{n}$. Classical p-value tests, by contrast, do not require a minimum relevant effect size for validity, even though such an effect size may still be needed for design or power calculations. GROW therefore forces the researcher to state explicitly which effects are worth detecting, a requirement that may feel unfamiliar at first, but can also lead to more scientifically meaningful tests.

To analyse this criterion, it is helpful to first fix the alternative. Theorem 1 of Grünwald et al. (2024) requires full support for Q and a finite infimum of $D(Q\|P_{W_0})$ over the null-model Bayes marginals. Under those conditions, once the alternative has been fixed, either as a single distribution Q or as a chosen mixture Q over the alternative parameter, the corresponding optimal e-variable has the likelihood-ratio form

$$E^* = \frac{q(Y)}{p_0^*(Y)}.$$

Here P_0^* is the **Reverse Information Projection (RIPr)** of Q onto the null-model Bayes marginals. In the examples used in this chapter, this projection is attained by an ordinary probability distribution. In general, Theorem 1 allows P_0^* to be a sub-probability. For composite alternatives, the choice of mixture is itself part of the design, and GROW corresponds to choosing this mixture so as to maximise the worst-case growth rate over the alternative. For a simple null $\mathcal{P} = \{P\}$, this RIPr is just P itself, so we recover the ordinary likelihood ratio. For the composite-null examples below, the RIPr identifies the null distribution that is hardest to distinguish from the chosen alternative, and betting

against this least favourable null yields an e-variable that remains valid uniformly over the whole null. Figure 3.1 illustrates the corresponding geometric picture.

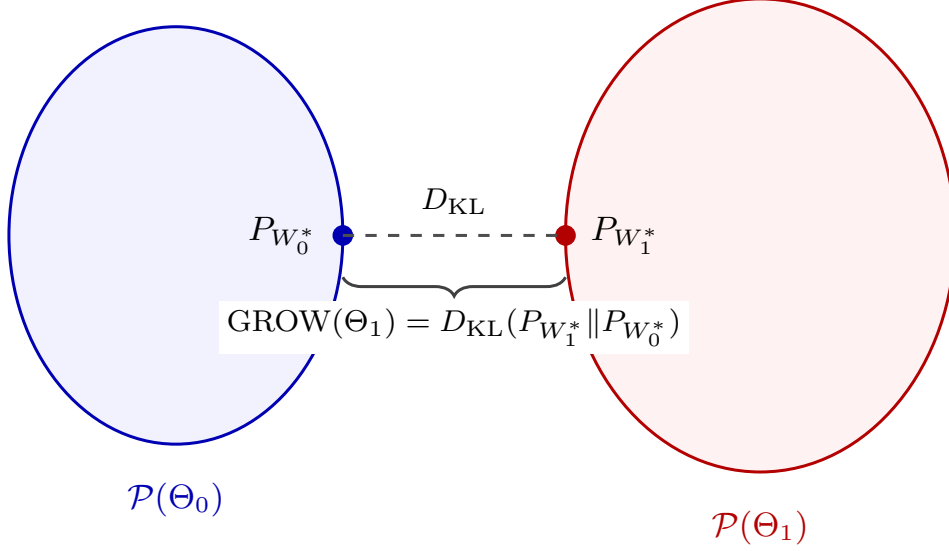


Figure 3.1: Joint information projection (adapted from Figure 1 of Grünwald et al. (2024)). The figure shows the two distributions $P_{W_1}^*$ and $P_{W_0}^*$ forming the least-favourable pair, that is, the pair that minimises the KL divergence between the alternative and null sets. In the resulting least-favourable pair, the GROW e-variable is the likelihood ratio between these two distributions.

For one-dimensional models with separated null and alternative hypotheses and a monotone likelihood ratio in a statistic, Grünwald et al. (2024) obtain a particularly clean boundary-point result in Proposition 3. One-parameter exponential families are an important special case, treated in their Example 4 in the mean-value parameterisation.

Proposition 3.1 (GROW under a monotone likelihood ratio). *Following Proposition 3 of Grünwald et al. (2024), suppose $\{P_\theta : \theta \in \Theta\}$ is a one-parameter model on a connected ordered parameter space and there exists a one-dimensional statistic $T = t(Y)$ such that the family has a monotone likelihood ratio in T . Fix separation points $\delta^- < \delta^+$ in Θ . For testing $\Theta_0 = \{\theta : \theta \leq \delta^-\}$ against $\Theta_1 = \{\theta : \theta \geq \delta^+\}$, the GROW e-variable is*

$$E^* = \frac{p_{\delta^+}(Y)}{p_{\delta^-}(Y)},$$

the likelihood ratio between the two boundary points. For an iid one-parameter exponential family in the mean-value parameterisation, Example 4 verifies this condition for the joint sample $Y^n = (Y_1, \dots, Y_n)$, with $p_\theta(Y^n) = \prod_{i=1}^n p_\theta(Y_i)$.

This result is striking, because even though both hypotheses are composite, the optimal e-variable reduces to a simple likelihood ratio. In the Gaussian location family with known variance (the setting of the safe z-test), this corresponds to testing $H_0 : \mu \leq 0$ against $H_1 : \mu \geq \delta^+$ with $\delta^- = 0$, which yields the GROW e-variable

$$E^* = \frac{p_{\delta^+}(Y^n)}{p_0(Y^n)}.$$

3.1.3 Nuisance parameters and the safe t-test

The GROW result above assumes that the only unknown quantity is the parameter of interest. In a z-test, the variance is known, so this works cleanly. In a t-test, however, the variance σ^2 is also unknown. This introduces a nuisance parameter, which is not itself the object of inference but still affects the distribution of the data and therefore the validity of the e-variable.

To see the difficulty, suppose we tried to apply GROW directly by constructing a likelihood-ratio e-variable as if $\sigma^2 = 1$ were known. An e-variable must satisfy $\mathbb{E}_P[E] \leq 1$ for every distribution in the null model, including every admissible value of σ . If the true variance were different, say $\sigma^2 = 100$, then the resulting statistic would generally no longer be calibrated against the whole null and might fail to be a valid e-variable. In the language of Section 2.4, the bet would be calibrated to the wrong game.

The solution developed in Grünwald et al. (2024) (Section 4.3) is to restrict attention to e-variables that are **invariant under rescaling of the data**. An invariant e-variable depends on the data only through scale-free statistics, whose distribution does not depend on σ , so the nuisance parameter effectively drops out and the GROW criterion can be applied within this restricted class. The **REGROW** (relative growth-rate optimal in worst-case) criterion plays a complementary role. It compares the growth rate of a candidate e-variable to the oracle GRO growth rate that would be available if σ were known, and Grünwald et al. (2024) use it to justify the invariance restriction, because the scale-invariant GROW e-variable is also worst-case optimal in the REGROW sense among all e-variables, not only the invariant ones (Grünwald et al. 2024; Pérez-Ortiz et al. 2024).

The key technical step is to reduce the data to a scale-free statistic whose distribution no longer depends on σ . In the t-test setting, this can be done by normalising the pooled data vector by a common scale factor, so that the nuisance parameter cancels out. As discussed in Grünwald et al. (2024) (Section 4.3), this coarsening step is mathematically equivalent to using the scale-invariant right-Haar prior $\pi(\sigma) \propto 1/\sigma$ on the scale parameter. The resulting e-variable therefore has a simple closed-form representation as a Bayes factor with right-Haar prior (Grünwald et al. 2024, sec. 4.3; Berger et al. 1998). Thus, a construction familiar from objective Bayesian analyses of the t-test also receives a frequentist optimality justification in the safe-testing framework. For the applied researcher, the practical point is that **safestats** implements this construction internally, so the resulting e-value can be used just like the other safe-test e-values and retains the same validity under optional stopping.

3.1.4 Optional stopping and optional continuation

The key practical consequence of using e-values for testing is **safety under optional stopping and optional continuation**. If $E^{(1)}, E^{(2)}, \dots$ are conditional e-variables from successive studies (formally defined in Section 4.3), meaning that each $E^{(j)}$ remains an e-variable conditional on the information available before study j , then their running product

$$\mathcal{E}^{(m)} = \prod_{j=1}^m E^{(j)}$$

forms a test supermartingale (Definition 4.1) under the null, that is, a nonnegative process whose conditional expectation cannot increase under the null. Consequently, Ville's

inequality (formalised as Theorem 4.1 in the next chapter) implies (Grünwald et al. 2024)

$$P(\exists m : \mathcal{E}^{(m)} \geq 1/\alpha) \leq \alpha$$

for all $P \in \mathcal{P}$. This guarantee holds regardless of the rule used to decide whether to continue collecting data, whether to start a new study, or when to stop. As Ly et al. (2025) emphasise, this means that the evidence threshold $1/\alpha$ can be monitored continuously during data collection, and crossing it at any time constitutes a valid rejection at level α .

3.2 Safe z-test

We begin with the simplest case, testing a normal mean with known variance. This is the safe analogue of the classical one-sided z-test. Although the assumption of known variance is rarely realistic, the safe z-test illustrates the main ideas in their simplest form before we move to the more practical t-test.

3.2.1 Setup and the GROW e-variable

Consider $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\theta, 1)$. We test

$$H_0 : \theta \leq 0 \quad \text{against} \quad H_1 : \theta \geq \delta_{\min},$$

for a pre-specified minimum relevant effect size $\delta_{\min} > 0$. In the notation of Proposition 3.1, we set $\delta^+ = \delta_{\min}$ and $\delta^- = 0$. The resulting GROW e-variable is the likelihood ratio

$$E_n = \frac{p_{\delta_{\min}}(X^n)}{p_0(X^n)} = \exp\left(\delta_{\min} S_n - \frac{n\delta_{\min}^2}{2}\right),$$

where $S_n = \sum_{i=1}^n X_i$. This is exactly the Gaussian likelihood ratio from Section 2.6, evaluated at $\mu = \delta_{\min}$.

Because the Gaussian likelihood factorises over observations, the process $(E_n)_{n \geq 1}$ can be written as a running product of one-step likelihood ratios,

$$E_n = \prod_{i=1}^n \exp\left(\delta_{\min} X_i - \frac{\delta_{\min}^2}{2}\right).$$

Under the boundary null $\theta = 0$, this process is a nonnegative martingale, and under the composite null $H_0 : \theta \leq 0$ it is a nonnegative supermartingale. Ville's inequality, formalised as Theorem 4.1 in the next chapter, therefore guarantees type-I error control under arbitrary stopping.

3.2.2 Implementation with **safestats**

The **safestats** package provides functions for designing and executing safe z-tests. The workflow consists of two steps. First, create a design object that specifies the test parameters. Second, apply the test to data.

```
# Design a safe z-test: meanDiffMin = 0.3, alpha = 0.05, power = 0.8
# Note: designSafeZ uses 'meanDiffMin' (not deltaMin), since sigma is known
# capture.output suppresses the txtProgressBar output from the simulations
```

```

set.seed(2026)
invisible(capture.output(
  design_z <- designSafeZ(meanDiffMin = 0.3, alpha = 0.05, beta = 0.2,
                           alternative = "greater")
))
print(design_z)

```

Safe One Sample Z-Test Design

```

              nPlan±2se = 103±6.600024
              nMean±2se = 61±1.936307
minimal mean difference = 0.3
              alternative = greater
              power: 1 - beta = 0.8
              parameter: phiS = 0.3
                      alpha = 0.05
decision rule: e-value > 1/alpha = 20

```

Timestamp: 2026-08-27 20:56:42 CEST

Note: If it is only possible to look at the data once, then `nPlan` = 131.

The design object provides several useful planning quantities. Using internal simulation under the design assumptions ($\alpha = 0.05$, power = 0.8, $\delta_{\min} = 0.3$), `safestats` reports the sequential planning horizon, the expected realised sample size, and the test-defining parameter. Let E_n denote the running e-value and define

$$\tau_\alpha = \inf\{n \geq 1 : E_n \geq 1/\alpha\}.$$

We briefly interpret the main components.

parameter: phiS = 0.3.

This is the safe test defining parameter ϕ_S . In the present z-test setting, it corresponds to the alternative value used in the likelihood ratio, so the package output is consistent with the GROW construction from Section 3.1.

decision rule: e-value > 1/alpha = 20.

This confirms that the rejection threshold is $1/\alpha = 20$, the standard Markov threshold from Proposition 2.1.

nPlan = 103.

If we monitor the data sequentially, this is the planned monitoring horizon calibrated so that, under the design alternative,

$$P_{\theta=\delta_{\min}}(\tau_\alpha \leq n_{\text{Plan}}) = P_{\theta=\delta_{\min}}(\exists n \leq n_{\text{Plan}} : E_n \geq 1/\alpha) \approx 0.8.$$

Thus, for the present design, `nPlan` means that the safe z-test crosses the threshold 20 by this sample size with probability approximately 0.8 when $\theta = \delta_{\min} = 0.3$.

nMean = 61.

This is the expected realised sample size under the same design alternative and monitoring

rule, approximately

$$\mathbb{E}_{\theta=\delta_{\min}}[\min(\tau_{\alpha}, n_{\text{Plan}})].$$

Because the test may stop as soon as the e-value crosses 20, the expected realised sample size is typically smaller than the planned monitoring horizon.

Batch note: `nPlan = 131` (single look).

If we commit to a fixed design and analyse the data only once at the end, the relevant power statement is no longer a first-crossing probability. Instead, the single-look sample size is calibrated so that

$$P_{\theta=\delta_{\min}}(E_{n_{\text{batch}}} \geq 1/\alpha) \approx 0.8.$$

For the present design, this gives 131 observations. For comparison, the classical one-sided z-test with known variance requires a sample size of

$$n = \left\lceil \frac{(z_{\alpha} + z_{\beta})^2}{\delta_{\min}^2} \right\rceil,$$

where $z_{\alpha} = \Phi^{-1}(1 - \alpha)$ and $z_{\beta} = \Phi^{-1}(1 - \beta)$ are the standard normal quantiles entering the usual fixed-horizon power calculation. With $\alpha = 0.05$, $\beta = 0.2$, and $\delta_{\min} = 0.3$, this evaluates to 69 observations. In this batch-style comparison, the safe test therefore requires roughly $1.9\times$ more data, reflecting the cost of retaining anytime validity without actually using early stopping.

Importantly, these design quantities are planning tools rather than binding requirements. Unlike classical fixed-horizon p-values, which lose their nominal error control under unplanned repeated looks, the safe z-test remains valid no matter when data collection stops.

3.2.3 Hand-coded vs `safestats` comparison

We now verify that the `safestats` implementation agrees with the hand-coded safe z-test e-variable derived above. Concretely, this is the same Gaussian likelihood ratio as in Section 2.6, specialised to the present GROW setup with $\delta_{\min} = 0.3$. We simulate data under the alternative $\theta^{\dagger} = 0.3$ and compute both versions.

```
theta_true <- 0.3
delta_min <- 0.3
n <- 150
set.seed(42)
x <- rnorm(n, mean = theta_true, sd = 1)

# Hand-coded GROW e-variable (running product of per-observation LRs)
log_e_hand <- cumsum(delta_min * x - delta_min^2 / 2)

# safestats sequential computation for verification
# safestats requires at least 2 observations; we start at 3 to avoid
# edge-case warnings with very small samples
e_safestats <- numeric(n)
for (i in 3:n) {
  result_i <- safeZTest(x[1:i], designObj = design_z, alternative =
    ↪ "greater")
  e_safestats[i] <- result_i$eValue
}
```

```
# Verify agreement (compare on log scale, skip first 2 where safestats = 0)
log_e_safestats <- log(e_safestats)
max_diff <- max(abs(log_e_hand[3:n] - log_e_safestats[3:n]), na.rm = TRUE)
cat("Max |log(E_hand) - log(E_safestats)| for n = 3,...,150:",
    formatC(max_diff, digits = 2, format = "e"), "\n")
```

Max |log(E_hand) - log(E_safestats)| for n = 3,...,150: 7.11e-15

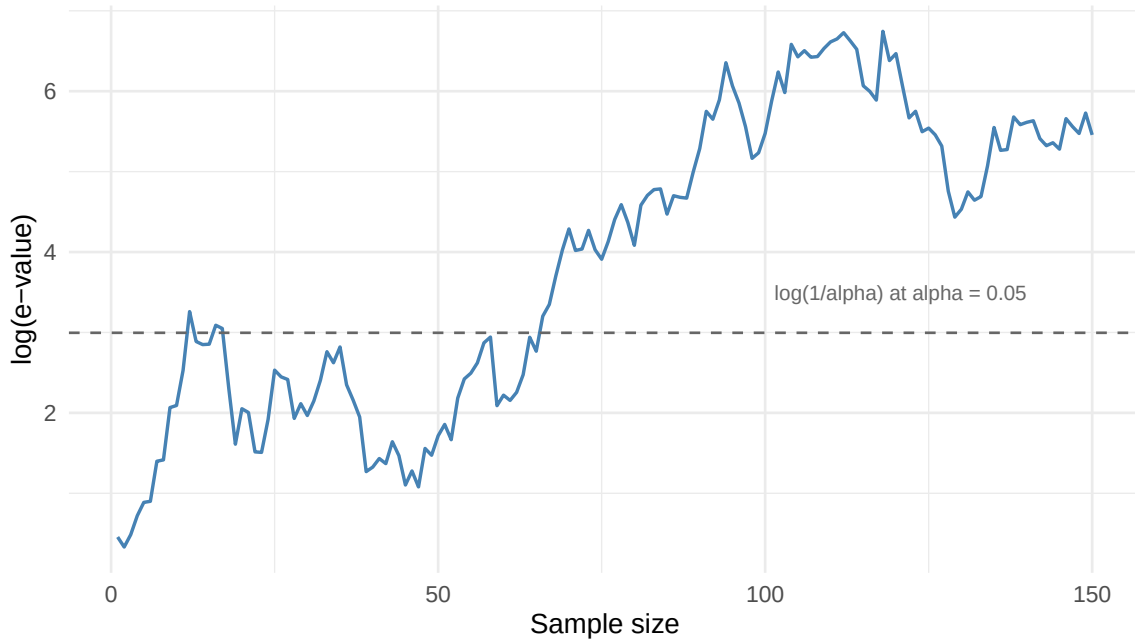


Figure 3.2: Sequential log e-value path for the safe z-test under the alternative $\theta^\dagger = 0.3$. The evidence crosses the rejection threshold $\log(1/\alpha) \approx 3$ well before $n = 100$. The hand-coded GROW formula and the `safestats` implementation agree to within numerical tolerance, with the maximum absolute difference printed above the figure.

Figure 3.2 shows the sequential path. For this simulated dataset, the maximum discrepancy between the hand-coded formula and `safestats` is $7.11\text{e-}15$, that is, within numerical tolerance. This shows that the package implementation matches the hand-coded likelihood-ratio e-variable up to numerical error. Both curves would therefore overlap visually, so we plot only the hand-coded path. Since the true effect equals the design alternative ($\theta^\dagger = \delta_{\min} = 0.3$), the expected log e-value increases by $\delta_{\min}^2/2 = 0.045$ per observation, giving the path a positive average drift under the alternative.

3.2.4 Comparison with the classical z-test

The key difference between the safe z-test and the classical z-test is their behaviour under repeated testing. We illustrate this with a simulation under the null.

```
n_max <- 200
n_sim <- 5000
delta_min <- 0.3

# Track cumulative false positive rates under continuous monitoring
```

```

fp_classical <- numeric(n_max)
fp_safe <- numeric(n_max)

set.seed(123)
for (sim in 1:n_sim) {
  x <- rnorm(n_max, mean = 0, sd = 1)
  s_cumsum <- cumsum(x)

  ever_rejected_classical <- FALSE
  ever_rejected_safe <- FALSE

  for (nn in 2:n_max) {
    # Classical z-test p-value (one-sided)
    z_stat <- s_cumsum[nn] / sqrt(nn)
    p_val <- 1 - pnorm(z_stat)
    if (p_val < 0.05 && !ever_rejected_classical) {
      ever_rejected_classical <- TRUE
    }
    if (ever_rejected_classical) fp_classical[nn] <- fp_classical[nn] + 1

    # Safe z-test e-value
    log_e <- delta_min * s_cumsum[nn] - nn * delta_min^2 / 2
    if (log_e >= log(20) && !ever_rejected_safe) {
      ever_rejected_safe <- TRUE
    }
    if (ever_rejected_safe) fp_safe[nn] <- fp_safe[nn] + 1
  }
}

fp_classical <- fp_classical / n_sim * 100
fp_safe <- fp_safe / n_sim * 100

df_fpr <- data.frame(
  n = rep(2:n_max, 2),
  fpr = c(fp_classical[2:n_max], fp_safe[2:n_max]),
  method = rep(c("Classical z-test (monitor p < 0.05)",
                  "Safe z-test (monitor E >= 20)"), each = n_max - 1)
)

```

This simulation illustrates what Ville's inequality guarantees theoretically. The safe z-test remains controlled at the nominal 5% level under continuous monitoring, whereas the classical z-test does not. In this simulation, the estimated false positive rate of the safe test stays close to and below the 5% boundary, while the classical z-test reaches 32.4% by $n = 200$. If monitoring were continued indefinitely, the classical rejection probability would converge to 1. Indeed, under the null, the partial sums satisfy the law of the iterated logarithm,

$$\limsup_{n \rightarrow \infty} \frac{S_n}{\sqrt{2n \log(\log n)}} = 1 \quad \text{almost surely,}$$

where $\log(\log n)$ denotes the logarithm applied twice. Hence the repeatedly monitored z-statistic $S_n/\sqrt{n}$ exceeds any fixed critical value at some time, indeed infinitely often, almost surely.

This difference arises because a classical p-value test is valid only when conducted once at

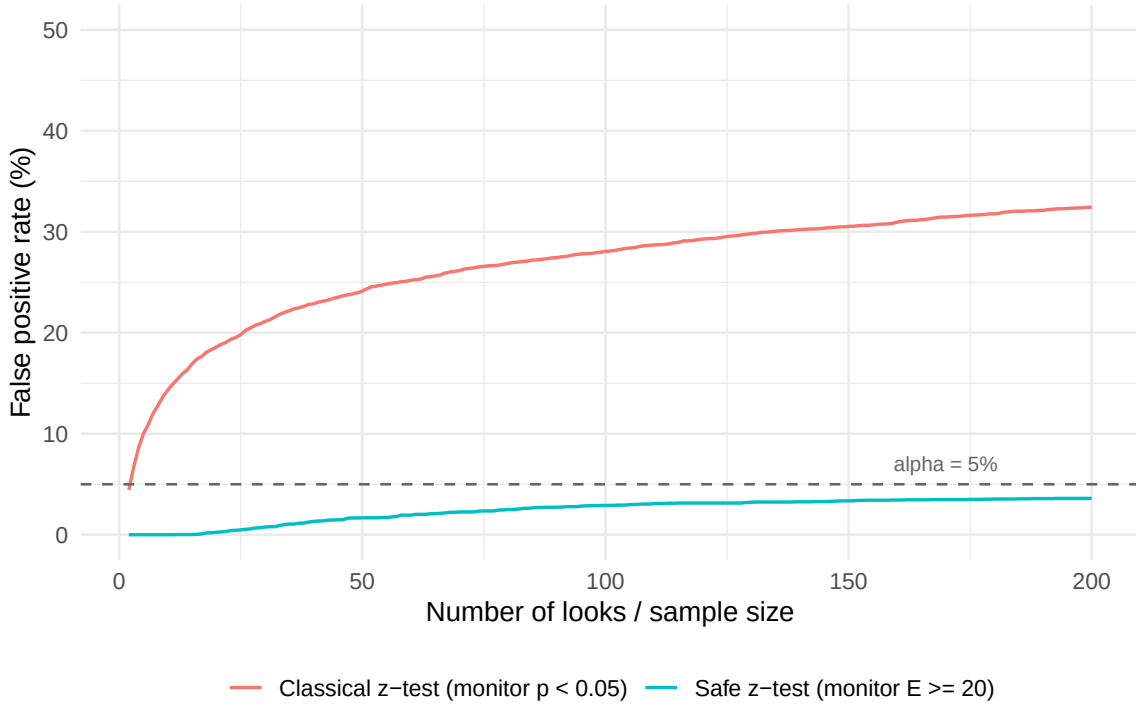


Figure 3.3: False positive rates under continuous monitoring. The classical z-test (monitoring $p < 0.05$) rapidly inflates beyond the nominal 5%. The safe z-test (monitoring $E \geq 20$) stays close to the nominal 5% level in this simulation.

a pre-specified sample size. If we repeatedly check whether $p < 0.05$ during data collection, the probability of eventually crossing that threshold becomes much larger than α . As Ly et al. (2025) emphasise (their Figure 2), this inflation is caused by repeated monitoring, not by any defect of the fixed-horizon z-test itself. At any fixed n , the classical test controls type-I error, but the probability of ever rejecting under continuous monitoring can become arbitrarily close to 1.

3.3 Safe t-test

The t-test is arguably the most widely used statistical test. Its safe analogue is therefore one of the most important examples in this chapter. The main challenge is the unknown variance σ^2 , which acts as a nuisance parameter.

3.3.1 One-sample derivation and two-sample implementation

The invariant derivation in Grünwald et al. (2024) is most transparent for the one-sample model

$$Y_1, \dots, Y_n \stackrel{\text{iid}}{\sim} N(\delta\sigma, \sigma^2),$$

where δ is the standardised mean and the unknown scale σ is a nuisance parameter. The null is $\delta = 0$, while the alternative is specified through a prior or a minimum relevant effect size.

The key insight of Grünwald et al. (2024) (Section 4.3, building on Berger et al. (1998) and Lai (1976)) is that the scale parameter can be removed by a **coarsening** of the data.

Let $Y = (Y_1, \dots, Y_n)$ and define $V = Y/|Y_1|$. Because every coordinate is multiplied by the same σ , the distribution of V depends on δ but not on σ . The resulting e-variable can be written as a likelihood ratio on the coarsened data,

$$E^* = \frac{p'_{W[\delta]}(V)}{p'_0(V)},$$

where p'_0 is the density of V under the null and $p'_{W[\delta]}$ is its density under a prior $W[\delta]$ on the effect size. As shown in Grünwald et al. (2024) (Equation 29), this coincides with the Bayes factor obtained by using the right-Haar prior $\pi(\sigma) \propto 1/\sigma$ on the scale parameter in both models.

Combining Theorems 1 and 2 of Grünwald et al. (2024) yields the optimality result. Theorem 2 is a special case of the group-invariance results of Pérez-Ortiz et al. (2024). For a fixed effect-size distribution $W[\delta]$ satisfying the finite- $(2 + \epsilon)$ -moment condition in Theorem 2, a condition satisfied by the finite-support GROW tuning distribution used below, the resulting e-variable is GROW relative to the class of proper product priors on (δ, σ) under which δ and σ are independent and the marginal on δ equals $W[\delta]$. This gives a frequentist optimality justification for a construction that also appears in objective Bayesian analyses of the t-test.

The applied example below uses an independent two-sample problem with a common unknown variance. Its equal-variance t-statistic is

$$T = \frac{\bar{X}_1 - \bar{X}_2}{S_p \sqrt{1/n_1 + 1/n_2}},$$

where S_p is the pooled standard deviation. Under the standardised effect $\delta = (\mu_1 - \mu_2)/\sigma$, this statistic has a noncentral t-distribution with $n_1 + n_2 - 2$ degrees of freedom and noncentrality parameter $\sqrt{n_{\text{eff}}} \delta$, where

$$n_{\text{eff}} = \left(\frac{1}{n_1} + \frac{1}{n_2} \right)^{-1}.$$

The `safestats` implementation evaluates the corresponding e-value through the central and noncentral t densities (Turner et al. 2022). The one-sample coarsening therefore provides the theoretical motivation for handling the unknown scale, while `designSafeT()` and `safeTTest()` provide the two-sample implementation used below.

3.3.2 E-process types for the safe t-test

Ly et al. (2025) (Section 1.2.5) discuss four types of e-processes for the safe t-test, each based on a different tuning distribution over the effect-size parameter. These distributions are selected and optimised as design choices rather than interpreted as Bayesian prior beliefs:

1. **grow**: a two-point tuning distribution at $\pm\delta_{\min}$. This is the GROW-optimal choice in the worst-case sense, so it is tuned to accumulate evidence as fast as possible at the boundary $|\delta| = \delta_{\min}$.
2. **mom**: a non-local moment tuning distribution with two modes near $\pm\delta_{\min}$. Ly et al. (2025) recommend this as a good compromise between efficiency and robustness to misspecification of $\delta_{\min}$.

3. **eGauss**: a Gaussian tuning distribution on the effect size, with scale chosen as a function of $\delta_{\min}$.
4. **eCauchy**: a Cauchy tuning distribution on the effect size. This is the heaviest-tailed option and the most robust to misspecification among the four types.

All four define valid e-processes under optional stopping, but they differ in how aggressively they accumulate evidence and how sensitive they are to the choice of $\delta_{\min}$. Ly et al. (2025) (Table 1) compare their resulting planned sample sizes. The **safestats** package implements the safe t-test through **designSafeT()**, which reports the test-defining parameter **deltaS**. For a two-sided test, **deltaS** determines the equal-weight numerator mixture of noncentral t densities with noncentrality parameters $+\sqrt{n_{\text{eff}}} \delta_S$ and $-\sqrt{n_{\text{eff}}} \delta_S$, while the denominator is the central t density. It therefore plays a role analogous to ϕ_S in the safe z-test. In the GROW design below, **deltaS** equals the stated **deltaMin**.

3.3.3 Applied example: safe t-test on simulated data

We simulate a two-sample experiment to illustrate the safe t-test workflow, designing the test, monitoring the evidence sequentially, and comparing the e-value path with the classical p-value path.

Design. We design a two-sample safe t-test with a minimum relevant standardised effect size of $\delta_{\min} = 0.5$, conventionally described as a medium effect under Cohen’s benchmarks (Cohen 1992), at level $\alpha = 0.05$ and target power 0.8.

```
design_t <- designSafeT(deltaMin = 0.5, alpha = 0.05, beta = 0.2,
                      testType = "twoSample", seed = 42)
```

Safe Two Sample T-Test Design

```
n1Plan±2se, n2Plan±2se = 94±5.060965, 94±5.060965
n1Mean±2se, n2Mean±2se = 57±1.675445, 57±1.675445
minimal standardised mean difference = 0.5
alternative = twoSided
power: 1 - beta = 0.8
parameter: deltaS = 0.5
alpha = 0.05
decision rule: e-value > 1/alpha = 20
```

Timestamp: 2026-08-27 20:56:47 CEST

Note: If it is only possible to look at the data once, then **n1Plan** = 111 and **n2Plan** = 111.

The design object reports a sequential planning horizon of 94 observations per group and an expected realised sample size of approximately 57 under the design alternative. If the safe test is evaluated only once, the corresponding requirement is 111 observations per group, whereas the classical two-sided equal-variance t-test requires 64 observations per group. The safe design therefore trades a larger monitoring horizon and single-look requirement

for the possibility of earlier stopping when evidence accumulates quickly. This is a design-specific comparison rather than a general claim that the safe test is more sample-efficient, and Section 3.3.4 compares fixed-sample power systematically. As with the safe z-test, these design quantities are planning tools rather than binding requirements. Optional stopping does not invalidate the test, but the design still determines the intended power properties.

Data generating process. We simulate $n = 120$ independent observations per group from Gaussian distributions with a common standard deviation $\sigma = 2$:

$$X_{1,i} \stackrel{\text{iid}}{\sim} N(4.5, 4), \quad X_{2,i} \stackrel{\text{iid}}{\sim} N(3.5, 4), \quad i = 1, \dots, 120.$$

The true standardised effect size is $\delta = (\mu_1 - \mu_2)/\sigma = (4.5 - 3.5)/2 = 0.5$, so the simulation is calibrated to the design value $\delta_{\min} = 0.5$.

```
set.seed(123)
n_per_group <- 120
sigma <- 2
mu1 <- 4.5
mu2 <- 3.5

x_group1 <- rnorm(n_per_group, mean = mu1, sd = sigma)
x_group2 <- rnorm(n_per_group, mean = mu2, sd = sigma)
n_start <- 10
```

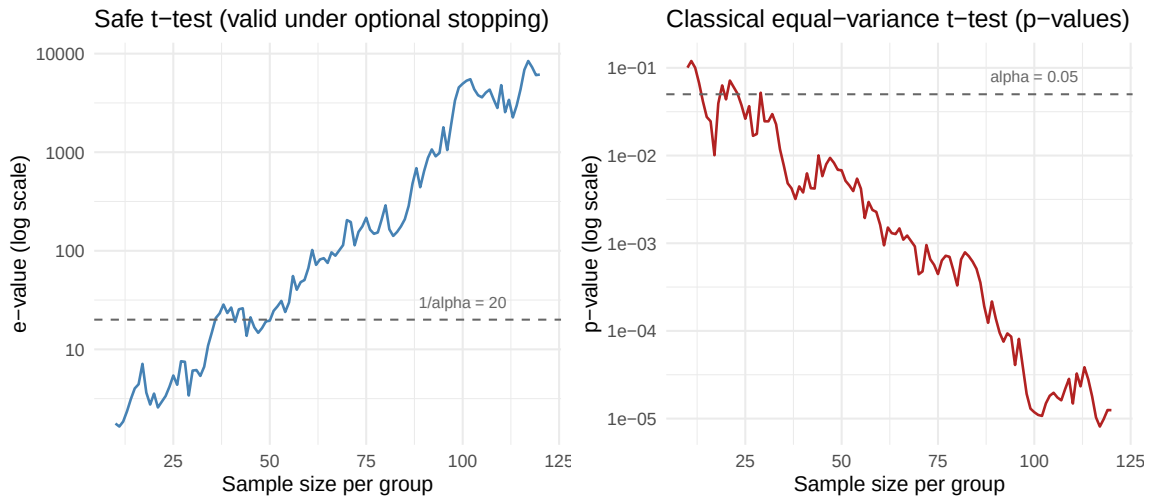


Figure 3.4: Safe t-test applied to a single simulated realisation with $\delta = 0.5$, equal to the design value $\delta_{\min}$. Left: sequential e-value path with rejection threshold $1/\alpha = 20$. Right: classical equal-variance t-test p-value recomputed at each sample size, shown for descriptive comparison only.

In the realisation shown in Figure 3.4, the e-value first crossed the threshold of 20 at $n = 36$ per group, well below the planned maximum of 94 per group. This illustrates the possibility of valid early stopping when evidence accumulates quickly, although it is only a single simulated path rather than a general guarantee.

The right-hand panel shows the classical equal-variance p-value recomputed at each sample size for descriptive comparison. Repeated monitoring of classical p-values is not anytime-

valid, so crossing 0.05 at an interim look would not justify rejection, as we demonstrated empirically in Figure 3.3.

3.3.4 Power comparison: safe t-test vs classical t-test

The flexibility of the safe t-test comes at a cost in fixed-sample settings. To illustrate this trade-off, we run Monte Carlo simulations comparing the safe t-test (tuned to $\delta_{\min} = 0.5$) with the classical equal-variance two-sample t-test. For the power simulations in this subsection, we sweep directly over the standardised effect δ and generate data from $N(\delta/2, 1)$ and $N(-\delta/2, 1)$, so the common standard deviation is 1 by construction and the mean difference is exactly δ . This is a change of parameterisation relative to the illustrative example in Section 3.3.3, which used $\sigma = 2$ and expressed δ through the raw mean difference. Because both the classical t-test and the safe t-test are formulated in terms of the standardised effect size, the two setups are equivalent up to a common rescaling whenever δ is the same. Sweeping directly over δ is therefore the more convenient way to display power curves.

Fixed-sample power

We first compare rejection rates when both tests analyse the same data at a single, pre-determined sample size. There is no sequential monitoring and no early stopping, only the fixed-sample setting that favours the classical test most strongly.

```
set.seed(2026)
n_sim_power <- 500
delta_grid_power <- seq(0, 1.0, by = 0.1)
n_grid_power <- c(30, 60, 120)
alpha <- 0.05

results_power <- expand.grid(
  delta = delta_grid_power,
  n_per_group = n_grid_power,
  power_classical = NA_real_,
  power_safe = NA_real_
)

for (r in seq_len(nrow(results_power))) {
  d <- results_power$delta[r]
  nn <- results_power$n_per_group[r]
  rej_c <- 0L
  rej_s <- 0L

  for (sim in seq_len(n_sim_power)) {
    x1 <- rnorm(nn, mean = d / 2, sd = 1)
    x2 <- rnorm(nn, mean = -d / 2, sd = 1)

    if (t.test(x1, x2, var.equal = TRUE)$p.value < alpha) {
      rej_c <- rej_c + 1L
    }

    safe_res <- safeTTest(x1, x2, designObj = design_t, pilot = FALSE)
    if (safe_res$eValue >= 1 / alpha) {
      rej_s <- rej_s + 1L
    }
  }
}
```

```

    }
  }

  results_power$power_classical[r] <- rej_c / n_sim_power
  results_power$power_safe[r] <- rej_s / n_sim_power
}

```

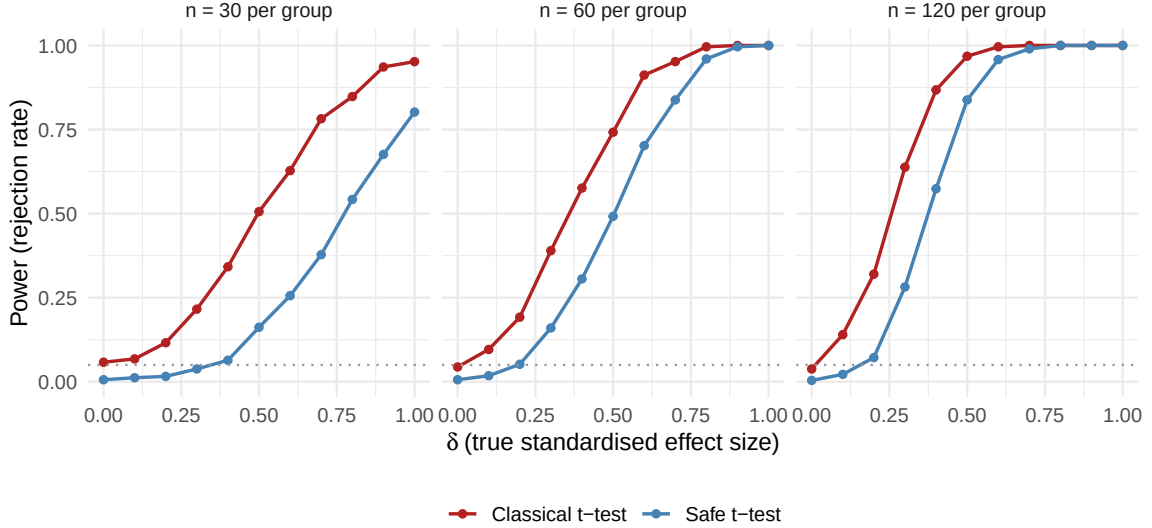


Figure 3.5: Monte Carlo power curves for the classical equal-variance t-test and the safe t-test (tuned to $\delta_{\min} = 0.5$) at three fixed sample sizes. Each point is based on 500 simulated experiments. In this simulation, the safe t-test is less powerful at every fixed n considered, reflecting the cost of anytime validity.

As Figure 3.5 shows, at every fixed sample size considered in this simulation, the classical equal-variance t-test is more powerful than the safe t-test. For $n = 30$ per group, the gap is substantial, and even at $n = 120$ the safe test still has noticeably lower rejection probability at the same true effect size. This is the fixed-sample price of anytime validity. If the test is used only once at a predetermined sample size, the flexibility of optional stopping is not exploited, while the evidential threshold $1/\alpha$ remains in force. The qualitative pattern is the same as in the z-test setting, where Grünwald et al. (2024) (Equation 32) derive a precise constant-factor result for the Gaussian location model. For the safe t-test with its more complex nuisance-parameter handling, the size of the gap depends on the chosen safe design, including the design value $\delta_{\min}$.

Power under optional stopping

The comparison becomes more nuanced once we allow sequential monitoring. In the present design, data are collected gradually and the safe t-test may reject as soon as the e-value exceeds $1/\alpha = 20$ at one of the periodic inspection times. We illustrate this by simulating up to $n_{\max} = 200$ observations per group and checking the e-value every 10 observations (Grünwald et al. 2024; Ly et al. 2025).

Figure 3.6 shows how the distribution of realised sample sizes changes with the true effect size. For $\delta = 0.3$, which is smaller than the design value $\delta_{\min} = 0.5$, many runs do not cross

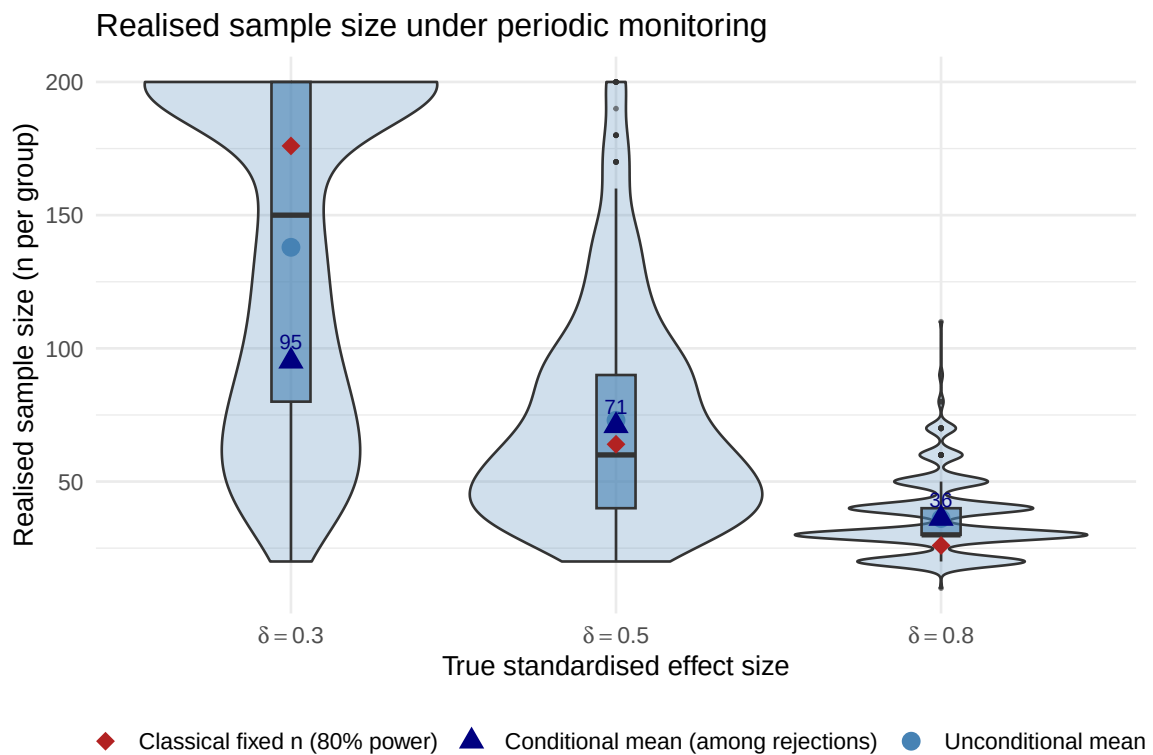


Figure 3.6: Realised sample sizes under the sequential safe t-test for three true effect sizes, based on 500 simulations per effect size. Monitoring is performed every 10 observations up to $n_{\max} = 200$ per group; runs that do not reject by $n_{\max}$ are recorded at 200. Blue circles mark the unconditional mean realised sample size. Dark-blue triangles mark the conditional mean realised sample size among runs that reject. Red diamonds mark the fixed sample size the classical equal-variance t-test requires for 80% power.

the rejection threshold before the administrative cap $n_{\max} = 200$, creating a large point mass at the cap. For larger effects, the distribution shifts downward, so the safe test more often rejects early and the realised sample size is frequently well below the maximum.

The blue circles show the unconditional mean realised sample size, while the red diamonds show the fixed sample size the classical equal-variance t-test requires for 80% power. The dark-blue triangles show the conditional mean realised sample size among runs that do reject. The red diamonds are reference values rather than an equal-power comparison, since the safe procedure's probability of rejecting by $n_{\max}$ varies with the true effect size. This distinction matters because the unconditional mean includes non-rejections capped at $n_{\max}$, whereas the conditional mean averages only over rejecting runs. Consequently, the unconditional mean is necessarily at least as large as the conditional mean among rejecting runs, even when successful rejections tend to occur early. For $\delta = 0.5$ and $\delta = 0.8$, the conditional mean stopping times are approximately 71 and 36 per group, respectively, well below the administrative cap $n_{\max} = 200$, showing that successful runs often terminate early. At $\delta = 0.3$ the rejection rate is much lower, so the corresponding triangle averages only 296 out of 500 simulated runs and should be read as indicative rather than precise.

This is in line with the broader discussion in Grünwald et al. (2024) and Ly et al. (2025). Safe tests typically pay a price in fixed-sample settings, but when sequential monitoring is allowed they may stop early under the alternative, so performance depends strongly on which criterion is used for comparison.

3.3.5 Optional stopping validity: simulation study

We illustrate the type-I error behaviour of the safe t-test under continuous monitoring by simulation.

```
n_max <- 100
n_sim <- 2000

fp_classical_t <- numeric(n_max)
fp_safe_t <- numeric(n_max)

# Design object for the safe t-test
invisible(capture.output(
  design_t_null <- designSafeT(deltaMin = 0.5, alpha = 0.05, beta = 0.2,
                                testType = "twoSample", seed = 1)
))

set.seed(42)
for (sim in 1:n_sim) {
  x1 <- rnorm(n_max, mean = 0, sd = 1)
  x2 <- rnorm(n_max, mean = 0, sd = 1)

  ever_rej_classical <- FALSE
  ever_rej_safe <- FALSE

  for (nn in 10:n_max) {
    # Classical equal-variance t-test
    tt <- t.test(x1[1:nn], x2[1:nn], var.equal = TRUE)
    if (tt$p.value < 0.05 && !ever_rej_classical) {
      ever_rej_classical <- TRUE
    }
  }
}
```

```

    }
    if (ever_rej_classical) fp_classical_t[nn] <- fp_classical_t[nn] + 1

    # Safe t-test
    safe_result <- safeTTest(x1[1:nn], x2[1:nn],
                           designObj = design_t_null, pilot = FALSE)
    if (safe_result$eValue >= 20 && !ever_rej_safe) {
      ever_rej_safe <- TRUE
    }
    if (ever_rej_safe) fp_safe_t[nn] <- fp_safe_t[nn] + 1
  }
}

fp_classical_t <- fp_classical_t / n_sim * 100
fp_safe_t <- fp_safe_t / n_sim * 100

df_fpr_t <- data.frame(
  n = rep(10:n_max, 2),
  fpr = c(fp_classical_t[10:n_max], fp_safe_t[10:n_max]),
  method = rep(c("Classical t-test (monitor p < 0.05)",
                  "Safe t-test (monitor E >= 20)"), each = n_max - 9)
)

ggplot(df_fpr_t, aes(x = n, y = fpr, color = method)) +
  geom_line(linewidth = 0.7) +
  geom_hline(yintercept = 5, linetype = "dashed", color = "grey40") +
  annotate("text", x = 80, y = 7, label = "alpha = 5%", size = 3,
          color = "grey40") +
  labs(x = "Number of looks / sample size per group",
       y = "False positive rate (%)",
       color = NULL) +
  coord_cartesian(ylim = c(0, 50)) +
  theme(legend.position = "bottom")

```

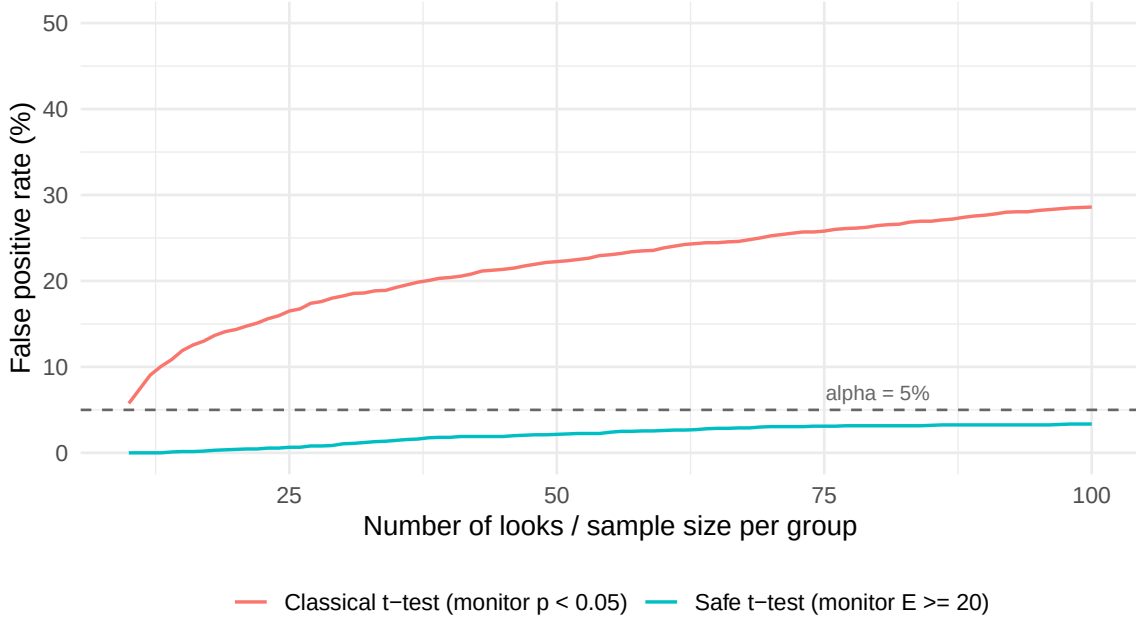



Figure 3.7: False positive rates under continuous monitoring for the safe t-test (blue) and the classical equal-variance t-test (red), based on 2000 simulations under the null ($\delta = 0$).

Figure 3.7 shows the same pattern as for the z-test. Under repeated monitoring, the classical equal-variance t-test shows substantial false positive inflation, whereas the safe t-test remains controlled at the nominal 5% level. In this simulation, the estimated false positive rate of the safe t-test stays close to and below the 5% boundary.

This is the practical consequence of the e-process property formalised in Definition 4.2. Because the safe t-test produces an anytime-valid e-value process, Ville’s inequality guarantees type-I error control under optional stopping. That is precisely what makes e-value-based tests suitable for experiments in which interim monitoring is desirable.

3.4 Safe tests for 2×2 tables

The 2×2 contingency table is ubiquitous in biomedical research. Classical analyses typically rely on Fisher’s exact test or Pearson’s chi-squared test. The 2×2 case provides an important binary-data example of how the GRO/REGROW framework extends beyond Gaussian location models (Grünwald et al. 2024, Examples 2, 6, 7; Turner et al. 2024).

3.4.1 Setup

Consider a 2×2 table where $Y_i \in \{0, 1\}$ is the outcome and $X_i \in \{a, b\}$ is the group indicator, with n_a and n_b observations in each group. A typical example is a clinical trial comparing a treatment (a) to a placebo (b), where $Y_i = 1$ indicates recovery. Let $\mu_{1|a}$ and $\mu_{1|b}$ denote the probabilities of outcome 1 in groups a and b , respectively. The null hypothesis is

$$H_0 : \mu_{1|a} = \mu_{1|b},$$

that is, the outcome distribution is the same in both groups. In the Bernoulli setting, equality of the outcome distributions is equivalent to equality of the success probabilities.

3.4.2 The simple-alternative case

A useful starting point is the case in which the alternative is fixed, so that $(\mu_{1|a}, \mu_{1|b})$ is treated as known. In this setting, Grünwald et al. (2024) (Example 2, Equation 14) show that the GRO e-variable takes the likelihood-ratio form

$$E^* = \frac{p_{\mu_{1|a}, \mu_{1|b}}(Y | x)}{p_{\mu^\circ}(Y | x)},$$

where

$$\mu^\circ = \frac{n_a \mu_{1|a} + n_b \mu_{1|b}}{n_a + n_b}$$

is the pooled success probability under the null. Here $p_{\mu_{1|a}, \mu_{1|b}}(Y | x)$ is the conditional likelihood given the fixed group labels x , and $p_{\mu^\circ}(Y | x)$ is the null likelihood under the Bernoulli model with common success probability μ° .

Thus, in the simple-alternative case, the reverse information projection of the chosen alternative onto the null is the Bernoulli model with common success probability μ° . Grünwald et al. (2024) note that this particular form of point-prior RIPr, at a weighted average of the group probabilities, is specific to contingency tables.

3.4.3 Composite alternatives and sequential testing

Unlike the simple-alternative case, where the alternative probabilities are known and the RIPr has a closed form, the composite case requires handling unknown group-specific success probabilities while maintaining anytime validity. As discussed in Grünwald et al. (2024) (Section 4.1, with Examples 6 and 7), the 2×2 case can be handled within the REGROW/GROW framework by separating nuisance and effect parameters. For practical sequential testing, the exact k -sample construction developed by Turner et al. (2024) yields e-variables that allow optional stopping and can be implemented sequentially. This is the basis of the 2×2 functionality in **safestats**.

3.4.4 R implementation

We illustrate the safe test for two proportions using the **safestats** package on a simulated dataset. We simulate $n = 500$ observation pairs from two groups with success probabilities $\mu_a = 0.5$ and $\mu_b = 0.3$. Thus, group a has the higher success probability by 0.2, and equivalently $\mu_b - \mu_a = -0.2$, which is the sign convention used by **designSafeTwoProportions()**.

We explicitly set all four beta hyperparameters supplied to **designSafeTwoProportions()** to 1. With **alternativeRestriction = "difference"** and $\delta = -0.2$, these hyperparameters tune the mixing distribution along the restricted alternative $\mu_{1|b} - \mu_{1|a} = -0.2$. They should not be described as independent Beta(1, 1) priors on two unrestricted group probabilities. This is a user-specified choice. For balanced blocks in **safestats** 0.8.7, omitting the hyperparameters instead sets all four to 0.18, the package's REGRET-optimised default. Other choices affect the speed of evidence accumulation but not the null-validity guarantee.

Final 2x2 table:

	Y=1	Y=0
Group A	235	265
Group B	145	355

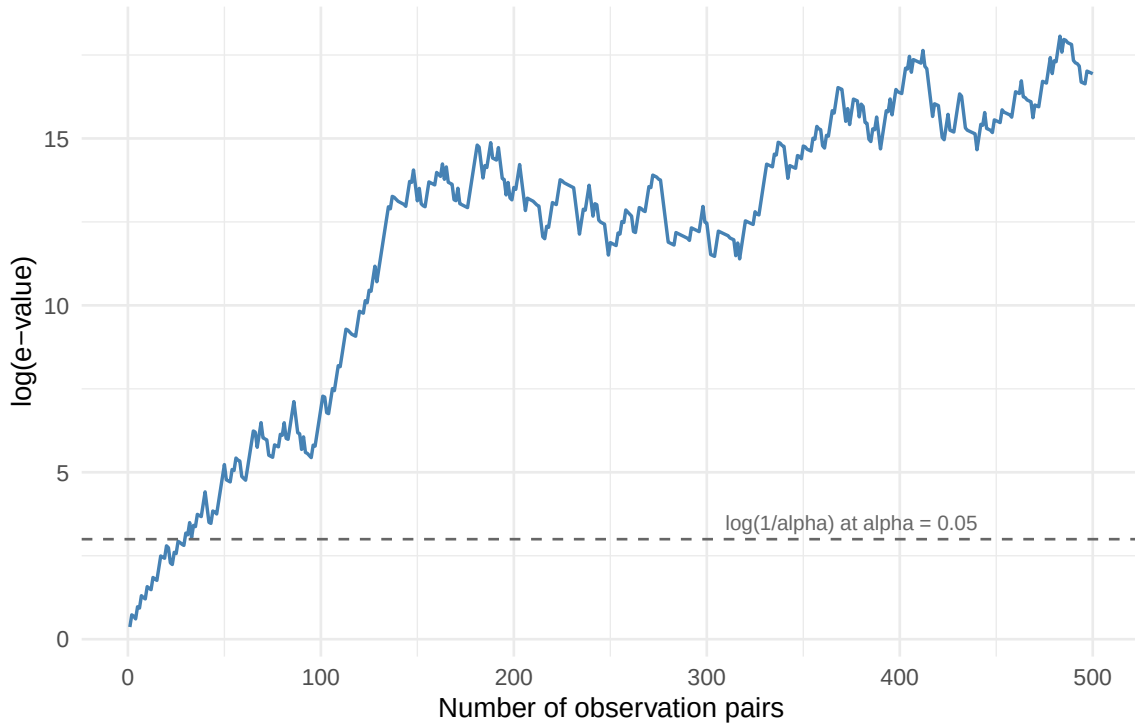


Figure 3.8: Sequential log e-value path for a simulated 2×2 table with $\mu_a = 0.5$ and $\mu_b = 0.3$, computed using `safeTwoProportionsTest()` with design alternative $\delta = \mu_b - \mu_a = -0.2$. The dashed line marks the rejection threshold $\log(1/\alpha) \approx 3$.

```
Fisher's exact test p-value:    < 1e-4
Final safestats e-value:      2.27e+07
Reject at alpha = 0.05?      Yes
```

In the simulated path shown in Figure 3.8, the e-value first crossed the rejection threshold of 20 at observation pair 30. The classical one-sided Fisher’s exact test also rejects the null on the complete dataset, but as with the other tests in this chapter, only the e-value path guarantees valid type-I error control under optional stopping. The `safeTwoProportionsTest()` implementation is based on the beta-binomial prior construction of Turner et al. (2024) and is designed here for the directional alternative $\delta = \mu_b - \mu_a = -0.2$, using a blockwise one-from-each-group update scheme.

3.5 Safe logrank test

The logrank test is a standard method for comparing survival curves, especially in clinical trials. It is also the setting where the benefits of safe testing are most compelling, because clinical trials involve patients, making early stopping both ethically important and practically desirable. The anytime-valid logrank test of ter Schure et al. (2024) is implemented in the `safestats` package (Turner et al. 2022).

3.5.1 Why survival analysis needs safe tests

In a two-arm clinical trial comparing a treatment to a control, the typical design involves periodic interim analyses where a Data Safety Monitoring Board (DSMB) reviews the accumulating evidence. With classical methods, the nominal type-I error guarantee generally relies on a pre-specified monitoring design using group-sequential boundaries such as O'Brien–Fleming (O'Brien and Fleming 1979) or alpha-spending approaches such as Lan–DeMets (Lan and DeMets 1983). If the schedule changes, because of slower-than-expected enrolment, for example, the analysis requires appropriate adjustment or re-design.

Safe logrank tests remove the need to pre-specify a monitoring schedule for validity, although a maximum number of events may still be chosen for design and power planning. The analyst must still specify the testing direction and a clinically relevant design alternative such as $\text{HR}_{\min}$. The freedom concerns the monitoring schedule rather than the model, since the guarantee still requires proportional hazards and censoring that is independent of survival time. Because the e-process remains valid at any stopping time with respect to its filtration, the DSMB can review the accumulating evidence at arbitrary times and frequencies, and recommend stopping whenever the evidence is compelling. We explore this in depth in the clinical trial case study in Chapter 6.

3.5.2 Setup

Consider a two-arm trial with n subjects, where subject i is assigned to treatment ($Z_i = 1$) or control ($Z_i = 0$). Let T_i be the survival time and C_i the censoring time, so we observe $\tilde{T}_i = \min(T_i, C_i)$ and the event indicator $D_i = \mathbf{1}\{T_i \leq C_i\}$. Under the proportional hazards model, the hazard function for subject i is

$$h(t \mid Z_i) = h_0(t) \exp(\theta Z_i),$$

where $h_0(t)$ is the unspecified baseline hazard and the treatment effect is parameterised by the log hazard ratio $\theta = \log(\text{HR})$. The null hypothesis is $H_0 : \theta = 0$ (no treatment effect). In ter Schure et al. (2024), the treatment effect is often parameterised directly by the hazard ratio rather than its logarithm. Here we use the equivalent Cox-model notation $\theta = \log(\text{HR})$.

3.5.3 The safe logrank e-process

The safe logrank test of ter Schure et al. (2024) is constructed for sequential survival monitoring under a proportional hazards model and yields an e-process that remains valid under arbitrary stopping. Evidence is updated at event times through conditional likelihood contributions based on the current risk sets. At each event, the test contrasts how compatible the observed outcome is with the null and with a clinically relevant alternative. As in other safe tests, the design requires specifying a clinically relevant alternative, expressed through a hazard ratio $\text{HR}_{\min}$. We refer the reader to ter Schure et al. (2024) for the formal construction and its theoretical properties, and focus on the practical workflow below.

In practice, the workflow is to choose the testing direction, specify a clinically relevant hazard ratio $\text{HR}_{\min}$, create the design object with `designSafeLogrank()`, evaluate the e-value at interim looks using `safeLogrankTest()`, and reject once $E \geq 1/\alpha$.

3.5.4 Illustration: event-by-event evidence accumulation

To convey the basic logic, we show a hand-coded event-by-event likelihood-ratio path on simulated survival data and verify that it agrees with the package-based `safeLogrankTest()`. In the present continuous-time simulation with exponentially distributed event times, ties occur with probability zero, so the hand-coded path reproduces the event-by-event safe logrank update for the design alternative implemented by `safeLogrankTest()` in the basic two-arm no-ties setting, where each observed event contributes the ratio of its Cox partial-likelihood term under the design hazard ratio and under the null. In real applications, tied event times and other complications make the package implementation preferable. In the presence of ties, the noncentral-hypergeometric numerator approximates the alternative conditional likelihood, while the null conditional likelihood remains exact, so null validity and type-I error control are therefore preserved (ter Schure et al. 2024).

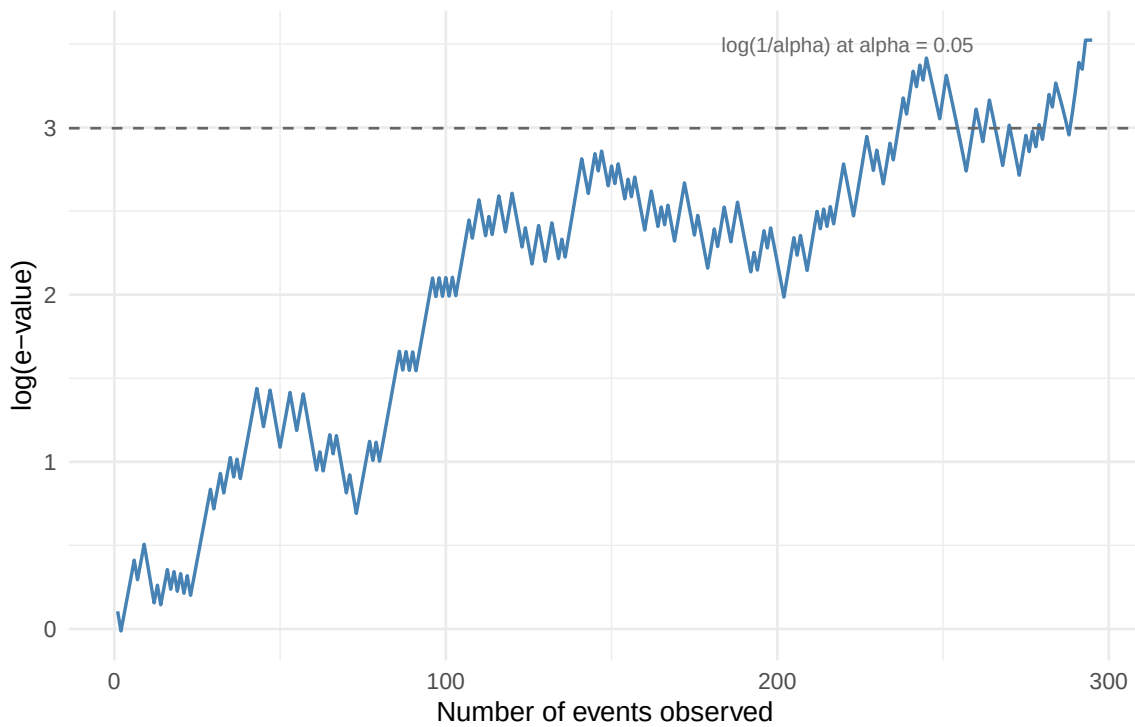


Figure 3.9: Sequential log e-value path for the safe logrank test on simulated survival data with true $HR = 0.7$, designed for $HR_{\min} = 0.8$. The hand-coded path and the `safeLogrankTest()` path agree to within numerical tolerance, so only the hand-coded path is shown.

Classical one-sided logrank test:

z-statistic: -2.813
p-value: 2.45e-03

Final e-value (hand-coded): 3.39e+01

Reject at alpha = 0.05? Yes

Max $|\log(E_{\text{hand}}) - \log(E_{\text{safestats}})|$: 1.76e-12

As Figure 3.9 shows, the hand-coded path and the reconstructed `safeLogrankTest()` path agree to within numerical tolerance in this no-ties simulation, so only the hand-coded path is plotted. The classical one-sided logrank test also rejects the null on the complete dataset, but only the e-value path guarantees valid type-I error control under arbitrary optional stopping, whereas classical group-sequential methods require a prespecified monitoring design.

Note that the test is designed for a minimum clinically relevant hazard ratio of 0.8, while the true data-generating hazard ratio is 0.7. This separation between design alternative and truth mirrors the practice discussed throughout this chapter.

3.5.5 Advantages for clinical trials

The safe logrank test offers three concrete advantages for clinical trials, as discussed in ter Schure et al. (2024):

1. **No pre-specified monitoring schedule.** The e-value can be computed after every event, and the decision to stop can be made at any time. This contrasts with group sequential methods, which typically rely on a pre-specified monitoring plan and corresponding alpha-spending or boundary calculations.
2. **Optional continuation across studies.** E-values from separate looks or studies can be combined multiplicatively when each new contribution is conditionally valid given the preceding information under the same global null. Under this condition, the running product remains valid, as formalised in Chapter 4. Fresh data are necessary, but do not by themselves establish conditional validity.
3. **Ethical early stopping.** Because the e-value process is anytime-valid, a DSMB can support earlier stopping decisions while preserving valid type-I error control under continuous monitoring.

These advantages are demonstrated in the full clinical trial case study in Chapter 6, where we compare naive repeated testing with e-value monitoring.

3.6 A worked example: the exponential rate

The safe tests so far have either been Gaussian (the z- and t-tests) or have required a dedicated construction (the 2×2 and logrank tests). To show that the framework is not tied to any one of these settings, we now work through a fresh example from first principles, namely a safe test for the rate of an exponential distribution. The exponential is a natural choice. It is the canonical model for waiting times and lifetimes, so it connects to the survival setting of Section 3.5. It is also a one-parameter exponential family with monotone likelihood ratio, so Proposition 3.1 applies and we can derive the GROW e-variable, verify its validity over the composite null, and compute its growth rate in closed form by hand. This makes it a useful consolidation of the ideas in this chapter and in Chapter 2.

3.6.1 Setup

Suppose we observe lifetimes $X_1, X_2, \dots \stackrel{\text{iid}}{\sim} \text{Exp}(\lambda)$, with density $p_\lambda(x) = \lambda e^{-\lambda x}$ for $x \geq 0$. The scientifically natural parameter here is the **mean lifetime** $\eta = 1/\lambda$, since reliability and survival questions are typically phrased in terms of how long things last rather than

how often they fail. A larger mean lifetime corresponds to a smaller rate and represents a more durable or longer-surviving population. Consider a context in which η_0 is a known baseline mean lifetime and a worthwhile improvement would lengthen it to some $\eta_1 > \eta_0$. As with the other safe tests, we separate null and alternative by a pre-specified minimum relevant effect, here the gap between η_0 and η_1 . We test

$$H_0 : \eta \leq \eta_0 \quad \text{against} \quad H_1 : \eta \geq \eta_1, \quad \eta_1 > \eta_0.$$

Writing $\lambda_0 = 1/\eta_0$ and $\lambda_1 = 1/\eta_1$, the same hypotheses read $H_0 : \lambda \geq \lambda_0$ against $H_1 : \lambda \leq \lambda_1$ in terms of the rate, with $\lambda_1 < \lambda_0$. Stating the hypotheses in η keeps the direction of the alternative aligned with the direction of scientific interest, namely that longer lifetime is the improvement. It also matches the orientation of Proposition 3.1, where the alternative sits on the larger side of the parameter. The exponential density is most compactly expressed in λ , so the derivations below are carried out in the rate.

3.6.2 The GROW e-variable

The family $\{\text{Exp}(\lambda) : \lambda > 0\}$ is a one-parameter exponential family, and the mean lifetime $\eta = 1/\lambda$ is a monotone reparameterisation of the rate. The likelihood ratio between two rates is monotone in the sufficient statistic $S_n = \sum_{i=1}^n X_i$, since increasing the mean lifetime (decreasing the rate) stochastically increases the lifetimes and therefore S_n . This is exactly the monotone-likelihood-ratio structure required by Proposition 3.1, and the property is preserved under the monotone reparameterisation from λ to η , so the proposition applies equally with the family reindexed by mean lifetime. Taking the mean lifetime as the parameter, with null boundary η_0 and alternative boundary η_1 , the proposition tells us that the GROW e-variable is the likelihood ratio between the two boundary distributions. In terms of the corresponding rates $\lambda_1 = 1/\eta_1$ and $\lambda_0 = 1/\eta_0$,

$$E_n = \frac{p_{\lambda_1}(X^n)}{p_{\lambda_0}(X^n)} = \prod_{i=1}^n \frac{\lambda_1 e^{-\lambda_1 X_i}}{\lambda_0 e^{-\lambda_0 X_i}} = \left(\frac{\lambda_1}{\lambda_0}\right)^n \exp((\lambda_0 - \lambda_1) S_n).$$

Because $\lambda_0 > \lambda_1$, the coefficient $\lambda_0 - \lambda_1$ is positive, so E_n is increasing in the total lifetime S_n , so a long total lifetime is evidence for the longer mean lifetime, and the e-value grows accordingly. This is the exponential analogue of the Gaussian likelihood-ratio e-variable from Section 2.6, with the total lifetime S_n playing the role of the Gaussian sum.

3.6.3 Validity across the composite null

It is worth checking by hand that E_n is a genuine e-variable for the *whole* composite null, not merely at the boundary. Since the observations are independent and the same one-step factor is used at each time point, it suffices to compute the expectation of a single factor under an arbitrary true rate λ . For $\lambda > \lambda_0 - \lambda_1$, which holds throughout the null and ensures the integral below converges, standard evaluation of $\int_0^\infty e^{-cx} dx = 1/c$ gives

$$\mathbb{E}_\lambda \left[\frac{\lambda_1 e^{-\lambda_1 X}}{\lambda_0 e^{-\lambda_0 X}} \right] = \frac{\lambda_1}{\lambda_0} \lambda \int_0^\infty e^{-(\lambda_1 - \lambda_0 + \lambda)x} dx = \frac{\lambda_1 \lambda}{\lambda_0 (\lambda - \lambda_0 + \lambda_1)}.$$

Two features of this expression matter. First, at the boundary $\lambda = \lambda_0$ it equals exactly 1, so each factor is an *exact* e-variable there. Second, a short rearrangement shows that the expression is at most 1 if and only if $\lambda \geq \lambda_0$, that is, if and only if $\eta \leq \eta_0$, precisely the null. The expectation therefore increases as the rate falls below λ_0 (the mean lifetime

rising into the alternative) and decreases as the rate grows beyond it. For still smaller rates $\lambda \leq \lambda_0 - \lambda_1$ the expectation may be infinite, but this does not affect e-validity, which only requires the expectation bound under H_0 .

The boundary rate λ_0 is thus the *least favourable* null distribution, the hardest point in the null to distinguish from the alternative, and the e-variable is calibrated against it. This is a concrete, hand-verifiable instance of the reverse information projection discussed abstractly in Section 3.1. For the boundary alternative $\text{Exp}(\lambda_1)$, the reverse information projection onto the null set $\{\text{Exp}(\lambda) : \lambda \geq \lambda_0\}$ is the boundary distribution $\text{Exp}(\lambda_0)$.

Finally, since each factor satisfies $\mathbb{E}_{\lambda_0}[\cdot | \mathcal{F}_{i-1}] = 1$ at the boundary and ≤ 1 throughout the null, it is a conditional e-variable in the sense of Definition 4.3. The running product E_n is therefore a test supermartingale under every null rate, so Ville's inequality (Theorem 4.1) delivers anytime validity without any further work. This is the common backbone of Section 4.6, instantiated for the exponential.

3.6.4 Growth rate equals the KL divergence

How quickly does the evidence accumulate when the alternative is true? Following the e-power discussion of Section 2.9, the relevant quantity is the expected log-increment per observation under the alternative. With data generated at the design rate λ_1 , a direct calculation using $\mathbb{E}_{\lambda_1}[X] = 1/\lambda_1$ gives

$$\mathbb{E}_{\lambda_1}[\log E_1] = \log \frac{\lambda_1}{\lambda_0} + (\lambda_0 - \lambda_1) \mathbb{E}_{\lambda_1}[X] = \frac{\lambda_0}{\lambda_1} - 1 - \log \frac{\lambda_0}{\lambda_1} = \text{KL}(\text{Exp}(\lambda_1) \parallel \text{Exp}(\lambda_0)).$$

Writing $r = \lambda_0/\lambda_1 = \eta_1/\eta_0 > 1$, this is $r - 1 - \log r$, which is strictly positive whenever $r \neq 1$. The e-process therefore has positive expected log-growth under the design alternative, accumulating evidence at the rate of the KL divergence between the alternative and null boundary distributions. This makes explicit the log-optimality of the likelihood ratio established in Theorem 2.1, now specialised to the exponential, where the optimal growth rate is the KL divergence, and every term in it is available in closed form.

3.6.5 Implementation

We implement the test by hand, computing the running log e-value as a cumulative sum of one-step log factors. No special package is needed, which is part of the point. Once the one-step factor is identified, the rest is the generic supermartingale construction.

```
# Hypotheses stated through the mean lifetime eta = 1 / lambda
eta0 <- 1.0          # baseline mean lifetime (null boundary)
eta1 <- 1 / 0.7      # alternative boundary: longer mean lifetime
lambda0 <- 1 / eta0  # null boundary rate
lambda1 <- 1 / eta1  # alternative boundary rate (lower rate)
lambda_true <- lambda1 # data generated at the alternative boundary

alpha <- 0.05
log_threshold <- log(1 / alpha)

n <- 200
set.seed(2026)
x <- rexp(n, rate = lambda_true)
```



```
# One-step log factor:  $\log(\lambda_{\text{bda1}} / \lambda_{\text{bda0}}) + (\lambda_{\text{bda0}} - \lambda_{\text{bda1}}) * x_i$ 
log_factor <- log(lambda1 / lambda0) + (lambda0 - lambda1) * x
log_e <- cumsum(log_factor)

# Closed-form per-step growth rate (KL divergence) for reference
kl <- lambda0 / lambda1 - 1 - log(lambda0 / lambda1)
cat("Per-step growth rate KL(Exp(lambda1) || Exp(lambda0)):",
    round(kl, 4), "\n")
```

Per-step growth rate KL(Exp(lambda1) || Exp(lambda0)): 0.0719

```
first_cross <- which(log_e >= log_threshold)
first_cross <- if (length(first_cross) == 0) NA_integer_ else first_cross[1]
cat("First crossing of log(1/alpha):", first_cross, "\n")
```

First crossing of log(1/alpha): 14

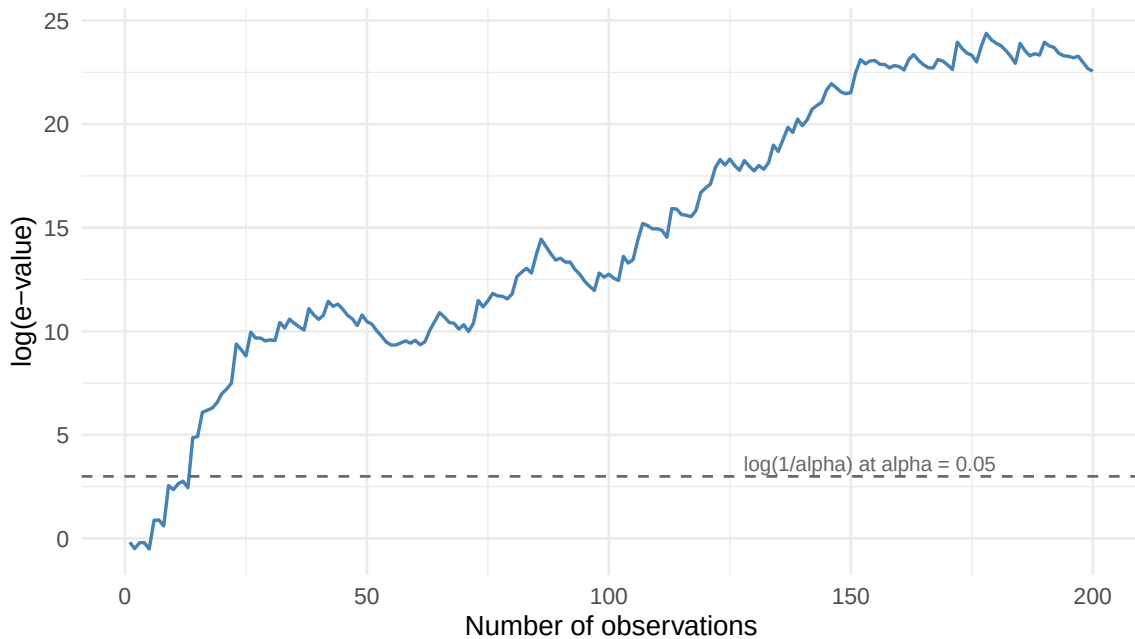


Figure 3.10: Sequential log e-value path for the safe exponential-rate test under the alternative boundary (mean lifetime $\eta_1 = 1/0.7$, rate $\lambda_1 = 0.7$), with null boundary $\eta_0 = 1$ ($\lambda_0 = 1$). The dashed line marks the rejection threshold $\log(1/\alpha) \approx 3$ at $\alpha = 0.05$. Under the alternative the path has positive expected drift equal to the KL divergence per observation.

Figure 3.10 shows the resulting log e-value path under the design alternative.

To illustrate anytime validity, we check the type-I error at the least-favourable null rate $\lambda = \lambda_0$, where the process is an exact test martingale. We simulate many data streams, monitor the e-value after every observation up to $n_{\max} = 500$, and record how often it ever crosses the threshold.

```

n_max <- 500
n_sim <- 5000
set.seed(123)

ever_rejected <- replicate(n_sim, {
  x0 <- rexp(n_max, rate = lambda0) # data simulated under the null
  ↪ boundary
  log_e0 <- cumsum(log(lambda1 / lambda0) + (lambda0 - lambda1) * x0)
  any(log_e0 >= log_threshold)
})

p_hat <- mean(ever_rejected)
cat("Estimated P(reject by n_max under the null boundary):",
    round(p_hat, 4),
    "(Ville bound:", alpha, ")\n")

```

Estimated P(reject by n_max under the null boundary): 0.0328 (Ville bound: 0.05)

In this simulation, the estimated probability of ever crossing the threshold by $n_{\max} = 500$ is compatible with the nominal 5% level, up to Monte Carlo error. Ville's inequality gives the actual guarantee, $P_\lambda(\sup_n E_n \geq 1/\alpha) \leq \alpha$ for every $\lambda \geq \lambda_0$, and the simulation is an illustration rather than the source of the bound. As with the z-test in Figure 3.3, repeatedly monitoring a classical fixed-horizon test can inflate its type-I error well beyond α , whereas the safe test avoids this by construction.

3.6.6 Remark: a confidence sequence for the rate

The same inversion principle can be used to construct a confidence sequence for λ (equivalently for the mean lifetime η), in the manner developed in the next chapter (Section 4.5). For each candidate rate $\ell > 0$, one can test the point null $H_0 : \lambda = \ell$ using a mixture likelihood-ratio e-process

$$M_n^{(\ell)} = \int \frac{p_{\lambda'}(X^n)}{p_\ell(X^n)} \pi(d\lambda'),$$

where a gamma mixing distribution over λ' is conjugate for the exponential rate model and gives a closed form. The set $C_n = \{\ell > 0 : M_n^{(\ell)} < 1/\alpha\}$ is then an anytime-valid confidence sequence for the rate. We do not pursue the derivation here. The point is that the exponential example follows the same general backbone of Section 4.6 (construct an e-process, apply Ville's inequality, and invert the resulting family of tests), and provides a natural parametric companion to the Gaussian confidence sequence of Section 4.5.3.

3.7 Summary and comparison

This chapter showed that several classical tests admit safe e-value analogues that retain type-I error control under arbitrary stopping while often preserving good efficiency. The following table summarises the key properties.

Table 3.1: Summary of safe tests for classical problems.

Test	Nuisance	Design principle	Function	Reference
z-test	None (σ known)	Exact GROW	<code>safeZTest</code>	Grünwald et al. (2024)
t-test	Unknown σ	REGROW/GROW, right-Haar prior	<code>safeTTest</code>	Grünwald et al. (2024)
2×2 table	Marginal probabilities	Motivated by GROW/REGROW	<code>safeTwoProportionsTest</code>	Turner et al. (2024)
Logrank	Baseline hazard	Conditional likelihood	<code>safeLogrankTest</code>	ter Schure et al. (2024)

The common pattern across these tests is:

1. **Validity comes from the e-process property.** If the running e-value process is valid under the null, in the sense formalised in the next chapter, then Ville’s inequality guarantees type-I error control under any stopping rule.
2. **Efficiency is guided by growth-rate optimality.** The GROW and REGROW criteria of Grünwald et al. (2024) provide a principled way to choose e-variables that accumulate evidence efficiently, where these criteria are available. The degree of formal optimality varies across settings. The z-test admits an exact GROW result, the t-test has strong optimality properties via the Haar prior, and the 2×2 and logrank constructions are motivated by the same principles but involve additional practical considerations.
3. **The cost is often a larger worst-case sample size.** The price of optional-stopping robustness is often a larger planned sample size, and fixed-horizon power can be lower than that of a classical test designed for the same setting. As shown in the t-test simulations of Section 3.3.4, whether this cost is offset by early stopping depends on the true effect size, the design parameters, and the monitoring frequency.
4. **Software support is available.** The `safestats` R package (Turner et al. 2022) provides implementations of all four tests discussed in this chapter, with design functions that compute planned sample sizes and tuning parameters.

In the next chapter, we move from these test-specific constructions to the general sequential framework behind them, showing how e-processes and confidence sequences provide a unified language for anytime-valid inference beyond the individual examples studied here.

Chapter 4

E-Processes and Anytime-Valid Inference

In Chapter 3, we repeatedly used the fact that safe tests remain valid under optional stopping, so a researcher may monitor the e-value during data collection and stop once it crosses $1/\alpha$ while still controlling the type-I error at level α . In that chapter, however, this was treated as a property of particular tests such as the safe z-test, safe t-test, safe 2×2 test, and safe logrank test. This chapter steps back from those examples and develops the general sequential machinery that makes such guarantees possible.

The two central ideas are **e-processes**, which extend e-values from a single analysis time to an evolving stream of evidence, and **confidence sequences**, which extend fixed-sample confidence intervals to procedures that remain valid under continuous monitoring. Together, they provide the general framework for anytime-valid inference.

4.1 The sequential setting

The tests in Chapter 3 were applied to data that arrived sequentially. At each time step n , a new observation X_n was observed and the evidence was updated. The key feature of optional stopping is that the decision to stop may depend on the information accumulated so far, but not on future observations. To state this precisely, we need a small amount of sequential vocabulary (Ramdas and Wang 2025, chap. 7).

A **filtration** $(\mathcal{F}_n)_{n \geq 0}$ is an increasing sequence of sub- σ -algebras of $\mathcal{F}$:

$$\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \dots$$

Intuitively, $\mathcal{F}_n$ represents the information available by time n . In the canonical case, $\mathcal{F}_0 = \{\emptyset, \Omega\}$ and $\mathcal{F}_n = \sigma(X_1, \dots, X_n)$, the σ -algebra generated by the first n observations, so $\mathcal{F}_n$ then contains exactly the yes-or-no statements that can be answered from the data observed so far, such as whether the sample mean is positive or whether the running e-value has already crossed a rejection threshold. More generally, $\mathcal{F}_n$ may contain further information available by time n , such as treatment assignments, external randomisation, or observations from other data streams.

A random variable τ taking values in $\{0, 1, 2, \dots\} \cup \{\infty\}$ is a **stopping time** with respect to $(\mathcal{F}_n)$ if the event $\{\tau \leq n\}$ belongs to $\mathcal{F}_n$ for every $n \geq 0$. In words, whether the

experiment has stopped by time n must be decidable using only the information available by time n , never future observations. The convention $\tau = \infty$ means that the analyst never stops.

Example. In the safe z-test of Section 3.2 at $\alpha = 0.05$, one possible rule is to stop and reject as soon as the running e-value reaches the threshold $1/\alpha = 20$. The corresponding stopping time is

$$\tau = \inf\{n \geq 1 : E_n \geq 20\}, \quad \inf \emptyset := \infty.$$

This is a stopping time with respect to $\mathcal{F}_n = \sigma(X_1, \dots, X_n)$, because after observing the first n data points we can determine whether the threshold has been crossed at or before time n .

By contrast, a rule such as “stop at time n if the next observation X_{n+1} exceeds 2” is not a stopping time with respect to $(\mathcal{F}_n)$, because it depends on information that is not yet available at time n .

A process $(Y_n)_{n \geq 0}$ is **adapted** to $(\mathcal{F}_n)$ if Y_n is $\mathcal{F}_n$ -measurable for each n . Intuitively, this means that the value of the process at time n can be computed from the information available by time n . All the running e-value paths in Chapter 3 are adapted because the value at time n is a function of the first n observations.

4.2 Test supermartingales and e-processes

In Section 2.2, we defined an e-variable as a nonnegative random variable with expectation at most 1 under the null. The sequential analogue replaces this single random variable by a process that evolves over time as new data arrive. The key requirement is that this running measure of evidence should remain properly calibrated under the null, not only at one fixed sample size but throughout the monitoring process (Ramdas and Wang 2025).

Definition 4.1 (Test supermartingale (adapted from Ramdas and Wang 2025, Definition 7.3(i))). A nonnegative process $(M_n)_{n \geq 0}$ adapted to $(\mathcal{F}_n)$ is a **test supermartingale** for $\mathcal{P}$ if, for every $P \in \mathcal{P}$,

$$\mathbb{E}_P[M_0] \leq 1 \quad \text{and} \quad \mathbb{E}_P[M_{n+1} \mid \mathcal{F}_n] \leq M_n \quad \text{for all } n \geq 0.$$

If, in addition,

$$\mathbb{E}_P[M_{n+1} \mid \mathcal{F}_n] = M_n \quad \text{for all } n \geq 0 \text{ and all } P \in \mathcal{P},$$

the process is a **test martingale**. Throughout this thesis, we use the common normalisation $M_0 = 1$ almost surely.

Throughout this chapter, equalities and inequalities involving conditional expectations are understood to hold P -almost surely for each $P \in \mathcal{P}$.

The supermartingale condition says that, under the null, the process cannot be expected to increase from one step to the next, conditional on everything observed so far. Intuitively, once the first n observations have been seen, the current value M_n already reflects all evidence available at that time, so under the null there is no systematic way to grow this evidence further in expectation. In the betting language from Section 2.4, the process tracks Skeptic’s wealth, and the condition is the same one that made Forecaster willing

to offer the bet in the first place: under the null, Forecaster expects not to lose money on the next round, given the information currently available (Ramdas and Wang 2025).

A closely related notion captures the property that is most relevant for sequential testing.

Definition 4.2 (E-process (adapted from Ramdas and Wang 2025, Definition 7.3(ii))). A nonnegative process $(E_n)_{n \geq 0}$ adapted to $(\mathcal{F}_n)$ is an **e-process** for $\mathcal{P}$ if, for every $P \in \mathcal{P}$ and every stopping time τ with respect to $(\mathcal{F}_n)$,

$$\mathbb{E}_P[E_\tau] \leq 1,$$

where $E_\infty = \limsup_{n \rightarrow \infty} E_n$.

An e-process is therefore a process whose value at any stopping time is itself a valid e-variable. This is the formal meaning of optional-stopping safety, so even if the analyst decides to stop in a data-dependent way, as long as the rule is a valid stopping time, the resulting value can still be thresholded at $1/\alpha$ without invalidating type-I error control (Ramdas and Wang 2025).

The distinction between the two notions is useful. A test supermartingale is defined through a one-step conditional expectation inequality, whereas an e-process is defined directly through validity at every stopping time. Thus test supermartingales give a local, step-by-step calibration condition, while e-processes express the global property that sequential testing actually relies on. The following proposition makes the link precise.

Proposition 4.1 (Test supermartingales are e-processes (Ramdas and Wang 2025)). *Every test supermartingale for $\mathcal{P}$ is an e-process for $\mathcal{P}$.*

This follows from Definition 7.3 and Fact 7.5 of Ramdas and Wang (2025), which imply that every test supermartingale satisfies the optional-stopping property at arbitrary stopping times. Hence, for every stopping time τ with respect to $(\mathcal{F}_n)$,

$$\mathbb{E}_P[M_\tau] \leq \mathbb{E}_P[M_0] \leq 1 \quad \text{for all } P \in \mathcal{P}.$$

Thus the stopped value M_τ is a valid e-variable under every $P \in \mathcal{P}$, so (M_n) is an e-process. In other words, it is the one-step conditional supermartingale inequality, rather than fixed-time calibration alone, that makes a test supermartingale safe under optional stopping.

The converse need not hold in full generality: there are testing problems for which useful e-processes exist even though no nontrivial test supermartingales are available, so e-processes can form a strictly broader class; see Ramdas et al. (2023) and Ramdas and Wang (2025) for discussion. For the present thesis, however, this distinction is mostly technical rather than substantive: the safe tests we use are all constructed as test supermartingales, and the e-process property then follows automatically.

4.3 Building e-processes from conditional e-variables

The running e-value paths in Chapter 3 were all constructed multiplicatively. This reflects a general principle: sequential evidence is often built from a sequence of one-step factors, where each new factor is chosen using the information available so far and then evaluated on fresh data. The right one-step notion is a **conditional e-variable**.

This definition is naturally motivated by the betting interpretation of sequential testing. At each step, before the next observation is seen, the analyst chooses a nonnegative payoff based on the past. Under the null, its conditional expected value given the past must be at most 1. In the language of Grünwald et al. (2024), this is a conditional e-variable given $\mathcal{F}_{i-1}$; in the language of Ramdas and Wang (2025), it is the one-step bet underlying a test supermartingale.

Definition 4.3 (Conditional e-variable (adapted from Grünwald et al. 2024; Ramdas and Wang 2025)). A nonnegative random variable E_i is a **conditional e-variable** for $\mathcal{P}$ given $\mathcal{F}_{i-1}$ if E_i is $\mathcal{F}_i$ -measurable and

$$\mathbb{E}_P[E_i \mid \mathcal{F}_{i-1}] \leq 1 \quad \text{for all } P \in \mathcal{P}.$$

A conditional e-variable is therefore a valid one-step e-variable once the past has been fixed. Its usefulness is practical as well as conceptual. It allows the analyst to adapt the next step to the data already seen, while preserving validity under the null. In the constructions used here, the rule defining E_i may be chosen using $\mathcal{F}_{i-1}$ and is then evaluated when the new information at step i arrives. This local conditional validity is exactly what makes sequential updating possible (Grünwald et al. 2024; Ramdas and Wang 2025).

The key payoff from the definition is that conditional e-variables multiply.

Proposition 4.2 (Product construction (adapted from Grünwald et al. 2024; Ramdas and Wang 2025)). *If $E_1, E_2, \dots$ are nonnegative random variables such that each E_i is a conditional e-variable for $\mathcal{P}$ given $\mathcal{F}_{i-1}$, then the running product*

$$M_n = \prod_{i=1}^n E_i$$

is a test supermartingale for $\mathcal{P}$, with $M_0 = 1$, and hence an e-process for $\mathcal{P}$.

Proof. The empty product gives $M_0 = 1$. For $n \geq 0$, each factor E_i with $i \leq n$ is $\mathcal{F}_i$ -measurable and hence $\mathcal{F}_n$ -measurable. Therefore, M_n is nonnegative and $\mathcal{F}_n$ -measurable, and

$$\mathbb{E}_P[M_{n+1} \mid \mathcal{F}_n] = \mathbb{E}_P[M_n E_{n+1} \mid \mathcal{F}_n] = M_n \mathbb{E}_P[E_{n+1} \mid \mathcal{F}_n] \leq M_n.$$

Thus (M_n) is a test supermartingale, and an e-process by Proposition 4.1. $\square$

This result is useful because it turns a **local** condition, validity one step ahead given the past, into a **global** sequential evidence process. In practice, this is how most concrete e-processes are built. One first constructs a valid one-step factor, then multiplies these factors over time. In that sense, conditional e-variables are not an optional extra piece of theory: they are one of the main construction devices behind sequential safe tests (Grünwald et al. 2024; Ramdas and Wang 2025).

This construction captures the common sequential backbone behind the tests in Chapter 3. In the **safe z-test**, each factor is a one-step Gaussian likelihood ratio. In the **safe logrank test**, each event contributes a conditional factor based on the current risk set. In the **MLE plug-in** strategy from Section 2.10, where under the null the X_i are iid with density p , the estimate $\hat{Q}_{i-1}$ with density $\hat{q}_{i-1}$ is constructed from $\mathcal{F}_{i-1}$, so X_i still has conditional density p given the past and the factor $\hat{q}_{i-1}(X_i)/p(X_i)$ has conditional expectation one. In more elaborate examples such as the **safe t-test** and **safe tests for 2×2 tables**, the

same principle applies, even though the one-step factors may be expressed after coarsening the data or integrating out nuisance parameters (Grünwald et al. 2024; Turner et al. 2024; ter Schure et al. 2024; Ramdas et al. 2023). For dependent observations, the denominator would instead need to be the appropriate null conditional density given the past.

The product construction also formalises **optional continuation**, that is, the valid accumulation of evidence across separate stages or studies. Suppose study 1 produces an e-variable $E^{(1)}$ for a null hypothesis $\mathcal{P}$, and let $\mathcal{G}$ denote the information generated by that study, so that $E^{(1)}$ is $\mathcal{G}$ -measurable. A second study then collects fresh data and produces $E^{(2)}$, a conditional e-variable for the same null given $\mathcal{G}$. By Proposition 4.2, the product $E^{(1)}E^{(2)}$ is again a valid e-variable for $\mathcal{P}$. Repeating the argument shows that evidence can be accumulated over any number of stages, provided each new factor is conditionally valid given everything that came before, under the same null. In practice this is achieved by basing each new factor on fresh data. This is one of the main advantages of the e-value framework in the optional continuation setting (Grünwald et al. 2024; Ramdas and Wang 2025).

This is the sequential formalisation of the optional continuation idea already used informally in Chapter 3: one may continue the experiment, or even start a new one, without invalidating the evidence.

4.4 Ville's inequality and anytime-valid testing

We now state the theorem that turns the supermartingale property into an anytime-valid testing guarantee. Ville's inequality is the sequential analogue of Markov's inequality: Markov controls threshold crossings at one fixed analysis time, whereas Ville controls the probability of **ever** crossing a threshold along the monitored path (Ramdas and Wang 2025; Ramdas et al. 2023).

Theorem 4.1 (Ville's inequality (Ville 1939; Ramdas and Wang 2025)). *Let $(M_n)_{n \geq 0}$ be a nonnegative supermartingale with respect to $(\mathcal{F}_n)$ under P , with $\mathbb{E}_P[M_0] \leq 1$. Then for any $c > 0$,*

$$P\left(\sup_{n \geq 0} M_n \geq c\right) \leq \frac{1}{c}.$$

In particular, taking $c = 1/\alpha$, for any $\alpha \in (0, 1]$,

$$P\left(\exists n \geq 0 : M_n \geq \frac{1}{\alpha}\right) \leq \alpha.$$

The theorem is stated here in its classical supermartingale form because that is the form proved below. The same crossing bound holds for every e-process by applying its defining stopped-value guarantee to the first-crossing time (Ramdas and Wang 2025, Fact 7.6).

Proof. Define the stopping time

$$\tau = \inf\{n \geq 0 : M_n \geq c\},$$

with $\tau = \infty$ if the set is empty. For any fixed N , the stopping time $\tau \wedge N$ is bounded. Since the stopped process $(M_{n \wedge (\tau \wedge N)})_{n \geq 0}$ is again a nonnegative supermartingale,

$$\mathbb{E}_P[M_{\tau \wedge N}] \leq \mathbb{E}_P[M_0] \leq 1.$$

Moreover,

$$M_{\tau \wedge N} \geq c \mathbf{1}_{\{\tau \leq N\}},$$

because on the event $\{\tau \leq N\}$ we have $M_{\tau \wedge N} = M_\tau \geq c$, while on $\{\tau > N\}$ both sides are nonnegative. Taking expectations yields

$$cP(\tau \leq N) \leq \mathbb{E}_P[M_{\tau \wedge N}] \leq 1.$$

Hence $P(\tau \leq N) \leq 1/c$ for every N . Letting $N \rightarrow \infty$ and using continuity of probability from below,

$$P(\tau < \infty) = \lim_{N \rightarrow \infty} P(\tau \leq N) \leq \frac{1}{c}.$$

Since

$$\{\tau < \infty\} = \{\exists n \geq 0 : M_n \geq c\},$$

the argument proves

$$P(\exists n \geq 0 : M_n \geq c) \leq \frac{1}{c}.$$

To obtain the stated supremum bound, fix any $c' \in (0, c)$. If $\sup_{n \geq 0} M_n \geq c$, then $M_n \geq c'$ for some finite n , and hence, by the same argument applied at level c' ,

$$P\left(\sup_{n \geq 0} M_n \geq c\right) \leq P(\exists n \geq 0 : M_n \geq c') \leq \frac{1}{c'}.$$

Letting $c' \uparrow c$ gives

$$P\left(\sup_{n \geq 0} M_n \geq c\right) \leq \frac{1}{c},$$

as required. $\square$

Applying the theorem with $c = 1/\alpha$ shows that a sequential test which rejects as soon as $M_n \geq 1/\alpha$ has type-I error at most α . This is the basic mechanism behind anytime-valid testing with test supermartingales, and hence with the e-processes derived from them (Ramdas and Wang 2025; Grünwald et al. 2024).

We can now formalise the notion that was used informally throughout Chapter 3.

Definition 4.4 (Anytime-valid test). A sequential test is **anytime-valid** at level α for $\mathcal{P}$ if, for every $P \in \mathcal{P}$,

$$P(\exists n \geq 1 : \text{the null hypothesis is rejected at time } n) \leq \alpha.$$

This is stronger than fixed-horizon validity: a classical fixed-sample test controls error only at the single sample size for which it was designed, whereas an anytime-valid test controls the probability of **ever** rejecting along the monitored path.

By Ville's inequality, any rule of the form “reject the null whenever $M_n \geq 1/\alpha$ ” is anytime-valid whenever (M_n) is a test supermartingale. The same guarantee also holds if one evaluates the process at any stopping time τ with respect to $(\mathcal{F}_n)$ and rejects when $M_\tau \geq 1/\alpha$.

This clarifies the relationship to the group-sequential methods discussed in Section 3.5. The original O'Brien–Fleming procedure (O'Brien and Fleming 1979) fixes the maximum

number and spacing of interim analyses. Lan–DeMets alpha-spending (Lan and DeMets 1983) is more flexible: the spending function is specified in advance, but future decision times and the total number of looks need not be known. Anytime-valid tests go further by controlling type-I error uniformly over arbitrary valid stopping times, including stopping rules based on the accumulated evidence itself. The price of that extra flexibility is often reduced fixed-sample efficiency, as seen earlier in Section 2.11 and Section 3.3.4. We already saw the error-control side of this contrast empirically in Figure 3.3 and Figure 3.7.

4.5 Confidence sequences

The tests in Chapter 3 and the theory above answer a binary question: should the null hypothesis be rejected? Estimation asks a richer question: which parameter values remain plausible given the data? Just as e-processes extend e-values to the sequential setting, confidence sequences extend fixed-sample confidence intervals to procedures that remain valid under continuous monitoring.

4.5.1 Why fixed-sample intervals fail under monitoring

A classical $(1 - \alpha)$ confidence interval C_n satisfies $P_\theta(\theta \in C_n) \geq 1 - \alpha$ for each fixed, pre-specified sample size n , where P_θ denotes the law of the data when the parameter of interest takes value θ . If the analyst reports C_n only once at that predetermined sample size, this guarantee is valid. But if such intervals are recomputed and inspected repeatedly as data accumulate, this pointwise guarantee does not imply simultaneous coverage over time. In particular, the probability that the interval excludes the true parameter at least once along the monitored path can substantially exceed α . As a consequence, fixed-sample intervals also do not in general retain their nominal coverage at arbitrary data-dependent stopping times (Howard et al. 2021; Ramdas et al. 2023, sec. 2.8). This is the estimation analogue of repeated testing: the relevant quantity is no longer coverage at one fixed look, but the probability of ever excluding the truth during monitoring.

To illustrate this distinction, we simulate iid $N(0, 1)$ observations and compare repeatedly checked fixed-sample Gaussian z -intervals with a Gaussian mixture confidence sequence for the mean. For the confidence sequence, we use the half-width formula derived later in Section 4.5.3. The parameter ρ is a tuning parameter of the Gaussian mixture construction; here we set $\rho = 1$ for illustration.

```
n_max_cov <- 5000
n_sim_cov <- 2000
alpha <- 0.05
rho <- 1

# Precompute half-widths (independent of simulated data)
ci_hw <- qnorm(1 - alpha / 2) / sqrt(1:n_max_cov)
cs_hw <- sqrt(2 * (1 + (1:n_max_cov) * rho) /
              ((1:n_max_cov)^2 * rho) *
              (log(1 / alpha) + 0.5 * log(1 + (1:n_max_cov) * rho)))

set.seed(123)
ever_miss_ci <- numeric(n_max_cov)
ever_miss_cs <- numeric(n_max_cov)
```

```

for (sim in 1:n_sim_cov) {
  x <- rnorm(n_max_cov, mean = 0, sd = 1)
  xbar <- cumsum(x) / (1:n_max_cov)

  # At n = 1 the fixed-sample interval
  # misses with probability exactly alpha, which is the reference point
  # against which the later looks degrade
  miss_ci <- which(abs(xbar) > ci_hw)[1]
  miss_cs <- which(abs(xbar) >= cs_hw)[1]

  if (!is.na(miss_ci)) {
    ever_miss_ci[miss_ci:n_max_cov] <-
      ever_miss_ci[miss_ci:n_max_cov] + 1
  }
  if (!is.na(miss_cs)) {
    ever_miss_cs[miss_cs:n_max_cov] <-
      ever_miss_cs[miss_cs:n_max_cov] + 1
  }
}

ever_miss_ci <- ever_miss_ci / n_sim_cov * 100
ever_miss_cs <- ever_miss_cs / n_sim_cov * 100

plot_idx <- unique(c(seq(1, 500, by = 1), seq(500, n_max_cov, by = 10)))

df_cov <- data.frame(
  n = rep(plot_idx, 2),
  pct_miss = c(ever_miss_ci[plot_idx], ever_miss_cs[plot_idx]),
  method = rep(c("Repeated fixed-sample CI",
                  "Confidence sequence"), each = length(plot_idx))
)

ggplot(df_cov, aes(x = n, y = pct_miss, color = method)) +
  geom_line(linewidth = 0.7) +
  geom_hline(yintercept = 5, linetype = "dashed", color = "grey40") +
  annotate("text", x = n_max_cov * 0.8, y = 8,
    label = "alpha == 5*%", parse = TRUE,
    size = 3, color = "grey40") +
  scale_color_manual(values = c(
    "Confidence sequence" = "steelblue",
    "Repeated fixed-sample CI" = "firebrick")) +
  labs(x = "Number of looks / sample size",
    y = "% ever excluding true mean by time n",
    color = NULL) +
  coord_cartesian(ylim = c(0, 65)) +
  theme(legend.position = "bottom")

```

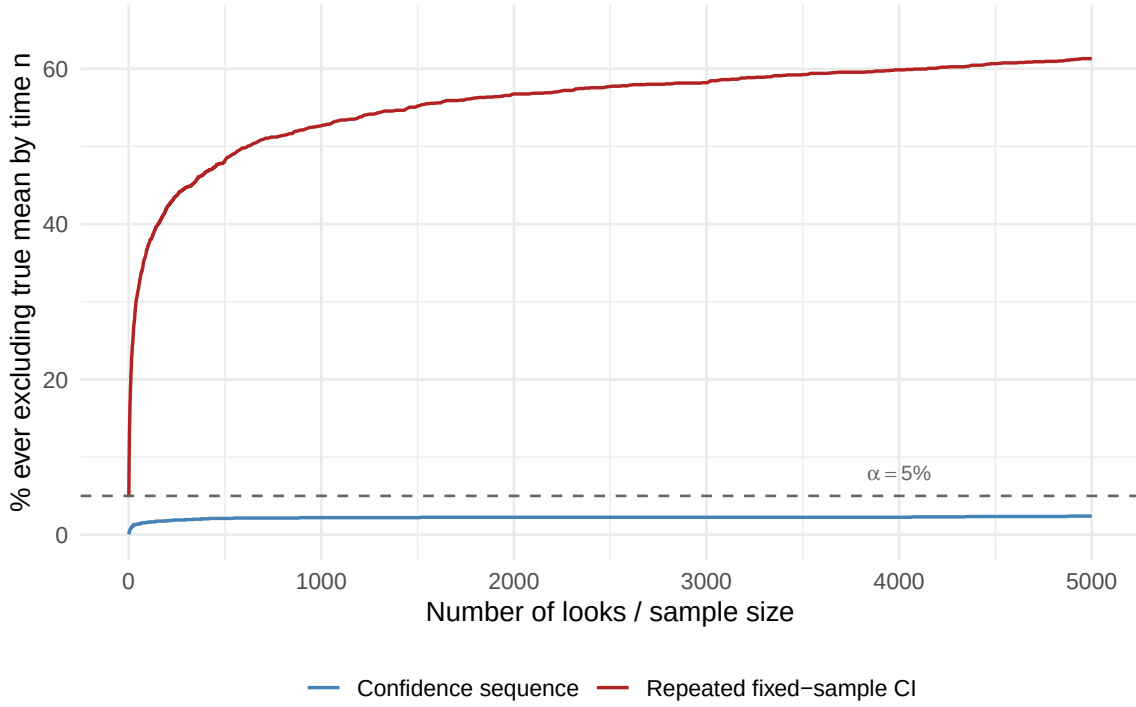


Figure 4.1: Probability of ever excluding the true mean by time n under continuous monitoring, for repeatedly checked fixed-sample Gaussian z -intervals and a Gaussian mixture confidence sequence, both nominally at 95%. Both quantities are cumulative and therefore nondecreasing. The repeated fixed-sample intervals increasingly fail to cover the true parameter somewhere along the path, whereas the confidence-sequence curve remains below the nominal 5% error bound in this simulation. At $n = 1$, the fixed-sample interval has theoretical miss probability exactly α .

Figure 4.1 shows the probability of having excluded the true mean at least once by time n . This is the relevant pathwise quantity under continuous monitoring. Repeated fixed-sample intervals, like repeated pointwise tests, do not control it. Confidence sequences are designed precisely to do so: their guarantee is time-uniform, meaning that the true parameter remains inside the sequence of intervals simultaneously over time with probability at least $1 - \alpha$.

4.5.2 Definition and inversion

Definition 4.5 (Confidence sequence (Lai 1976; Howard et al. 2021; Ramdas et al. 2023, sec. 2.8)). In a model indexed by $\theta \in \Theta$, a **confidence sequence** at level $1 - \alpha$ is a sequence of confidence sets $(C_n)_{n \geq 1}$ such that, for every $\theta \in \Theta$,

$$P_\theta(\forall n \geq 1 : \theta \in C_n) \geq 1 - \alpha.$$

Equivalently,

$$P_\theta(\exists n \geq 1 : \theta \notin C_n) \leq \alpha.$$

The phrase “for all n ” is the crucial difference from an ordinary confidence interval. The parameter must be covered at every sample size simultaneously, not just at one pre-chosen

time. As a result, a confidence sequence remains valid under continuous monitoring and therefore also at arbitrary data-dependent stopping times.

Confidence sequences can be constructed by inverting test supermartingales. For each candidate value θ_0 , suppose we have a test supermartingale $(M_n^{(\theta_0)})$ for testing $H_0 : \theta = \theta_0$. Define

$$C_n = \left\{ \theta_0 : M_n^{(\theta_0)} < 1/\alpha \right\}.$$

Then (C_n) is a confidence sequence at level $1 - \alpha$. Indeed, if the true parameter value is θ , then

$$P_\theta(\exists n \geq 1 : \theta \notin C_n) = P_\theta(\exists n \geq 1 : M_n^{(\theta)} \geq 1/\alpha) \leq \alpha,$$

where the equality restates the definition of C_n , and the inequality is the second display of Theorem 4.1 applied to the test supermartingale associated with the true parameter value. Thus, the true parameter can leave the confidence set at some time only if its own test supermartingale ever crosses the threshold $1/\alpha$, and the probability of this happening is at most α .

Note that (C_n) need not be nested: because the e-process value $M_n^{(\theta_0)}$ can rise above $1/\alpha$ at one time and fall below it later, a candidate θ_0 can leave C_n and subsequently re-enter $C_{n'}$ for some $n' > n$. For applications where a monotonically shrinking sequence is desired, one can take the running intersection

$$\tilde{C}_n = \bigcap_{i \leq n} C_i = \left\{ \theta_0 : \max_{i \leq n} M_i^{(\theta_0)} < 1/\alpha \right\},$$

which is nested by construction and inherits the same simultaneous coverage guarantee. In the Gaussian example below, the pointwise sets C_n are the more convenient objects to derive and plot.

4.5.3 A Gaussian mixture confidence sequence

To make the construction concrete, we derive a confidence sequence for the mean of Gaussian observations with known variance. This uses the same method of mixtures as in Section 2.10, here with a Gaussian mixing distribution, now applied pointwise to each candidate value θ_0 .

Suppose $X_1, X_2, \dots \stackrel{\text{iid}}{\sim} N(\theta, 1)$ with unknown mean θ . For a candidate value θ_0 , define

$$S_n^{(\theta_0)} = \sum_{i=1}^n (X_i - \theta_0) = n(\bar{X}_n - \theta_0).$$

For a fixed local alternative $\theta_0 + \lambda$, let

$$L_n^{(\theta_0, \lambda)} = \exp\left(\lambda S_n^{(\theta_0)} - \frac{n\lambda^2}{2}\right).$$

This is the likelihood ratio of $N(\theta_0 + \lambda, 1)$ against $N(\theta_0, 1)$ after n observations. Setting $S_0^{(\theta_0)} = 0$, as the empty sum requires, gives $L_0^{(\theta_0, \lambda)} = 1$, so the process $(L_n^{(\theta_0, \lambda)})_{n \geq 0}$ is a test martingale under P_{θ_0} for each fixed λ .

The statistic $S_n^{(\theta_0)}$ measures the cumulative deviation of the data from the candidate value θ_0 . If $\bar{X}_n$ stays close to θ_0 , then $S_n^{(\theta_0)}$ remains small and little evidence accumulates against that candidate. If $\bar{X}_n$ drifts away from θ_0 , then the evidence against θ_0 increases.

Since the direction and size of the deviation from θ_0 are unknown, we average this likelihood ratio over a symmetric Gaussian mixing distribution $\lambda \sim N(0, \rho)$ with variance $\rho > 0$, where ρ is a fixed tuning parameter chosen in advance. Because the mixing distribution is symmetric around zero, positive and negative deviations from θ_0 are handled within the same construction, yielding a two-sided mixture (see Ramdas et al. 2023, sec. 5.1; Howard et al. 2021):

$$M_n^{(\theta_0)} = \int_{-\infty}^{\infty} L_n^{(\theta_0, \lambda)} \frac{1}{\sqrt{2\pi\rho}} \exp\left(-\frac{\lambda^2}{2\rho}\right) d\lambda.$$

This is the same mixture strategy used for composite hypotheses in Section 2.10, now applied pointwise for each candidate value θ_0 . The tuning parameter ρ determines how widely the mixture spreads its mass over possible deviations from θ_0 . In Section 5.1 of Ramdas et al. (2023) the centred Gaussian mixing distribution is written as $N(0, \tilde{\rho}^2)$, so $\tilde{\rho}^2 = \rho$ in the notation used here.

Combining the quadratic terms in λ gives

$$M_n^{(\theta_0)} = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\rho}} \exp\left(\lambda S_n^{(\theta_0)} - \frac{1}{2}\left(n + \frac{1}{\rho}\right)\lambda^2\right) d\lambda.$$

Write $\sigma_*^2 = 1/(n + 1/\rho)$ and $\mu^* = \sigma_*^2 S_n^{(\theta_0)}$. Then the exponent in the integrand can be rewritten as

$$\lambda S_n^{(\theta_0)} - \frac{1}{2\sigma_*^2}\lambda^2 = -\frac{1}{2\sigma_*^2}(\lambda - \mu^*)^2 + \frac{(S_n^{(\theta_0)})^2}{2(n + 1/\rho)},$$

so the integral becomes

$$M_n^{(\theta_0)} = \frac{1}{\sqrt{2\pi\rho}} \exp\left(\frac{(S_n^{(\theta_0)})^2}{2(n + 1/\rho)}\right) \int_{-\infty}^{\infty} \exp\left(-\frac{(\lambda - \mu^*)^2}{2\sigma_*^2}\right) d\lambda.$$

The remaining integral is that of an unnormalised Gaussian with variance $\sigma_*^2 = 1/(n + 1/\rho)$, so it evaluates to $\sqrt{2\pi\sigma_*^2} = \sqrt{2\pi/(n + 1/\rho)}$. Multiplying by the prefactor $1/\sqrt{2\pi\rho}$ gives

$$\frac{1}{\sqrt{2\pi\rho}} \cdot \sqrt{\frac{2\pi}{n + 1/\rho}} = \frac{1}{\sqrt{\rho(n + 1/\rho)}} = \frac{1}{\sqrt{1 + n\rho}}.$$

Thus

$$M_n^{(\theta_0)} = \frac{1}{\sqrt{1 + n\rho}} \exp\left(\frac{(S_n^{(\theta_0)})^2}{2(n + 1/\rho)}\right).$$

Since $1/(n + 1/\rho) = \rho/(1 + n\rho)$, this is equivalently

$$M_n^{(\theta_0)} = \frac{1}{\sqrt{1 + n\rho}} \exp\left(\frac{\rho (S_n^{(\theta_0)})^2}{2(1 + n\rho)}\right).$$

Substituting $S_n^{(\theta_0)} = n(\bar{X}_n - \theta_0)$ gives the closed form

$$M_n^{(\theta_0)} = \frac{1}{\sqrt{1 + n\rho}} \exp\left(\frac{n^2\rho}{2(1 + n\rho)} (\bar{X}_n - \theta_0)^2\right).$$

This is in fact a test martingale under $\theta = \theta_0$. For each fixed λ , the process $(L_n^{(\theta_0, \lambda)})_{n \geq 0}$ is a test martingale under P_{θ_0} , and integrating these processes against a fixed proper mixing distribution preserves the martingale property, by Tonelli's theorem for conditional expectations, which applies because the integrand is nonnegative. It is therefore also an e-process under P_{θ_0} by Proposition 4.1. Since $M_n^{(\theta_0)}$ increases with $|\bar{X}_n - \theta_0|$, candidate means further from the running sample mean yield larger e-values. Thus, when we invert the process, the surviving values of θ_0 will be precisely those that remain sufficiently close to $\bar{X}_n$.

To invert the e-process at time n , require $M_n^{(\theta_0)} < 1/\alpha$ and take logarithms:

$$-\frac{1}{2} \log(1 + n\rho) + \frac{n^2 \rho}{2(1 + n\rho)} (\bar{X}_n - \theta_0)^2 < \log(1/\alpha).$$

Rearranging gives

$$(\bar{X}_n - \theta_0)^2 < \frac{2(1 + n\rho)}{n^2 \rho} \left[\log(1/\alpha) + \frac{1}{2} \log(1 + n\rho) \right].$$

Thus the half-width is

$$w_n = \sqrt{\frac{2(1 + n\rho)}{n^2 \rho} \left[\log\left(\frac{1}{\alpha}\right) + \frac{1}{2} \log(1 + n\rho) \right]} \quad (4.1)$$

Equivalently,

$$C_n = (\bar{X}_n - w_n, \bar{X}_n + w_n).$$

No $\alpha/2$ appears here because the symmetric Gaussian mixture already defines a two-sided test: positive and negative deviations from θ_0 are handled within the same e-process, so inverting the threshold $1/\alpha$ directly yields a two-sided confidence sequence.

The form of this interval is familiar, but its width differs from that of a classical fixed-sample Gaussian interval. A standard fixed-sample interval has half-width proportional to $1/\sqrt{n}$, whereas the additional $\log(1 + n\rho)$ term here slows that decay. The consequences are worked out in Section 4.5.5.

This half-width formula will be used directly in the R implementation below.

4.5.4 Confidence sequences in R

We now implement the Gaussian mixture confidence sequence and compare it with the ordinary fixed-sample Gaussian confidence interval with known variance, evaluated at each sample size n .

```
theta_true <- 0.5
n_max <- 400
alpha <- 0.05
rho <- 1

set.seed(42)
x <- rnorm(n_max, mean = theta_true, sd = 1)
xbar <- cumsum(x) / (1:n_max)
```

```

cs_hw <- sqrt(2 * (1 + (1:n_max) * rho) / ((1:n_max)^2 * rho) *
             (log(1 / alpha) + 0.5 * log(1 + (1:n_max) * rho)))

ci_hw <- qnorm(1 - alpha / 2) / sqrt(1:n_max)

# Start plotting at n = 5 to avoid visually extreme early-sample behaviour
path_idx <- 5:n_max
df_cs <- data.frame(
  n = path_idx,
  xbar = xbar[path_idx],
  cs_lo = xbar[path_idx] - cs_hw[path_idx],
  cs_hi = xbar[path_idx] + cs_hw[path_idx],
  ci_lo = xbar[path_idx] - ci_hw[path_idx],
  ci_hi = xbar[path_idx] + ci_hw[path_idx]
)

ggplot(df_cs, aes(x = n)) +
  geom_ribbon(aes(ymin = cs_lo, ymax = cs_hi),
             fill = "steelblue", alpha = 0.2) +
  geom_line(aes(y = cs_lo), color = "steelblue", linewidth = 0.5) +
  geom_line(aes(y = cs_hi), color = "steelblue", linewidth = 0.5) +
  geom_line(aes(y = ci_lo), color = "firebrick",
            linetype = "dashed", linewidth = 0.4) +
  geom_line(aes(y = ci_hi), color = "firebrick",
            linetype = "dashed", linewidth = 0.4) +
  geom_line(aes(y = xbar), color = "grey30", linewidth = 0.4) +
  geom_hline(yintercept = theta_true, linetype = "dotted",
             color = "grey50") +
  annotate("text", x = n_max - 5, y = theta_true + 0.12,
           label = "theta == 0.5", parse = TRUE,
           hjust = 1, size = 3, color = "grey50") +
  labs(x = "Sample size",
       y = "Mean and interval bounds")

```

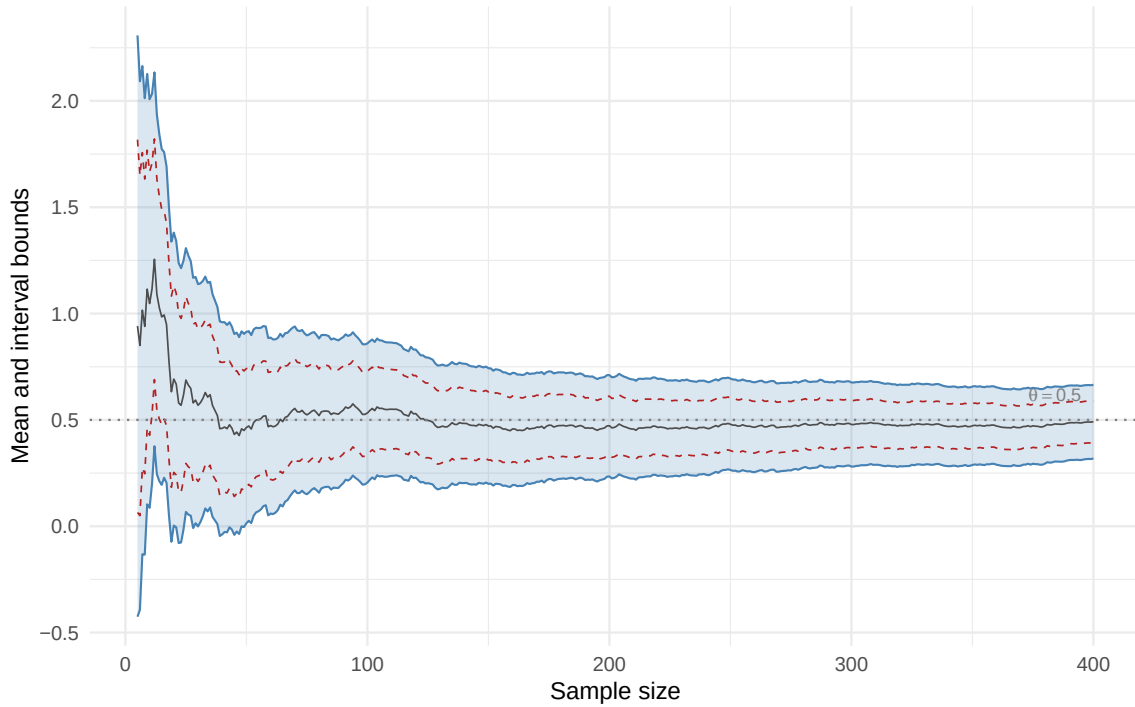



Figure 4.2: A Gaussian mixture confidence sequence (shaded blue band) and fixed-sample 95% Gaussian confidence intervals evaluated at each sample size (red dashed lines) for the same data stream. The grey solid line is the running sample mean $\bar{X}_n$ and the true mean is $\theta = 0.5$ (grey dotted line). The tuning parameter is fixed at $\rho = 1$ for illustration. Display starts at $n = 5$ for readability.

Figure 4.2 shows one realised Gaussian data stream. At early sample sizes, the confidence sequence is noticeably wider than the fixed-sample interval evaluated at the same n : this is the cost of simultaneous validity over time. As the sample size grows, both intervals contract and the absolute gap between them narrows, but the confidence sequence remains wider throughout, and the ratio of the two half-widths eventually grows slowly with n , as Section 4.5.5 makes precise.

The blue band retains its coverage guarantee under continuous monitoring, whereas the red dashed intervals have 95% coverage only pointwise at each fixed sample size.

For illustration, the figure fixes the tuning parameter at $\rho = 1$. The simultaneous coverage guarantee holds for any fixed choice of $\rho > 0$; changing ρ affects the width profile of the sequence over time, but not its validity. In that sense, ρ is a tuning parameter, analogous to the design choices used for safe tests in Chapter 3.

4.5.5 Width of the Gaussian mixture confidence sequence

The Gaussian mixture confidence sequence is wider than the corresponding fixed-sample Gaussian confidence interval because it must remain valid under continuous monitoring, not just at one pre-specified sample size. This extra width is the price of time-uniform validity. Starting from the exact half-width in Equation 4.1, a large- n approximation uses

$1 + n\rho \approx n\rho$, which gives

$$w_n \approx \sqrt{\frac{2[\log(1/\alpha) + \frac{1}{2}\log(n\rho)]}{n}} \asymp \sqrt{\frac{\log n}{n}}.$$

Here $f(n) \asymp g(n)$ means that the ratio $f(n)/g(n)$ is bounded above and below by positive constants for large n . Since $\log(1/\alpha)$ and $\frac{1}{2}\log \rho$ are constant in n , the dominant term for large n is $\frac{1}{2}\log n$, so the half-width shrinks to zero, but more slowly than the fixed-sample $n^{-1/2}$ rate.

By contrast, the fixed-sample Gaussian confidence interval has half-width

$$z_{1-\alpha/2}/\sqrt{n} \asymp 1/\sqrt{n}.$$

Thus, this Gaussian mixture confidence sequence incurs an additional factor of order $\sqrt{\log n}$ relative to the fixed-sample interval. The parameter ρ is a tuning parameter of the Gaussian mixture construction: it affects the finite-sample width, but not the asymptotic rate $\sqrt{\log n/n}$. Although the extra $\sqrt{\log n}$ factor diverges, it does so very slowly, so the resulting widening is limited over a broad range of sample sizes.

Sharper confidence sequences with smaller time-uniform penalties are available (see, e.g., Howard et al. 2021), but the Gaussian mixture construction is especially useful here because it is explicit, easy to derive from an e-process, and easy to invert. In a sequential setting such as a clinical trial with repeated interim looks, this means that the price of being allowed to monitor the interval continuously is a modest widening relative to the interval one would report at a single fixed analysis.

```
nn <- 20:2000
rho_values <- c(0.5, 1, 2)

df_ratio <- do.call(rbind, lapply(rho_values, function(rho) {
  cs_w <- sqrt(2 * (1 + nn * rho) / (nn^2 * rho) *
    (log(1 / alpha) + 0.5 * log(1 + nn * rho)))
  ci_w <- qnorm(1 - alpha / 2) / sqrt(nn)
  data.frame(
    n = nn,
    ratio = cs_w / ci_w,
    rho = paste0("rho == ", rho)
  )
}))

ggplot(df_ratio, aes(x = n, y = ratio, color = rho)) +
  geom_line(linewidth = 0.7) +
  scale_color_discrete(labels = function(l) parse(text = l)) +
  expand_limits(y = 1) +
  labs(x = "Sample size",
    y = "CS half-width / CI half-width",
    color = NULL) +
  theme(legend.position = "bottom")
```

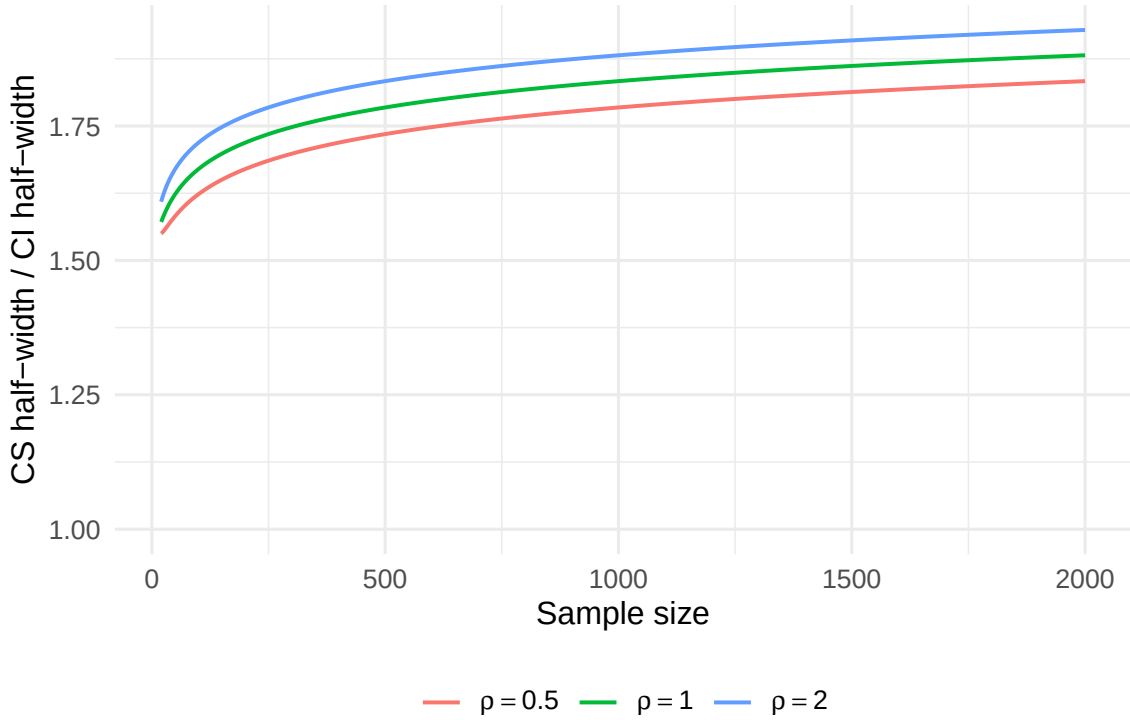


Figure 4.3: Ratio of the Gaussian mixture confidence-sequence half-width to the fixed-sample Gaussian confidence-interval half-width for three values of ρ . The widening factor grows slowly with n , illustrating the cost of requiring validity at all stopping times.

Figure 4.3 shows that the confidence-sequence half-width exceeds the fixed-sample half-width by a factor that increases only slowly with n . Over the plotted range, the ratio never exceeds 1.93 for any of the three values of ρ , illustrating that the cost of time-uniform validity is visible but not dramatic.

4.5.6 Beyond the Gaussian case

The Gaussian mixture confidence sequence above relied on the same simplifying assumption as the safe z-test in Section 3.2: the variance was treated as known. In more realistic settings, the same broad strategy still applies, but the construction of the underlying e-process becomes more delicate.

For a normal mean with unknown variance, the nuisance scale parameter prevents the simple known-variance likelihood-ratio construction from carrying over unchanged. The right-Haar-prior ideas that already appeared in the safe t-test in Section 3.3 are also relevant here, and Lai’s classical confidence-sequence construction can be understood from this perspective (Lai 1976). However, as Wang and Ramdas (2025) explain, Lai’s construction combines an improper right-Haar mixture over the scale parameter with an improper flat mixture over the mean μ . The resulting process is an extended nonnegative supermartingale rather than an ordinary integrable martingale, and therefore is not an e-process. They present two valid alternatives: a test martingale in a reduced scale-invariant filtration and an e-process in the canonical data filtration. We refer the reader to their paper for the full constructions.

Beyond Gaussian models, sub-Gaussian methods apply to observations that need not

be bounded, while empirical-Bernstein methods provide variance-adaptive nonparametric confidence sequences for bounded observations (Howard et al. 2021; Waudby-Smith and Ramdas 2024).

For the e-process-based constructions described above, the unifying principle is the same: construct a family of e-processes indexed by the candidate parameter value, obtain an anytime-valid test for each candidate by thresholding at $1/\alpha$, and then invert that family of tests to obtain a confidence sequence.

4.6 The common backbone of safe tests

We can now reinterpret the safe tests from Chapter 3 as instances of a common sequential template:

1. **Construct conditional e-variables.** At each time step, form a one-step e-variable that is conditionally valid given the past.
2. **Multiply them.** By Proposition 4.2, the running product is a test supermartingale.
3. **Apply Ville’s inequality.** Thresholding this test supermartingale at $1/\alpha$ gives an anytime-valid test.
4. **Invert.** If the resulting family of test supermartingales is indexed by a candidate parameter value, inverting the corresponding anytime-valid tests yields a confidence sequence.

This template is cleanest when one-step conditional e-variables are available directly in the chosen filtration. Some important constructions are slightly less direct: they require a suitable coarsening of the filtration or a different representation of the data before the sequential validity argument takes this product form.

Across the safe tests, the common backbone is therefore the same, while the main modelling work lies in step 1. In the safe z-test, the one-step factor is a Gaussian likelihood ratio; see Section 3.2. In the safe t-test, the construction only becomes valid after handling the nuisance scale parameter through the right-Haar-prior approach and its associated invariant structure (Grünwald et al. 2024); see Section 3.3. In the safe 2×2 test, the factor is based on the beta-binomial construction of Turner et al. (2024); see Section 3.4. In the safe logrank test, it is built from conditional risk-set contributions (ter Schure et al. 2024); see Section 3.5. Once an appropriate conditionally valid factor has been identified in the relevant filtration, the remaining supermartingale argument and the use of Ville’s inequality are generic.

This is also, in broad terms, the practical architecture of the `safestats` package (Turner et al. 2022). The `designSafe*` functions specify the design parameters of the safe procedure, while the `safe*Test` functions update the running evidence and compare it with the threshold $1/\alpha$.

Figure 4.4 summarises this four-step template.

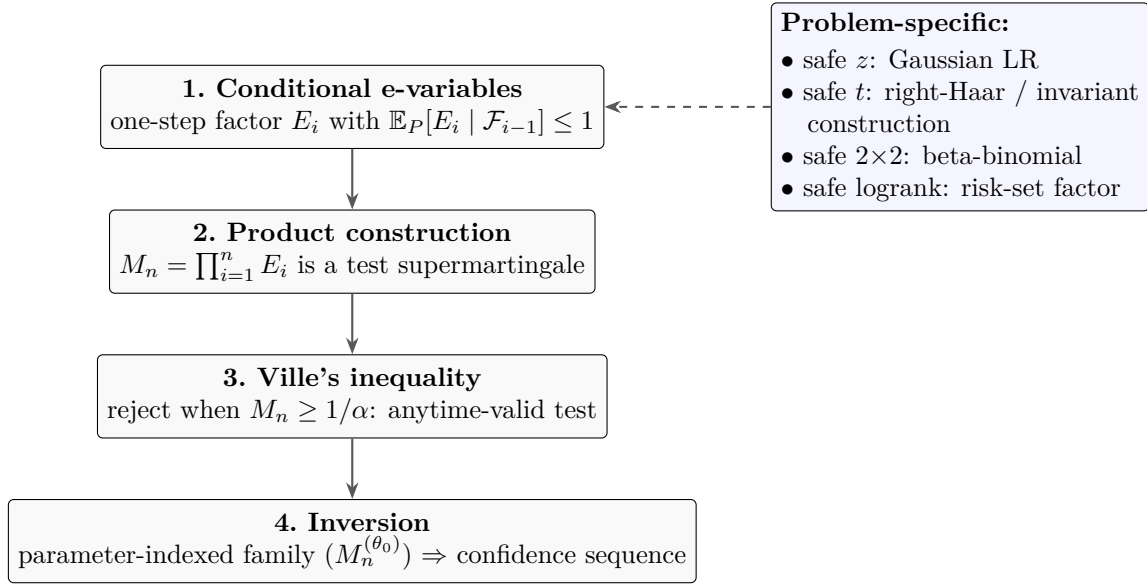


Figure 4.4: Common sequential backbone behind the safe tests of Chapter 3. Steps 2–4 are generic; only the construction of the one-step factor in step 1 is problem-specific. Inversion to a confidence sequence applies when a parameter-indexed family of such processes is available.

4.7 Summary

This chapter developed the sequential machinery that underlies the safe tests in Chapter 3.

1. **Test supermartingales and e-processes** extend e-values from single analyses to evolving evidence streams. In this chapter, test supermartingales were the main construction; by Proposition 4.1, every test supermartingale is also an e-process, meaning its value at every stopping time is a valid e-value.
2. **Conditional e-variables and the product construction** explain how sequential evidence is built. At each step, one chooses a conditionally valid e-variable based on the past, and the running product remains valid under the null. This is the basic mechanism behind adaptive sequential monitoring and optional continuation.
3. **Ville's inequality** is the key theorem behind anytime-valid testing. For a test supermartingale, it guarantees that the probability of ever crossing the threshold $1/\alpha$ is at most α , with no restriction on when the analyst looks. Separately, the value of the process at a stopping time with respect to the chosen filtration is itself a valid e-value.
4. **Confidence sequences** are the estimation analogue of anytime-valid tests. They are obtained by inverting a family of anytime-valid tests derived from e-processes and provide simultaneous coverage over all sample sizes.
5. **Safe tests share a common sequential backbone.** What changes from one testing problem to another is the construction of an appropriate e-process or test supermartingale, often via conditional e-variables and sometimes after a suitable coarsening or re-expression of the data; once such a process is available, Ville-based validity is generic.

In the next chapter, we turn from a single hypothesis to many. The same e-value framework extends naturally to multiple testing, where methods such as eBH and the e-Partitioning Principle provide tools for false discovery rate control with e-values.

Chapter 5

Multiple Testing with E-Values

The previous chapters considered one hypothesis at a time. Chapter 3 constructed e-values for classical testing problems, and the sequential machinery of Definition 4.2, together with Ville's inequality (Theorem 4.1), showed that e-processes may be monitored and stopped without losing their null validity. This chapter asks what happens when many such hypotheses are tested together.

Let

$$H_1, \dots, H_m$$

be a family of null hypotheses, and write $[m] = \{1, \dots, m\}$. A multiple-testing procedure returns a random rejection set

$$R \subseteq [m],$$

where $i \in R$ means that H_i is rejected. For a data-generating distribution P , define the set of true null hypotheses as

$$N_P = \{i \in [m] : P \in H_i\},$$

so that the false discoveries are the rejected true nulls, $R \cap N_P$.

The classical family-wise error rate controls the probability of making at least one false rejection. In many scientific settings this is too stringent. When testing many endpoints, biomarkers, or subgroups, the more relevant question is often what fraction of the reported discoveries is false on average. Controlling that quantity, rather than the probability of any false rejection, is the goal of the false discovery rate.

Definition 5.1 (False discovery proportion and false discovery rate (adapted from Benjamini and Hochberg 1995; Xu et al. 2025, Table 3)). For a rejection set $R \subseteq [m]$, the **false discovery proportion** under P is

$$\text{FDP}_P(R) = \frac{|R \cap N_P|}{|R| \vee 1},$$

where $a \vee b = \max(a, b)$, so the denominator $|R| \vee 1$ enforces the convention $\text{FDP}_P(\emptyset) = 0$. The **false discovery rate** is its expectation,

$$\text{FDR}_P(R) = \mathbb{E}_P[\text{FDP}_P(R)].$$

A procedure controls FDR at level α if $\text{FDR}_P(R) \leq \alpha$ for every P in the model.

Suppose that for each H_i we have a valid e-value E_i , meaning that

$$\mathbb{E}_P[E_i] \leq 1 \quad \text{whenever } P \in H_i.$$

The E_i 's may come from the safe tests of Chapter 3, or from stopped e-processes as in the previous chapter. In particular, if $E_{i,t}$ is an e-process for H_i and τ_i is a stopping time with respect to the filtration for which that e-process is valid, then E_{i,τ_i} is itself a valid e-value for H_i (Definition 4.2). These stopped e-values can therefore be passed to an FDR procedure. If stopping depends jointly on information from several hypothesis streams, however, the processes must be valid in the corresponding shared global filtration; validity only in separate local filtrations is not sufficient (Wang et al. 2025). This is the main bridge from anytime-valid inference to multiple testing.

This chapter focuses on FDR control with e-values. Section 5.1 introduces the eBH procedure of Wang and Ramdas (2022), the e-value counterpart of the Benjamini–Hochberg procedure of Benjamini and Hochberg (1995). Section 5.2 then presents the FDR-specialised partitioning formulation of the e-Closure Principle developed in Xu et al. (2025). The terms “e-Partitioning”, “eBH+”, “BY+”, and “Su+” are retained from the earlier preprint Goeman et al. (2025) for continuity with the derivations and simulation below; the merged paper calls the general framework e-Closure and presents corresponding closed or simultaneous versions of eBH, BY, and Su.

5.1 The eBH procedure

The eBH procedure of Wang and Ramdas (2022) is the e-value analogue of the Benjamini–Hochberg step-up procedure, obtained by running Benjamini–Hochberg on the reciprocals $1/E_1, \dots, 1/E_m$. Its input is a vector

$$E_1, \dots, E_m$$

of valid e-values, where E_i is valid for H_i .

Let $i_{[1]}, \dots, i_{[m]}$ be an ordering of the hypotheses such that

$$E_{[1]} \geq E_{[2]} \geq \dots \geq E_{[m]},$$

where $E_{[k]} = E_{i_{[k]}}$. For a target FDR level $\alpha \in (0, 1)$, define

$$\hat{k} = \max \left\{ k \in \{1, \dots, m\} : E_{[k]} \geq \frac{m}{\alpha k} \right\},$$

with the convention that $\hat{k} = 0$ if the set is empty. The eBH rejection set is the set of the $\hat{k}$ hypotheses with the largest e-values,

$$R_{\text{eBH}} = \begin{cases} \{i_{[1]}, \dots, i_{[\hat{k}]}\}, & \hat{k} \geq 1, \\ \emptyset, & \hat{k} = 0. \end{cases}$$

Ties can be broken in any deterministic way. The R implementation below uses the original order of the hypotheses as the tie-breaker.

The eBH rejection set has a key self-consistency property. Whenever $i \in R_{\text{eBH}}$, its e-value is at least the threshold evaluated at the size of the rejection set,

$$E_i \geq \frac{m}{\alpha |R_{\text{eBH}}|}.$$

To see this, recall that $|R_{\text{eBH}}| = \hat{k}$. Any rejected index i is one of the top $\hat{k}$, so its e-value is at least the $\hat{k}$ -th largest, $E_i \geq E_{[\hat{k}]}$. By the definition of $\hat{k}$, the $\hat{k}$ -th largest e-value clears its own threshold, $E_{[\hat{k}]} \geq m/(\alpha\hat{k})$, and $\hat{k} = |R_{\text{eBH}}|$ gives the stated bound.

It is convenient to rewrite this bound so that it holds for every index i , not only the rejected ones,

$$\frac{\mathbf{1}\{i \in R_{\text{eBH}}\}}{|R_{\text{eBH}}| \vee 1} \leq \frac{\alpha}{m} E_i.$$

For a rejected index $i \in R_{\text{eBH}}$ the left-hand side equals $1/|R_{\text{eBH}}|$, and rearranging the self-consistency bound above gives exactly this inequality. For an index $i \notin R_{\text{eBH}}$ the left-hand side is zero while the right-hand side is non-negative, so the inequality holds automatically. The value of this version is that it can be summed over any set of indices, which is precisely what the FDR proof uses. Summing it over $i \in N_P$ and taking expectations turns self-consistency into the bound $\text{FDR}_P(R_{\text{eBH}}) \leq \alpha$, with no assumption on the joint distribution of $(E_1, \dots, E_m)$, as shown next.

Example 5.1 (eBH on a small input). Take $m = 10$ and $\alpha = 0.05$, and suppose the e-values are

$$(E_1, \dots, E_{10}) = (0.4, 220, 1.1, 95, 0.8, 60, 0.6, 55, 45, 0.9).$$

Sorting in decreasing order gives the $E_{[k]}$ shown in Table 5.1, alongside the threshold $m/(\alpha k) = 200/k$, which is 200 at rank 1 and decreases to 20 at rank 10.

Table 5.1: eBH walkthrough at $m = 10$, $\alpha = 0.05$. Ranks 7 to 10 all have $E_{[k]} < 1$ and are omitted.

k	$E_{[k]}$	$200/k$	$E_{[k]} \geq 200/k$
1	220	200.0	yes
2	95	100.0	no
3	60	66.7	no
4	55	50.0	yes
5	45	40.0	yes
6	1.1	33.3	no

The largest k at which $E_{[k]} \geq 200/k$ is $\hat{k} = 5$, so eBH rejects the indices behind the five largest e-values, $R_{\text{eBH}} = \{2, 4, 6, 8, 9\}$.

Note that $\hat{k}$ is the *last* crossing, not the first. The e-values at ranks two and three fail their own positional thresholds, but are still rejected because the cohort of size five clears the cohort-size-five threshold of 40. Self-consistency holds at $|R_{\text{eBH}}| = 5$ even though some rejected hypotheses would not have cleared the bar in isolation. The procedure also remains conservative when only one or two e-values are very large. A value of 95 would not suffice as the second of only two discoveries, but it is rejected here because the five-hypothesis cohort clears the rank-five threshold of 40.

Proposition 5.1 (FDR control of eBH (Wang and Ramdas 2022)). *Suppose that E_i is a valid e-value for H_i , for each $i \in \{1, \dots, m\}$. Then the eBH procedure controls FDR at level α :*

$$\text{FDR}_P(R_{\text{eBH}}) \leq \alpha$$

for all data-generating distributions P , under arbitrary dependence among the e -values.

Proof. The argument follows Wang and Ramdas (2022). Fix a data-generating distribution P and let N_P be its set of true nulls. The false discoveries are the rejected true nulls, and their number is a sum of indicators,

$$|R_{\text{eBH}} \cap N_P| = \sum_{i \in N_P} \mathbf{1}\{i \in R_{\text{eBH}}\}.$$

Dividing by $|R_{\text{eBH}}| \vee 1$ and taking the expectation gives the false discovery rate,

$$\text{FDR}_P(R_{\text{eBH}}) = \mathbb{E}_P \left[\sum_{i \in N_P} \frac{\mathbf{1}\{i \in R_{\text{eBH}}\}}{|R_{\text{eBH}}| \vee 1} \right].$$

Now apply the rearranged self-consistency bound to each summand. For every i ,

$$\frac{\mathbf{1}\{i \in R_{\text{eBH}}\}}{|R_{\text{eBH}}| \vee 1} \leq \frac{\alpha}{m} E_i,$$

so, summing over $i \in N_P$ and using linearity of expectation,

$$\text{FDR}_P(R_{\text{eBH}}) \leq \frac{\alpha}{m} \sum_{i \in N_P} \mathbb{E}_P[E_i].$$

Each $i \in N_P$ is a true null, so $\mathbb{E}_P[E_i] \leq 1$, and there are $|N_P| \leq m$ such terms. Therefore

$$\text{FDR}_P(R_{\text{eBH}}) \leq \frac{\alpha}{m} \cdot |N_P| \leq \alpha.$$

The only property of the e -values used is the per-hypothesis bound $\mathbb{E}_P[E_i] \leq 1$ on the nulls, so the guarantee holds for any joint distribution of $(E_1, \dots, E_m)$, that is, under arbitrary dependence. $\square$

In R, the procedure can be implemented directly from the definition. The function returns a logical vector in the original order of the hypotheses.

```
e_bh <- function(e, alpha = 0.05) {
  stopifnot(is.numeric(e), all(e >= 0), alpha > 0, alpha < 1)
  m <- length(e)
  ord <- order(-e, seq_along(e))
  e_ord <- e[ord]
  k <- seq_len(m)
  eligible <- which(e_ord >= m / (alpha * k))
  k_hat <- if (length(eligible) == 0L) 0L else max(eligible)
  reject <- rep(FALSE, m)
  if (k_hat > 0L) {
    reject[ord[seq_len(k_hat)]] <- TRUE
  }
  reject
}
```

Applied to the input from Example 5.1, the function reproduces $R_{\text{eBH}} = \{2, 4, 6, 8, 9\}$.

```
e_bh(c(0.4, 220, 1.1, 95, 0.8, 60, 0.6, 55, 45, 0.9), alpha = 0.05)
```

```
[1] FALSE TRUE FALSE TRUE FALSE TRUE FALSE TRUE TRUE FALSE
```

Comparison with the BH procedure. The relationship between BH and eBH is exact after a reciprocal transformation. Define

$$\tilde{P}_i = \frac{1}{E_i},$$

with the convention $1/0 = \infty$. By Proposition 2.1, $\tilde{P}_i$ is a p-variable for H_i . Because reciprocation reverses the ordering,

$$E_{[1]} \geq \cdots \geq E_{[m]} \iff \tilde{P}_{(1)} \leq \cdots \leq \tilde{P}_{(m)},$$

and the two rank-wise criteria are equivalent:

$$E_{[k]} \geq \frac{m}{\alpha k} \iff \tilde{P}_{(k)} \leq \frac{\alpha k}{m}.$$

Thus, with the same deterministic tie-breaking, eBH returns exactly the rejection set obtained by applying the classical BH step-up rule of Benjamini and Hochberg (1995) to the reciprocal p-variables $\tilde{P}_1, \dots, \tilde{P}_m$.

This algorithmic identity does not mean that the usual BH guarantee extends to arbitrary p-values under arbitrary dependence. Reciprocal p-variables obtained from e-values satisfy the stronger condition

$$\mathbb{E}_P \left[\frac{1}{\tilde{P}_i} \right] = \mathbb{E}_P[E_i] \leq 1 \quad \text{whenever } P \in H_i.$$

This is much stronger than p-variable validity alone. A uniform p-variable U is exactly valid, but $\mathbb{E}[1/U] = \infty$. The proof of Proposition 5.1 uses precisely the stronger expectation bound: self-consistency bounds each false-discovery term by $\alpha E_i/m$, after which the null expectations can be summed without any assumption on the joint dependence of the e-values.

For general p-values, the ordinary BH thresholds are valid under independence and under suitable positive-dependence conditions such as the positive regression dependence on a subset (PRDS) condition of Benjamini and Yekutieli (2001), a form of positive dependence under which a small p-value at one hypothesis makes small p-values at the others no less likely, but they are not guaranteed under arbitrary dependence. One standard dependence-robust modification is the harmonic correction used by the BY procedure, revisited in Section 5.6. The arbitrary-dependence guarantee of eBH therefore comes from the stronger expectation constraint on its reciprocal p-variables, not from a different step-up rule.

5.2 The e-Partitioning Principle

The eBH procedure controls the FDR of one selected rejection set. A more flexible target is to control the expected worst-case false discovery proportion over a *collection* $\mathcal{R}$ of possible rejection sets:

$$\mathbb{E}_P \left[\max_{R \in \mathcal{R}} \frac{|R \cap N_P|}{|R| \vee 1} \right] \leq \alpha.$$

We assume that $\mathcal{R}$ is non-empty. This entails no loss of generality because adjoining $\emptyset$ does not increase the maximum FDP.

This is the *simultaneous FDR* criterion (Xu et al. 2025). The maximum inside the expectation is the worst false discovery proportion among all the sets the analyst might end up reporting, so bounding its expectation controls the FDR of every set in $\mathcal{R}$ at once. This is what makes the criterion useful in practice. With ordinary FDR control, a pre-specified procedure returns one data-dependent rejection set. Under simultaneous FDR control, a pre-specified procedure returns a data-dependent collection of admissible rejection sets, and after inspecting the data the analyst may choose a member of the realised collection, with the guarantee still applying to the selected set. If $\mathcal{R}$ contains only one set, the criterion reduces to the usual FDR criterion from Definition 5.1.

To control this worst-case quantity, it helps to notice what the false discovery proportion really depends on. For a fixed rejection set R , the FDP $|R \cap N_P|/(|R| \vee 1)$ depends on the unknown distribution P only through the set of true nulls N_P . If we knew N_P , we would know the FDP of every R exactly. This suggests organising the analysis around the possible values of N_P rather than around the individual hypotheses.

For each subset $S \subseteq [m]$, define the partitioning hypothesis

$$H_S = \{P : N_P = S\}.$$

Thus, H_S is the hypothesis that the set of true nulls is exactly S . This is a different kind of hypothesis from the individual H_i . Note that Xu et al. (2025) reserve H_S for the intersection hypothesis $\bigcap_{i \in S} H_i$ and write $\bar{H}_S$ for the partitioning hypothesis used here. This chapter retains H_S for the partitioning hypothesis throughout. Whereas H_i asks only whether hypothesis i is null and says nothing about the others, H_S pins down the entire null configuration at once. These hypotheses partition the model, meaning that they are mutually exclusive and exhaustive. Every P belongs to exactly one H_S , namely the one with $S = N_P$, so exactly one of them is true even though we do not know which.

The case $S = \emptyset$ does not require a partitioning e-value in the suite. If $N_P = \emptyset$, no false discoveries are possible for any R , and the constraint defining $\mathcal{R}_\alpha(\mathbf{E})$ below is automatically satisfied. A suite of partitioning e-values is a collection

$$\mathbf{E} = (e_S)_{S \subseteq [m], S \neq \emptyset},$$

where bold $\mathbf{E}$ distinguishes a suite of e-values, indexed by subsets, from a single e-value E_i . There is one e-value e_S for each candidate configuration S , and e_S measures the evidence against the claim that the true-null set is exactly S . Each e_S is an e-value for H_S , meaning that $e_S \geq 0$ and

$$\mathbb{E}_P[e_S] \leq 1 \quad \text{for all } P \in H_S.$$

There are $2^m - 1$ non-empty subsets, so a suite contains exponentially many e-values. This is harmless for the theory but matters for computation, and Section 5.4 returns to it. Given such a suite, define the e-Partitioning procedure

$$\mathcal{R}_\alpha(\mathbf{E}) = \left\{ R \subseteq [m] : \frac{|R \cap S|}{|R| \vee 1} \leq \alpha e_S \text{ for all non-empty } S \subseteq [m] \right\}.$$

To read this definition, interpret $|R \cap S|/(|R| \vee 1)$ as the false discovery proportion that R *would* have if S were the true-null set, since $R \cap S$ would then be the false discoveries. A

candidate R is admissible only if the corresponding inequality holds for every non-empty $S \subseteq [m]$. For a fixed data-generating distribution P , the proof below uses the single relevant constraint $S = N_P$.

A small example makes the constraint concrete. Take $m = 2$, so the non-empty subsets are $\{1\}$, $\{2\}$, and $\{1, 2\}$, and consider the candidate $R = \{1\}$. The three inequalities that R must satisfy are

$$\frac{|\{1\} \cap \{1\}|}{1} = 1 \leq \alpha e_{\{1\}}, \quad \frac{|\{1\} \cap \{2\}|}{1} = 0 \leq \alpha e_{\{2\}}, \quad \frac{|\{1\} \cap \{1, 2\}|}{1} = 1 \leq \alpha e_{\{1, 2\}}.$$

The middle inequality holds automatically, since index 1 cannot be a false discovery when the true-null set is $\{2\}$. The set $\{1\}$ is therefore admissible exactly when $e_{\{1\}} \geq 1/\alpha$ and $e_{\{1, 2\}} \geq 1/\alpha$, that is, when there is sufficient evidence against both configurations in which index 1 is null.

Proposition 5.2 (The e-Partitioning Principle (adapted from Xu et al. 2025, Theorem 14)). *A random collection $\mathcal{R} \subseteq 2^{[m]}$ (a data-dependent family of candidate rejection sets) controls simultaneous FDR at level α if and only if there exists a suite $\mathbf{E} = (e_S)_S$ of e-values for the partitioning hypotheses H_S such that*

$$\mathcal{R} \subseteq \mathcal{R}_\alpha(\mathbf{E}).$$

This is the FDR case of the e-Closure Principle. The suite condition above is exactly the e-collection condition of Definition 5 there, written separately for each partitioning hypothesis; the partitioning-hypothesis formulation is discussed in Section 9 of the same paper.

Proof. Following Xu et al. (2025), we prove the “if” direction here, which is the one the procedures later in the chapter rely on. Fix a distribution P . If $N_P = \emptyset$ there are no true nulls, so the FDP is zero for every rejection set and there is nothing to prove. Otherwise let $S = N_P$. The point is that once P is fixed, N_P is a single fixed subset of $[m]$, not a random one, so we may use it to pick out one particular constraint from the definition of $\mathcal{R}_\alpha(\mathbf{E})$. This is where the “for every S ” requirement pays off. Membership in $\mathcal{R}_\alpha(\mathbf{E})$ guarantees the bound for all candidate configurations, and we apply it at the fixed configuration $S = N_P$. For every $R \in \mathcal{R}_\alpha(\mathbf{E})$,

$$\text{FDP}_P(R) = \frac{|R \cap N_P|}{|R| \vee 1} = \frac{|R \cap S|}{|R| \vee 1} \leq \alpha e_S.$$

The first equality is the definition of the FDP, the second substitutes $S = N_P$, and the inequality is the defining constraint of $\mathcal{R}_\alpha(\mathbf{E})$ evaluated at this S . The bound αe_S on the right does not depend on R , so it bounds the FDP of every set in the collection at once, and hence bounds the maximum over the collection. Taking expectations,

$$\mathbb{E}_P \left[\max_{R \in \mathcal{R}_\alpha(\mathbf{E})} \text{FDP}_P(R) \right] \leq \alpha \mathbb{E}_P[e_S] \leq \alpha.$$

The first inequality pulls out the constant α and uses the pointwise bound just established. The second uses that e_S is an e-value for H_S together with $P \in H_S$, which holds because $S = N_P$, so $\mathbb{E}_P[e_S] \leq 1$. This is the step where the validity of the partitioning e-value enters. Since the maximum over any subcollection $\mathcal{R} \subseteq \mathcal{R}_\alpha(\mathbf{E})$ is no larger than the maximum over the whole collection, the same bound holds for every such $\mathcal{R}$. $\square$

The converse direction, that every simultaneous-FDR-controlling collection can be embedded in some e-Partitioning collection, is the deeper part of the principle. It is what turns the result into a full characterisation rather than a one-way sufficient condition, since it says that nothing is lost by studying procedures of the form $\mathcal{R}_\alpha(\mathbf{E})$. Any procedure that achieves simultaneous FDR control is, in principle, contained in $\mathcal{R}_\alpha(\mathbf{E})$ for an appropriate $\mathbf{E}$. The proof requires constructing the suite from a given collection and is given in Xu et al. (2025). The construction is direct, but the resulting suite is best understood as proof-of-concept rather than as a method-design tool. The procedures developed in the rest of this chapter proceed in the other direction, choosing a suite first and then forming the collection.

Compound e-values provide a complementary reading. A vector of nonnegative statistics $(\tilde{E}_1, \dots, \tilde{E}_m)$ consists of compound e-values if

$$\sum_{i \in N_P} \mathbb{E}_P[\tilde{E}_i] \leq m$$

for every data-generating distribution P . Individual components need not be ordinary e-values; validity is imposed through this joint bound over the true nulls. Ignatiadis et al. (2024) show that every FDR-controlling procedure can be reconstructed as eBH applied to suitable compound e-values. The e-Partitioning formulation works in the opposite direction. It starts from a chosen suite of e-values for the partitioning hypotheses H_S and produces the collection $\mathcal{R}_\alpha(\mathbf{E})$. The two perspectives meet because the suite can exploit dependence and structural information that is invisible to a single list of marginal e-values $E_1, \dots, E_m$.

5.3 eBH+ from averaged partition e-values

The eBH+ procedure, called closed eBH in Xu et al. (2025), is obtained by applying the e-Partitioning Principle to a particularly simple suite of partition e-values. Let $E_1, \dots, E_m$ be the per-hypothesis e-values from Section 5.1. For every non-empty $S \subseteq [m]$, define

$$e_S = \frac{1}{|S|} \sum_{i \in S} E_i.$$

The arithmetic mean is the natural symmetric way to combine the e-values indexed by S when no assumptions are made about their joint distribution. Importantly, the average is taken only over S . Under the partitioning hypothesis H_S , these are precisely the indices for which the individual e-value validity guarantees are available. Other valid suites give rise to different e-Partitioning procedures, and BY+ and Su+ in Section 5.6 arise from suites tailored to p-value inputs.

For instance, with $m = 3$ and per-hypothesis e-values $(E_1, E_2, E_3) = (6, 2, 1)$, the partition e-value for $S = \{1, 2\}$ is $e_{\{1,2\}} = (6+2)/2 = 4$, while for the singleton $S = \{1\}$ it is $e_{\{1\}} = 6$ and for the full set $S = \{1, 2, 3\}$ it is $e_{\{1,2,3\}} = (6+2+1)/3 = 3$. In this example the smaller additional e-values dilute the evidence contributed by E_1 . More generally, enlarging S may either lower or raise the average, depending on the e-values that are added.

To verify validity, fix $P \in H_S$. Then $P \in H_i$ for every $i \in S$, so $\mathbb{E}_P[E_i] \leq 1$ for all $i \in S$. By linearity of expectation,

$$\mathbb{E}_P[e_S] = \frac{1}{|S|} \sum_{i \in S} \mathbb{E}_P[E_i] \leq 1.$$

Thus e_S is an e-value for H_S .

Applying the e-Partitioning Principle to the suite $\mathbf{E} = (e_S)_S$ gives the eBH+ collection

$$\mathcal{R}_\alpha^{\text{eBH}+} = \left\{ R \subseteq [m] : \frac{|R \cap S|}{|R| \vee 1} \leq \frac{\alpha}{|S|} \sum_{i \in S} E_i \text{ for all non-empty } S \subseteq [m] \right\}.$$

By Proposition 5.2, this collection controls simultaneous FDR at level α . The implementation in Section 5.4 selects a maximum-cardinality member as one representative of the collection.

Proposition 5.3 (eBH+ uniformly improves eBH (adapted from Xu et al. 2025, Theorem 17)). *Let R_{eBH} denote the rejection set returned by eBH at level α . Then*

$$R_{\text{eBH}} \in \mathcal{R}_\alpha^{\text{eBH}+}$$

for every vector of e-values. Moreover, if $m \geq 2$, there are e-value vectors for which eBH rejects nothing, while $\mathcal{R}_\alpha^{\text{eBH}+}$ contains a non-empty rejection set.

Proof. Following Xu et al. (2025), we prove the containment by showing that R_{eBH} satisfies the eBH+ constraint at every non-empty S , which is what membership in $\mathcal{R}_\alpha^{\text{eBH}+}$ demands. If $R_{\text{eBH}} = \emptyset$ the constraint reads $0 \leq \alpha e_S$, which holds since e-values are non-negative, so assume $R_{\text{eBH}} \neq \emptyset$. Then $|R_{\text{eBH}}| \vee 1 = |R_{\text{eBH}}|$, and for a fixed non-empty S the inequality to be verified is

$$\frac{|R_{\text{eBH}} \cap S|}{|R_{\text{eBH}}|} \leq \frac{\alpha}{|S|} \sum_{i \in S} E_i.$$

The starting point is the self-consistency property of eBH, which says that every rejected index carries at least the threshold amount of evidence,

$$E_i \geq \frac{m}{\alpha |R_{\text{eBH}}|} \quad \text{for all } i \in R_{\text{eBH}}.$$

This bound is available only for rejected indices, so before using it we shrink the sum on the larger side to those indices. Fix a non-empty $S \subseteq [m]$. Then

$$\frac{\alpha}{|S|} \sum_{i \in S} E_i \geq \frac{\alpha}{|S|} \sum_{i \in R_{\text{eBH}} \cap S} E_i \geq \frac{\alpha}{|S|} \cdot |R_{\text{eBH}} \cap S| \cdot \frac{m}{\alpha |R_{\text{eBH}}|} = \frac{m}{|S|} \cdot \frac{|R_{\text{eBH}} \cap S|}{|R_{\text{eBH}}|} \geq \frac{|R_{\text{eBH}} \cap S|}{|R_{\text{eBH}}|}.$$

Reading the chain from left to right, the first inequality drops the indices outside R_{eBH} from the sum, which can only lower it because every $E_i \geq 0$, and leaves a sum over $R_{\text{eBH}} \cap S$ in which every index is rejected. The second inequality applies self-consistency to each of these $|R_{\text{eBH}} \cap S|$ terms, replacing each E_i by the common lower bound $m/(\alpha |R_{\text{eBH}}|)$. The equality cancels the factor α . The final inequality uses $|S| \leq m$, so that $m/|S| \geq 1$. The two ends of the chain are exactly the eBH+ constraint at $R = R_{\text{eBH}}$ for this S . Since S was an arbitrary non-empty subset, $R_{\text{eBH}} \in \mathcal{R}_\alpha^{\text{eBH}+}$. $\square$

The examples below show how this containment can be strict.

Example 5.2 (A four-hypothesis strict-improvement example). Take $m = 4$ hypotheses and test at level $\alpha = 0.05$. Suppose the four e-values are

$$(E_1, E_2, E_3, E_4) = (50, 30, 1, 1).$$

The first two e-values carry substantial evidence against their null hypotheses, while the last two are neutral.

First consider eBH. The e-values are already in decreasing order, and the eBH thresholds are

$$\frac{m}{\alpha k} = \frac{4}{0.05 k} = \frac{80}{k}.$$

Thus the rank-wise comparisons are

$$50 < 80, \quad 30 < 40, \quad 1 < \frac{80}{3}, \quad 1 < 20.$$

No rank satisfies the eBH condition, so eBH rejects nothing.

Now consider the candidate rejection set

$$R = \{1, 2\}.$$

We show that R is admissible for eBH+. The eBH+ condition requires

$$\frac{|R \cap S|}{|R| \vee 1} = \frac{|R \cap S|}{2} \leq \frac{\alpha}{|S|} \sum_{i \in S} E_i$$

for every non-empty $S \subseteq [4]$. There are three cases.

If S contains neither 1 nor 2, then $|R \cap S| = 0$ and the inequality is immediate.

If S contains both 1 and 2, then the sum over S contains the terms E_1 and E_2 , and dropping any further indices only lowers it because the remaining e-values are non-negative, so

$$\sum_{i \in S} E_i \geq E_1 + E_2 = 80 \quad \text{and} \quad |S| \leq 4,$$

which gives

$$\frac{\alpha}{|S|} \sum_{i \in S} E_i \geq \frac{0.05}{4} \cdot 80 = 1 = \frac{|R \cap S|}{2}.$$

If S contains exactly one of 1 and 2, two subcases arise. If $1 \in S$ but $2 \notin S$, then $\sum_{i \in S} E_i \geq E_1 = 50$ and $|S| \leq 3$, so

$$\frac{\alpha}{|S|} \sum_{i \in S} E_i \geq \frac{0.05}{3} \cdot 50 > \frac{1}{2}.$$

If $2 \in S$ but $1 \notin S$, then $\sum_{i \in S} E_i \geq E_2 = 30$ and $|S| \leq 3$, so

$$\frac{\alpha}{|S|} \sum_{i \in S} E_i \geq \frac{0.05}{3} \cdot 30 = \frac{1}{2}.$$

The eBH+ constraint therefore holds for every non-empty $S \subseteq [4]$, and hence

$$\{1, 2\} \in \mathcal{R}_\alpha^{\text{eBH+}}.$$

On this input, eBH rejects nothing, while eBH+ allows rejection of the two hypotheses with the largest e-values. The case analysis uses lower bounds that meet the required thresholds; the actual vector has slight slack because $E_3 = E_4 = 1$. For example, the full-set constraint reads $1 \leq 1.025$. The next example sets the remaining e-values to zero and turns the same mechanism into a general strict-improvement witness.

The following general witness is taken from the proof of Theorem 17 in Xu et al. (2025).

Example 5.3 (A general witness for strict improvement). The construction is chosen so that E_1 falls just below the eBH-at-rank-one threshold m/α , with the slack $1/(2\alpha)$ deposited in E_2 . Two mechanisms then combine to make the singleton $\{1\}$ admissible for eBH+ even though eBH rejects nothing. For sets S containing both indices, the pair sums to exactly $E_1 + E_2 = m/\alpha$, which clears the average threshold. For sets S containing 1 but not 2, excluding index 2 caps the size at $|S| \leq m - 1$, and E_1 alone is large enough to clear the threshold over this smaller denominator. The split of m/α into $m - 1/2$ and $1/2$ parts is chosen precisely so that both mechanisms work at once.

Let $m \geq 2$ be an integer and consider

$$E_1 = \frac{m - 1/2}{\alpha}, \quad E_2 = \frac{1}{2\alpha}, \quad \text{and set } E_i = 0 \text{ for all } i \geq 3$$

(the last condition is vacuous when $m = 2$). The descending order is $E_{[1]} = E_1$, $E_{[2]} = E_2$, and $E_{[k]} = 0$ for $k \geq 3$. The eBH condition is

$$E_{[k]} \geq \frac{m}{\alpha k}.$$

For $k = 1$, $E_{[1]} = (m - 1/2)/\alpha < m/\alpha$. For $k = 2$, $E_{[2]} = 1/(2\alpha) < m/(2\alpha)$ since $m \geq 2$. For $k \geq 3$, $E_{[k]} = 0$, so the eBH condition also fails. Thus eBH rejects nothing.

Now consider $R = \{1\}$. Because R is a singleton, the left-hand side of the eBH+ constraint simplifies to an indicator,

$$\frac{|\{1\} \cap S|}{|\{1\}| \vee 1} = \mathbf{1}\{1 \in S\}.$$

For S with $1 \notin S$ the left-hand side is zero and the constraint holds automatically, since e-values are non-negative. It therefore remains to show that $\alpha e_S \geq 1$ for every S containing 1, and there are two cases.

If $2 \in S$, then dropping the zero terms E_i with $i \geq 3$ leaves

$$\sum_{i \in S} E_i = E_1 + E_2 = \frac{m - 1/2}{\alpha} + \frac{1}{2\alpha} = \frac{m}{\alpha},$$

so

$$\alpha e_S = \frac{m}{|S|} \geq 1$$

since $|S| \leq m$.

If $2 \notin S$, then E_2 is excluded and all E_i with $i \geq 3$ are zero, so the sum reduces to

$$\sum_{i \in S} E_i = E_1 = \frac{m - 1/2}{\alpha}.$$

Moreover, S omits index 2, so it contains at most $m - 1$ indices, giving

$$\alpha e_S = \frac{m - 1/2}{|S|} \geq \frac{m - 1/2}{m - 1} > 1.$$

In both cases $\alpha e_S \geq 1 = \mathbf{1}\{1 \in S\}$, so the eBH+ constraint holds for every non-empty $S \subseteq [m]$, and therefore

$$\{1\} \in \mathcal{R}_\alpha^{\text{eBH+}}.$$

The example shows that eBH can reject nothing when the largest e-value falls just below the rank-one threshold m/α . The eBH+ collection can still contain a singleton rejection such as $\{1\}$, because the admissibility check also uses averages over partitioning sets S containing other hypotheses. Thus, e-values outside the proposed rejection set can help certify that this set satisfies the simultaneous FDR constraint.

5.4 Implementation and a small simulation

The function `e_bh()` from Section 5.1 returns one rejection set. This section adds two small helper functions for eBH+. The first checks whether a proposed set R belongs to the eBH+ collection $\mathcal{R}_\alpha^{\text{eBH+}}$. The second searches for a maximum-cardinality member of the collection, breaking ties by the total e-value in the candidate set. The search is deliberately exhaustive and is intended only for small teaching examples rather than for scalable computation.

Checking one proposed set requires enumerating all non-empty subsets $S \subseteq [m]$, so `ebh_plus_in_collection` accepts only small m . Searching for a maximum-cardinality admissible set additionally enumerates all $2^m - 1$ candidate rejection sets R , and checking each one costs $O(m2^m)$ through a matrix-vector product with m columns. The total is therefore $O(m4^m)$, which is why `ebh_plus_largest()` uses a default limit of `max_m = 12`, where 4^{12} is already about 1.7×10^7 .

Unlike `e_bh()`, the exhaustive eBH+ helpers restrict their inputs to finite e-values. This is a numerical requirement of the dense subset-matrix implementation, since the product `members %*% e` used for the right-hand side would create undefined $0 \times \infty$ terms for subsets that exclude an infinite coordinate. The limit $m \leq 20$ in `ebh_plus_in_collection` is a hard guard rather than a recommended working size, since the membership matrix has $m(2^m - 1)$ entries and becomes memory-intensive near the cap.

```
# Small tolerance for floating-point comparisons in matrix multiplications.
tol <- 100 * .Machine$double.eps

ebh_plus_in_collection <- function(R, e, alpha) {
  stopifnot(
    is.numeric(e),
    length(e) >= 1L,
    all(e >= 0),
    all(is.finite(e)),
    alpha > 0,
    alpha < 1,
    length(unique(R)) == length(R),
    all(R %in% seq_along(e))
  )

  m <- length(e)
  if (m > 20L) {
    stop("Exhaustive subset enumeration is intended only for small m.")
  }

  R <- as.integer(R)

  # Enumerate all non-empty subsets of [m] via binary masks: integer s in
```

```

# {1,...,2^m - 1} represents the subset {i : bit i of s is set}.
masks <- seq_len(2^m - 1L)
bits <- 2 ^ (seq_len(m) - 1L)
members <- outer(
  masks,
  bits,
  FUN = function(mask, bit) (mask %/% bit) %% 2L == 1L
)

subset_size <- rowSums(members)
in_R <- seq_len(m) %in% R

lhs <- as.numeric(members %>% as.integer(in_R)) /
  max(length(R), 1L)

rhs <- alpha * as.numeric(members %>% e) / subset_size

all(lhs <= rhs + tol)
}

ebh_plus_largest <- function(e, alpha, max_m = 12) {
  stopifnot(
    is.numeric(e),
    length(e) >= 1L,
    all(e >= 0),
    all(is.finite(e)),
    alpha > 0,
    alpha < 1,
    max_m >= 1,
    length(e) <= max_m
  )

  m <- length(e)

  # Enumerate all non-empty subsets of [m] via binary masks, as above.
  masks <- seq_len(2^m - 1L)
  bits <- 2 ^ (seq_len(m) - 1L)
  members <- outer(
    masks,
    bits,
    FUN = function(mask, bit) (mask %/% bit) %% 2L == 1L
  )

  subset_size <- rowSums(members)
  subset_e_sum <- as.numeric(members %>% e)
  rhs <- alpha * subset_e_sum / subset_size

  # Iterate candidates by decreasing cardinality, breaking ties by total
# e-value in the set, and return the first admissible candidate.
  candidate_order <- order(
    subset_size,
    subset_e_sum,
    decreasing = TRUE
  )
}

```

```

for (idx in candidate_order) {
  in_R <- members[idx, ]
  r_size <- subset_size[idx]

  lhs <- as.numeric(members %*% as.integer(in_R)) / r_size

  if (all(lhs <= rhs + tol)) {
    return(in_R)
  }
}

rep(FALSE, m)
}

```

The membership function returns a single logical value, while the search function returns a logical vector aligned with the input, matching the convention of `e_bh()` from Section 5.1. Applied to the input from Example 5.2, the implementations reproduce the worked calculation. The set $\{1, 2\}$ is admissible, and the maximum-cardinality search also returns $\{1, 2\}$. No larger set can be admissible here, since any candidate of size three or four contains index 3 or 4 and fails the constraint at the corresponding singleton S , where the left-hand side is at least $1/4$ but $\alpha E_3 = \alpha E_4 = 0.05$.

```

e_num <- c(50, 30, 1, 1)
ebh_plus_in_collection(R = c(1, 2), e = e_num, alpha = 0.05)

```

```
[1] TRUE
```

```
ebh_plus_largest(e = e_num, alpha = 0.05)
```

```
[1] TRUE TRUE FALSE FALSE
```

The function `ebh_plus_largest()` returns one admissible set, not the whole collection; if no non-empty candidate is admissible it returns the empty set, which always belongs to the collection. This is enough for the simulation below. Since eBH+ controls FDR simultaneously over all sets in the collection, any data-dependent rule that selects a member of the collection is covered by the simultaneous guarantee.

The simulation below uses `MASS::mvrnorm()` through an explicit namespace; the setup chunk checks that the `MASS` package is available.

We now adapt the positively dependent simulation setting from Goeman et al. (2025). There are $m = 8$ hypotheses, a null proportion $\pi_0 = 0.25$, and hence six non-null hypotheses. We take

$$\mu = (A, A, A, A, A, A, 0, 0),$$

so that $N_P = \{7, 8\}$. The covariance matrix is Toeplitz,

$$\Sigma_{ij} = 0.8^{|i-j|}.$$

For each signal strength

$$A \in \{0.125, 0.25, 0.375, 0.5\},$$

we generate $n = 100$ independent observations $Z^{(1)}, \dots, Z^{(n)} \sim N(\mu, \Sigma)$ and compute the product e-value

$$E_j = \prod_{\ell=1}^n \exp(AZ_j^{(\ell)} - A^2/2) = \exp\left(A \sum_{\ell=1}^n Z_j^{(\ell)} - \frac{nA^2}{2}\right).$$

The text of Goeman et al. (2025) states the single-observation factor $\exp(a_j Z_j - a_j^2/2)$; the accompanying simulation code aggregates it over the n repeated observations as a product of independent per-observation factors, and the same aggregation is used here. The oracle tuning $a = A$ is used for all hypotheses, so that eBH and eBH+ are compared on the same e-values and the comparison isolates the multiple-testing step rather than the e-value design step. The negatively dependent case is mentioned in Goeman et al. (2025) but is not reproduced here.

```
set.seed(20260505)

alpha <- 0.05
m <- 8
pi0 <- 0.25
m0 <- as.integer(m * pi0)
n_per <- 100
n_rep <- 1000
A_values <- c(0.125, 0.25, 0.375, 0.5)

true_nulls <- (m - m0 + 1L):m
false_nulls <- setdiff(seq_len(m), true_nulls)

Sigma <- 0.8 ^ abs(outer(seq_len(m), seq_len(m), "-"))

as_rejection_indices <- function(x) {
  if (is.logical(x)) {
    which(x)
  } else {
    as.integer(x)
  }
}

fdp <- function(R) {
  R <- as_rejection_indices(R)
  length(intersect(R, true_nulls)) / max(length(R), 1L)
}

power_value <- function(R) {
  R <- as_rejection_indices(R)
  length(intersect(R, false_nulls)) / length(false_nulls)
}

simulate_setting <- function(A) {
  mu <- rep(A, m)
  mu[true_nulls] <- 0

  out <- replicate(n_rep, {
    z_samples <- MASS::mvrnorm(n = n_per, mu = mu, Sigma = Sigma)
    z_sum <- colSums(z_samples)
  })
}
```

```

e_values <- exp(A * z_sum - n_per * A^2 / 2)

R_ebh <- e_bh(e_values, alpha = alpha)
R_plus <- ebh_plus_largest(e_values, alpha = alpha)

c(
  fdr_ebh = fdp(R_ebh),
  power_ebh = power_value(R_ebh),
  fdr_plus = fdp(R_plus),
  power_plus = power_value(R_plus)
)
})

data.frame(
  A = A,
  method = c("eBH", "eBH+"),
  fdr = c(mean(out["fdr_ebh", ]), mean(out["fdr_plus", ])),
  power = c(mean(out["power_ebh", ]), mean(out["power_plus", ]))
)
}

simulation_results <- do.call(
  rbind,
  lapply(A_values, simulate_setting)
)

simulation_results$method <- factor(
  simulation_results$method,
  levels = c("eBH", "eBH+")
)

max_fdr <- max(simulation_results$fdr)

fdr_compare <- merge(
  subset(simulation_results, method == "eBH", select = c(A, fdr, power)),
  subset(simulation_results, method == "eBH+", select = c(A, fdr, power)),
  by = "A",
  suffixes = c("_ebh", "_plus")
)

n_settings <- nrow(fdr_compare)
fdr_n_higher <- sum(fdr_compare$fdr_plus > fdr_compare$fdr_ebh)
power_n_higher <- sum(fdr_compare$power_plus > fdr_compare$power_ebh)

fdr_comparison_text <- if (fdr_n_higher == n_settings) {
  "at every signal strength"
} else {
  sprintf("at %d of the %d signal strengths", fdr_n_higher, n_settings)
}

power_comparison_text <- if (power_n_higher == n_settings) {
  "at every signal strength"
} else {
  sprintf("at %d of the %d signal strengths", power_n_higher, n_settings)
}

```

```

max_fdr_row <- simulation_results[which.max(simulation_results$fdr), ]
max_fdr_A <- max_fdr_row$A
max_fdr_method <- as.character(max_fdr_row$method)

```

The empirical FDR is the Monte Carlo average of

$$\frac{|R \cap N_P|}{|R| \vee 1},$$

and the empirical power is the Monte Carlo average of

$$\frac{|R \cap N_P^c|}{|N_P^c|}.$$

The power reported here is therefore the true-positive proportion, the fraction of non-null hypotheses that are rejected. In the simulation code accompanying Goeman et al. (2025), the plotted power instead divides the average total number of rejections, true and false, by the number of non-null hypotheses. The two definitions nearly coincide in this setting because false discoveries are rare, but the comparison with the published figure is qualitative rather than exact.

Table 5.2: Empirical FDR and power in the eBH+ simulation.

A	method	fdr	power
0.125	eBH	0.000	0.003
0.125	eBH+	0.001	0.009
0.250	eBH	0.001	0.361
0.250	eBH+	0.008	0.478
0.375	eBH	0.001	0.808
0.375	eBH+	0.005	0.858
0.500	eBH	0.000	0.960
0.500	eBH+	0.001	0.972

Table 5.2 and Figure 5.1 show that the maximum-cardinality eBH+ selection has higher empirical FDR than eBH at every signal strength, while also having higher empirical power at every signal strength. The maximum-cardinality selection rejects at least as many hypotheses as eBH in every realisation, because R_{eBH} belongs to the eBH+ collection by Proposition 5.3 and the search returns a member of maximal size. All empirical FDR estimates for both methods remain below the nominal level $\alpha = 0.05$; the largest is 0.008, attained by eBH+ at signal strength $A = 0.25$.

In the terminology of Xu et al. (2025) (Definition 16), the improvement in Proposition 5.3 is both a simultaneity improvement and a power improvement, and both are deterministic statements about the observed e-value vector: the eBH rejection set itself remains admissible, and the eBH+ collection can admit a strictly larger rejection set. This rejection-set comparison is distinct from the empirical true-positive-proportion comparison in the simulation. Both the eBH rejection set and the selected member of the eBH+ collection retain level- α FDR control.

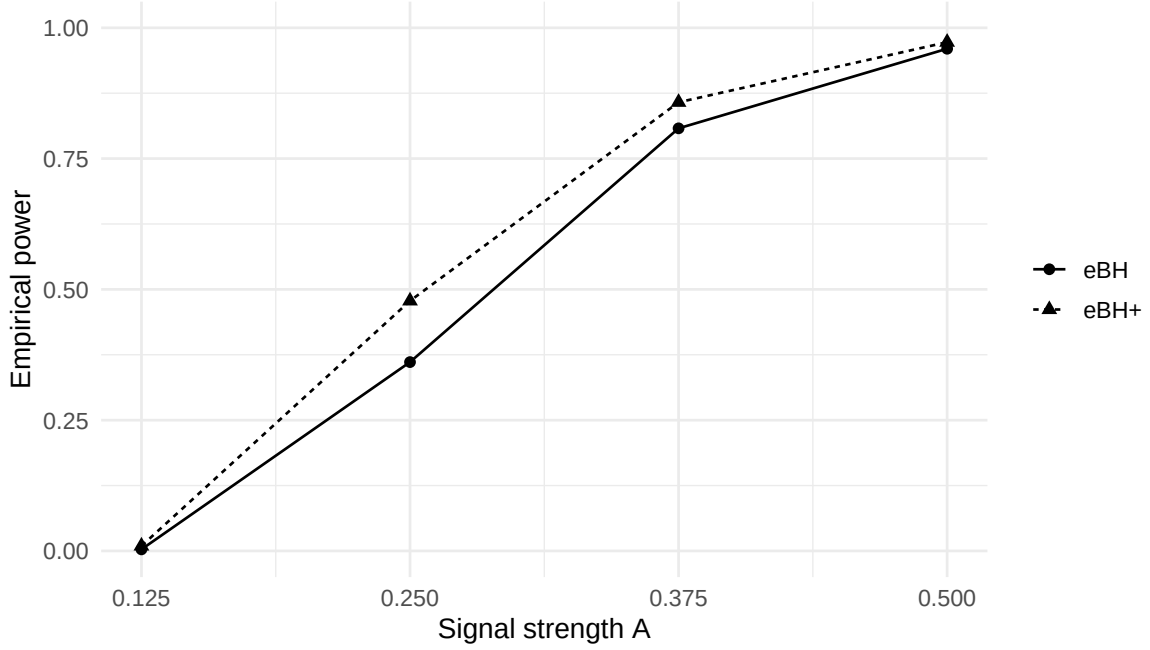


Figure 5.1: Empirical power of eBH and the maximum-cardinality eBH+ selection as a function of signal strength.

The exhaustive functions above are useful because they mirror the definition of $\mathcal{R}_\alpha^{\text{eBH+}}$ directly. They should not be used as a replacement for specialised software. For larger or non-didactic analyses, the CRAN package `eClosure` (Goeman 2026) implements the closed-eBH and closed-BY procedures described by Xu et al. (2025). For large m , polynomial shortcuts are available for eBH+. Section 10 of Xu et al. (2025) gives an $O(m^2 \log m)$ algorithm for finding the largest rejected set of the form $[r] = \{1, \dots, r\}$, the top r hypotheses after ordering by decreasing e-value, and an $O(m \log m)$ algorithm for checking a fixed set.

5.5 Post-hoc flexibility

The eBH procedure in Section 5.1 returns the single rejection set

$$R_{\text{eBH}} \subseteq [m].$$

By contrast, the e-Partitioning procedure in Section 5.2, and in particular eBH+ from Section 5.3, returns a whole collection of admissible rejection sets,

$$\mathcal{R}_\alpha(\mathbf{E}) \quad \text{or} \quad \mathcal{R}_\alpha^{\text{eBH+}}.$$

This distinction is practically important. The output is no longer just one set that must be reported exactly as produced by the algorithm. The researcher may choose any element of the collection after inspecting the data, while retaining the same simultaneous FDR guarantee.

This follows directly from Proposition 5.2. For a collection $\mathcal{R}$ with simultaneous FDR control, the guarantee is

$$\mathbb{E}_P \left[\max_{R \in \mathcal{R}} \text{FDP}_P(R) \right] \leq \alpha.$$

The bound therefore covers the worst false discovery proportion over every set in the collection. If an analyst later selects a particular $R \in \mathcal{R}$ after looking at the realised e-values, its FDP is bounded by the maximum over the collection. Post-hoc selection is not an extra operation outside the guarantee. It is one of the choices already covered.

This gives a different kind of flexibility from the optional stopping discussed in Chapter 4. There, the time at which the evidence is read may be data-dependent. Here, the realised collection and the final choice from that collection may both depend on the data.

One immediate use is interpretability. Suppose that $\mathcal{R}_\alpha^{\text{eBH+}}$ contains both a maximum-cardinality admissible set and a smaller set that is easier to explain scientifically. The maximum-cardinality set may be attractive if the only goal is to maximise the number of discoveries. The smaller set may be preferable if it corresponds to a coherent scientific unit. In Section 8.1, Xu et al. (2025) discuss settings where hypotheses correspond to voxels in neuroimaging or molecular markers in bioinformatics. In such applications, researchers often interpret discoveries as clusters or pathways rather than as isolated indices. Simultaneous FDR control permits such a set to be reported post-hoc, provided that it belongs to the admissible collection.

The crucial qualification is that the post-hoc flexibility extends only to elements of the collection. Sets outside $\mathcal{R}_\alpha(\mathbf{E})$ lack the simultaneous guarantee, and reporting them as FDR-controlled would be unjustified. Post-hoc *selection from the realised admissible collection* is covered by the guarantee. Post-hoc *modification outside that collection* is not. The collection defines that boundary precisely.

One consequence deserves emphasis, because it is easy to get backwards. The collection is not closed under taking subsets, so reporting fewer discoveries than an admissible set is not automatically safe. Shrinking R shrinks the denominator $|R| \vee 1$, while $|R \cap S|$ need not shrink in step, so a subset of an admissible set can fail a constraint that the larger set passed. For example, at $m = 3$ and $\alpha = 0.05$ with $E = (25, 25, 10)$, eBH rejects nothing, both $\{1, 2\}$ and $\{1, 2, 3\}$ are admissible, and none of the three singletons is. An analyst holding the admissible set $\{1, 2\}$ who reports only hypothesis 1, on the grounds that fewer claims must be more conservative, has left the collection and lost the guarantee. This is exactly the situation Xu et al. (2025) describe when they note that naively discarding low-effect-size findings from an FDR-significant set, as in volcano-plot filtering, may compromise FDR control.

A second consequence concerns the relation between FDR and FWER. For a singleton set $\{i\}$, the false discovery proportion is either zero or one:

$$\text{FDP}_P(\{i\}) = \mathbf{1}\{i \in N_P\}.$$

Thus, for a single reported singleton, its FDR is just the probability that this reported discovery is false. Xu et al. (2025) (Theorem 39 in Section 8.2; proof in Appendix H.9) give the corresponding family-wise statement. If a simultaneous FDR-controlled collection $\mathcal{R}$ contains singleton sets, define

$$R_{\text{sing}} = \{i \in [m] : \{i\} \in \mathcal{R}\}.$$

Then R_{sing} controls FWER at the same level, meaning that the probability that R_{sing} contains any true null is at most α . Importantly, R_{sing} itself need not be an element of $\mathcal{R}$. The proof is short enough to give in full.

Proof. Following the proof of Theorem 39 in Appendix H.9 of Xu et al. (2025), let $\mathcal{S}_{\text{sing}} = \{\{i\} : \{i\} \in \mathcal{R}\}$ be the singleton sets contained in the collection, and adopt the convention that a maximum over an empty set is zero. Then

$$\begin{aligned} \mathbb{P}_P(R_{\text{sing}} \cap N_P \neq \emptyset) &= \mathbb{E}_P \left[\max_{i \in R_{\text{sing}}} \mathbf{1}\{i \in N_P\} \right] \\ &= \mathbb{E}_P \left[\max_{R \in \mathcal{S}_{\text{sing}}} \text{FDP}_P(R) \right] \\ &\leq \mathbb{E}_P \left[\max_{R \in \mathcal{R}} \text{FDP}_P(R) \right] \\ &\leq \alpha. \end{aligned}$$

The first equality rewrites the event of at least one false rejection as a maximum of indicators over the rejected singletons. The second uses the singleton identity above, since $\text{FDP}_P(\{i\}) = \mathbf{1}\{i \in N_P\}$ and the indices in R_{sing} correspond exactly to the sets in $\mathcal{S}_{\text{sing}}$. The first inequality enlarges the maximum from $\mathcal{S}_{\text{sing}}$ to the full collection, using $\mathcal{S}_{\text{sing}} \subseteq \mathcal{R}$, and the final inequality is the simultaneous FDR guarantee. The argument only ever bounds maxima over sets that belong to $\mathcal{R}$, which is why R_{sing} itself never needs to be a member. $\square$

Since $\text{FDP}_P(R_{\text{sing}}) \leq \mathbf{1}\{R_{\text{sing}} \cap N_P \neq \emptyset\}$, family-wise control of R_{sing} also implies ordinary FDR control for this singleton-based rejection set, so the switch adds the stricter guarantee rather than trading one for the other. The switch from an FDR statement to an FWER statement can therefore be made after seeing the data, as long as it is based on singleton sets contained in the simultaneous collection.

In practice, a strong signal may therefore support both a broad FDR-controlled discovery set and a smaller list of individually protected discoveries. The broad set answers a discovery-oriented question about which collection of hypotheses can be reported while controlling the expected false discovery proportion. The singleton-based set answers a stricter question about which individual hypotheses can be reported while controlling the probability of any false individual discovery.

The practical gain from post-hoc flexibility is therefore distinct from any power advantage. When the realised collection contains several scientifically meaningful admissible sets, the analyst may choose among them after inspecting the data (for scientific interpretability, for a stricter error criterion, or for another transparent reporting goal) without leaving the simultaneous guarantee supplied by the e-Partitioning Principle.

5.6 Beyond eBH+: BY+ and Su+

The eBH+ procedure is one concrete instance of the e-Partitioning Principle. Once an existing FDR procedure can be represented through suitable partitioning e-values, the corresponding e-Partitioning collection often gives a uniform improvement. Table 5.3 summarises the three examples discussed in this chapter.

Table 5.3: Plus variants generated by the e-Partitioning Principle. The dependence assumption applies to the classical method and its plus variant alike.

Input	Classical method	Plus variant	Dependence assumption
e-values	eBH	eBH+	Arbitrary dependence
p-values	BY	BY+	Arbitrary dependence
p-values	Su	Su+	PRDN

The BY procedure of Benjamini and Yekutieli (2001) is BH run at the reduced level α/h_m , where

$$h_m = \sum_{k=1}^m \frac{1}{k}$$

is the m -th harmonic number. Equivalently, it rejects up to the largest rank r satisfying

$$p_{(r)} \leq \frac{r\alpha}{mh_m}.$$

This correction makes BY valid under arbitrary dependence of the p-values. Since h_m grows roughly like $\log m$, arbitrary dependence for p-values is paid for by a logarithmic penalty in the rejection threshold. Xu et al. (2025) construct the corresponding e-Partitioning collection from the p-values underlying BY. The resulting BY+ procedure returns a simultaneous FDR-controlled collection and uniformly improves BY for $m \geq 2$, in the same sense that eBH+ uniformly improves eBH (Theorem 28 of Xu et al. (2025)).

Xu et al. (2025) apply the same idea to the procedure of Su (2018). This method is valid under positive regression dependence on the null p-values (PRDN), a positive dependence condition imposed only on the p-values corresponding to true null hypotheses. This is weaker than the PRDS condition commonly used for BH, which also involves the p-values of false hypotheses. The Su procedure itself can be viewed as BH run at the reduced level α/ℓ_α , where $\ell_\alpha = -W_{-1}(-\alpha/e)$ and W_{-1} is the lower branch of the Lambert W function. For example, $\ell_{0.05} \approx 5.74$, so at nominal level 0.05 the Su procedure runs BH at roughly 0.0087, about 17% of the nominal level. The plus variant, Su+, retains the PRDN assumption and replaces the single classical rejection set by a simultaneous e-Partitioning collection, and Theorem 34 of Xu et al. (2025) gives the analogous uniform-improvement result for $\alpha \in (0, 1)$ and $m \geq 2$. Its role in this chapter is mainly conceptual. It shows that the e-Partitioning Principle is not tied to arbitrary-dependence procedures such as eBH or BY, but can also sharpen procedures that exploit positive dependence.

Taken together, eBH+, BY+, and Su+ illustrate a systematic pattern. The e-Partitioning Principle turns single-set FDR procedures into members of a larger, simultaneously valid collection, and this often creates room for post-hoc flexibility or larger rejection sets at the same nominal level. A further development not pursued here is the use of restricted combinations, where logical relationships between hypotheses can be incorporated through the partitioning hypotheses themselves.

5.7 Summary and practical guidance

This chapter has shown how the e-values constructed in Chapter 3 and Chapter 4 can be used as building blocks for multiple testing. The eBH procedure gives a direct FDR

method for one e-value per hypothesis and remains valid under arbitrary dependence. The e-Partitioning Principle then gives a broader view. Simultaneous FDR control is fully characterised by partitioning e-values, and this representation leads to constructive improvements such as eBH+, BY+, and Su+.

- If one valid e-value is available for each hypothesis and the goal is ordinary FDR control, eBH is the simplest default choice. Its main advantage is robustness. The guarantee does not require independence or any positive-dependence assumption among the e-values.
- If simultaneous FDR control, post-hoc selection, or potential power improvement is important, eBH+ is the natural next step, subject to the computational caveat below. It replaces the single eBH rejection set by the collection $\mathcal{R}_\alpha^{\text{eBH+}}$, allowing the rejection set to be chosen from the admissible collection after inspecting the results. Post-hoc selection means choosing a member of the collection, not shrinking one: the collection is not closed under taking subsets, so a subset of an admissible set need not itself be admissible.
- For larger or non-didactic analyses, exhaustive subset enumeration should be avoided. The CRAN package `eClosure` (Goeman 2026) implements closed-eBH and closed-BY procedures from Xu et al. (2025), while Section 10 of that paper gives polynomial algorithms for the prefix-set search and the membership check.
- If the input consists of p-values rather than e-values, BY+ and Su+ are the corresponding plus variants. BY+ applies under arbitrary dependence, whereas Su+ uses the PRDN condition carried over from the original Su procedure.
- E-values are not automatically more powerful than p-values in fixed-sample comparisons. Their practical advantage is strongest when robustness to dependence, optional stopping or continuation, or post-hoc selection is part of the analysis.

The conceptual link to Chapter 4 is that simultaneous FDR control plays a role analogous to anytime-validity in the multiple-testing setting. In the sequential setting of Chapter 4, validity was protected over an entire time path. Here it is protected over an entire collection of rejection sets. In both cases, flexible data-dependent decisions remain valid because what is controlled is the larger object, not just the final reported choice.

Chapter 6

Case study

6.1 Case study overview and design

This chapter brings together the methods developed earlier in two applied settings. The main example is a randomised clinical trial of antiretroviral treatments. It allows the evidence to be assessed at the end of follow-up, followed as events accumulate, and evaluated across several treatment and endpoint comparisons. A second example uses a collection of tuberculosis-vaccine studies to illustrate how evidence can be accumulated across separate studies. Together, the two examples show how the design choices and validity conditions from the earlier chapters enter an applied analysis.

The main dataset comes from study 175 of the AIDS Clinical Trials Group (ACTG 175), which randomised participants aged 12 or older with human immunodeficiency virus (HIV) and baseline CD4 counts between 200 and 500 cells per cubic millimetre to four antiretroviral regimens. The four regimens were zidovudine alone, didanosine alone, zidovudine with didanosine, or zidovudine with zalcitabine (Hammer et al. 1996). The analyses use the version of the trial data distributed in `speff2trial::ACTG175` (Juraska et al. 2022).

CD4 and CD8 T cells are involved in the immune response. CD4 T cells coordinate that response and are a primary target of HIV, while CD8 T cells help identify and destroy infected cells. The dataset records both counts at baseline and in a 20 ± 5 week assessment window.

The main ACTG analysis compares **ZDV only** with the three active arms pooled as **Other therapy**. The endpoint is the time to a decline in CD4 count of at least 50%, progression to AIDS, or death. Safe and classical logrank tests are applied at the end of follow-up, after which the event sequence is replayed to compare safe monitoring with repeated unadjusted logrank tests. CD4 and CD8 changes at 20 weeks are analysed with safe and classical t-tests, and event-free status at 96 weeks with a classical two-proportions comparison. The three active arms are also compared separately with zidovudine alone across the survival, CD4, and CD8 endpoints using eBH.

Because the package dataset has no reliable enrolment-order variable, the sequential analyses replay event time retrospectively rather than reconstructing the trial’s historical monitoring.

For the separate-study analysis, the BCG vaccine data in `metadat::dat.bcg` (Viechtbauer et al. 2026; Colditz et al. 1994) provide the two-by-two tables used to illustrate the product

construction in Proposition 4.2.

The observed datasets show how the methods are applied. They do not establish error control or power. Type-I error guarantees follow from the theoretical results, while the simulations estimate operating characteristics for the designs considered here. E-values are compared with $1/\alpha$ and p-values with α .

The following testing levels and design alternatives are used throughout the chapter.

```
alpha_single <- 0.05  # level for single safe tests
beta_design  <- 0.20  # design type-II error
alpha_fdr    <- 0.10  # FDR level for the multiple-testing family

hr_min       <- 0.75  # minimal relevant hazard ratio
delta_min_t  <- 0.30  # standardised mean difference for CD4/CD8 change
omega_min    <- 0.50  # working conditional odds ratio for the BCG analysis
```

6.2 ACTG 175 data, treatment arms, and endpoints

The package records treatment in two variables. `treat` compares zidovudine alone with the other three groups combined. `arms` retains the four original groups. Here, 0 is zidovudine alone, 1 zidovudine plus didanosine, 2 zidovudine plus zalcitabine, and 3 didanosine alone. The binary coding is used for the pooled comparisons, while the four-arm coding is used for the eBH analysis.

```
library(speff2trial)
library(survival)
library(dplyr)

data(ACTG175, package = "speff2trial")

actg <- ACTG175 |>
  mutate(
    treatment_binary = factor(
      treat,
      levels = c(0, 1),
      labels = c("ZDV only", "Other therapy")
    ),
    arm = factor(
      arms,
      levels = c(0, 1, 2, 3),
      labels = c("ZDV", "ZDV + ddI", "ZDV + ddC", "ddI")
    ),
    cd4_change_20 = cd420 - cd40,
    cd8_change_20 = cd820 - cd80
  )

actg_n <- nrow(actg)
original_trial_n <- 2467L

actg_n_fmt <- format(actg_n, big.mark = ",")
```

```
original_trial_n_fmt <- format(original_trial_n, big.mark = ",")
```

The package dataset contains 2,139 participants, compared with 2,467 in the original publication (Hammer et al. 1996). The package documentation does not explain which participants were omitted, so the results below apply to the available package sample. A causal interpretation also requires that these omissions did not introduce post-randomisation bias.

Table 6.1: Relationship between the binary and four-arm treatment variables.

	ZDV	ZDV + ddI	ZDV + ddC	ddI
ZDV only	532	0	0	0
Other therapy	0	522	524	561

The table confirms that `treat` places zidovudine alone in the reference group and combines the other three arms as `Other therapy`.

The survival endpoint is represented by `Surv(days, cens)`. The `speff2trial` documentation defines `days` as the time to a decline in CD4 count of at least 50%, progression to AIDS, or death (Juraska et al. 2022). This corresponds to the percentage-based composite endpoint reported for ACTG 175 (Hammer et al. 1996). The analyses therefore apply the endpoint encoded in the package to the reduced package sample, rather than reproducing the full published analysis.

The continuous endpoints are the changes in CD4 and CD8 count between baseline and the week-20 assessment window, documented in the package as 20 ± 5 weeks.

For the binary comparison, the composite endpoint is evaluated at 96 weeks. Patients who experience an event by that time have a known event outcome. Patients followed for at least 96 weeks without an earlier event have a known event-free outcome. Patients censored before 96 weeks have unknown status and are excluded.

```
t0 <- 96 * 7

actg <- actg |>
  mutate(
    event_by_t0 = cens == 1 & days <= t0,
    event_free_t0 = case_when(
      event_by_t0 ~ FALSE,
      days >= t0 ~ TRUE,
      TRUE ~ NA
    ),
    known_status_t0 = !is.na(event_free_t0)
  )

actg_bin <- actg |>
  filter(known_status_t0)

n_known_t0 <- nrow(actg_bin)
n_unknown_t0 <- sum(!actg$known_status_t0)
```

```
n_event_t0 <- sum(!actg_bin$event_free_t0)
n_event_free_t0 <- sum(actg_bin$event_free_t0)
```

Status at 96 weeks is known for 1,956 participants. Of these, 322 experienced the composite event and 1,634 were event-free. The remaining 183 participants were censored before 96 weeks. The fixed-horizon analysis is therefore restricted to participants with known status. Its interpretation depends on this exclusion not distorting the treatment comparison.

6.3 Main survival analysis with safe and classical logrank tests

The survival analysis compares the time to the composite event between `Other therapy` and `ZDV only`. Because the event is unfavourable, a hazard ratio below one favours `Other therapy`. Let HR denote this hazard ratio. The one-sided hypotheses are

$$H_0 : HR \geq 1 \quad \text{against} \quad H_1 : HR < 1.$$

The value $HR_{\min} = 0.75$ is the design alternative used for power planning. It is not the boundary of the alternative hypothesis and is not estimated from the ACTG data. The design targets level 0.05 and power 0.8 at $HR = HR_{\min}$.

```
n_zdv <- sum(actg$treatment_binary == "ZDV only")
n_other <- sum(actg$treatment_binary == "Other therapy")

set.seed(2026)

invisible(capture.output({
  design_lr <- safestats::designSafeLogrank(
    hrMin = hr_min,
    beta = beta_design,
    alpha = alpha_single,
    alternative = "less",
    exact = TRUE,
    m0 = n_zdv,
    m1 = n_other,
    pb = FALSE
  )

  safe_lr <- safestats::safeLogrankTest(
    survival::Surv(days, cens) ~ treatment_binary,
    data = actg,
    designObj = design_lr,
    exact = TRUE
  )
}))

safe_e_value <- unname(safe_lr$eValue)
safe_n_events <- sum(actg$cens == 1, na.rm = TRUE)
```


The safe test uses the observed group sizes and the design values above. At level 0.05, rejection requires an e-value of at least 20.

As in Section 3.5, the safe guarantee is relative to the conditional risk-set model and censoring assumptions used by the exact logrank construction. The analysis here is unstratified; the recorded `strat` variable is used only in the permutation diagnostic below.

For comparison, the classical logrank test is applied to the same endpoint and treatment contrast. It tests for a difference in the survival curves but does not provide an effect estimate. A Cox proportional hazards model is therefore also fitted to estimate the hazard ratio and its fixed-sample confidence interval. With `ZDV only` as the reference group, values below one favour `Other therapy`. The hazard-ratio estimate and its confidence interval rely on the proportional-hazards assumption.

```
classical_lr <- survival::survdiffr(
  survival::Surv(days, cens) ~ treatment_binary,
  data = actg
)

one_sided_logrank_p <- function(fit, beneficial_group) {
  idx <- grep(beneficial_group, names(fit$n), fixed = TRUE)
  observed <- as.numeric(fit$obs)
  expected <- as.numeric(fit$exp)
  variance <- unname(fit$var[idx, idx])
  z <- (observed[idx] - expected[idx]) / sqrt(variance)

  stats::pnorm(z)
}

classical_lr_p <- one_sided_logrank_p(
  classical_lr,
  beneficial_group = "Other therapy"
)

cox_lr <- survival::coxph(
  survival::Surv(days, cens) ~ treatment_binary,
  data = actg
)
```

The Kaplan–Meier estimator, the standard nonparametric estimator of the event-free probability under right-censoring, retains each participant in the risk set until their event or censoring time. Each downward step corresponds to one or more observed composite events. Figure 6.1 shows the estimated event-free probability over follow-up in the two groups. The curves describe the same comparison as the tests and are not used as an additional test.

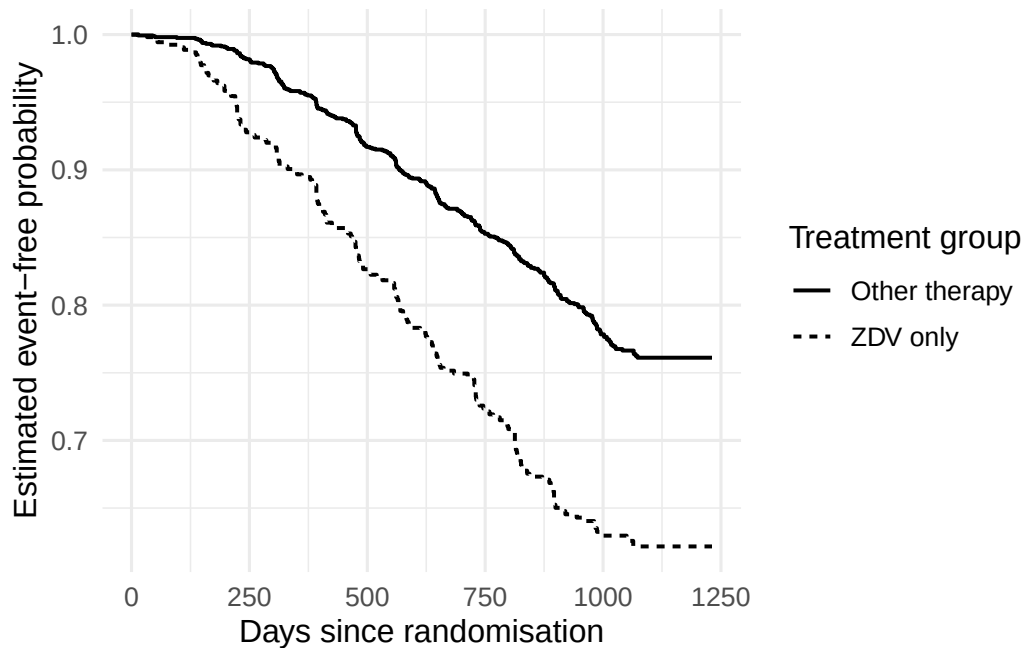


Figure 6.1: Kaplan–Meier curves for the ACTG 175 composite endpoint.

Table 6.2: Survival analysis comparing the combined active-treatment arms with zidovudine alone. The safe e-value threshold is $1/\alpha$.

One-sided logrank p-value	Cox HR (95% CI)	Safe e-value	Events
<0.001	0.535 (0.447, 0.641)	2.99e+06	521

Table 6.2 summarises the single-look survival results.

The analysis includes 521 composite events. The estimated hazard ratio of **Other therapy** relative to **ZDV only** is 0.535 (95% CI 0.447 to 0.641), so the estimated event hazard is lower in the combined active-treatment group. The one-sided logrank p-value is <0.001. The safe e-value is 2.99e+06, compared with the threshold 20.

6.4 Monitoring the safe logrank with optional stopping

The same safe logrank e-process is replayed over the ordered distinct event times to examine repeated monitoring. This retrospective replay does not reconstruct the trial’s historical monitoring.

`safeLogrankTest()` returns one log-evidence contribution for each distinct event time, with tied events at the same time handled within the same update. Cumulating these contributions gives the log e-process, and exponentiating gives the e-value after each update (Turner et al. 2022).

```
extract_log_e_path <- function(safe_obj) {
  cumsum(safe_obj$sumStats$logEValueLess)
}
```

```

safe_log_e_path <- extract_log_e_path(safe_lr)
safe_event_times <- safe_lr$sumStats$stopTimeVector

safe_monitor_path <- data.frame(
  update_index = seq_along(safe_log_e_path),
  event_time = safe_event_times,
  log_e_value = safe_log_e_path,
  e_value = exp(safe_log_e_path)
)

```

At level α , the safe rule rejects when the e-process first reaches $1/\alpha$.

```

safe_crossing_updates <- which(
  safe_monitor_path$log_e_value >= log(1 / alpha_single)
)

safe_crosses <- length(safe_crossing_updates) > 0L

```

The safe logrank path crosses the threshold after evidence update 27 (day 188) and ends at e-value $2.99\text{e}+06$. This crossing is specific to the observed data. The type-I error guarantee follows from the e-process property and Ville's inequality.

For comparison, a classical one-sided logrank test is recomputed at 30 interim times. At time τ_j , follow-up is administratively censored at τ_j , so only events observed by then enter the test.

```

event_times <- sort(unique(actg$days[actg$cens == 1]))
n_monitor_looks <- min(30L, length(event_times))

look_indices <- unique(round(
  seq(1, length(event_times), length.out = n_monitor_looks)
))

look_times <- event_times[look_indices]

classical_logrank_p_at <- function(data, tau) {
  data_tau <- data |>
    dplyr::mutate(
      time_tau = pmin(days, tau),
      event_tau = cens == 1 & days <= tau
    )

  lr_tau <- survival::survdiffr(
    survival::Surv(time_tau, event_tau) ~ treatment_binary,
    data = data_tau
  )

  one_sided_logrank_p(
    lr_tau,
    beneficial_group = "Other therapy"
  )
}

```

```

}

naive_look_path <- data.frame(
  look_index = seq_along(look_times),
  event_time = look_times,
  p_value = vapply(
    look_times,
    function(tau) classical_logrank_p_at(actg, tau),
    numeric(1)
  )
)

naive_crossing_indices <- which(
  naive_look_path$p_value <= alpha_single
)

naive_crosses <- length(naive_crossing_indices) > 0L

```

The classical p-value first falls below alpha at look 2 (day 137, $p = 0.00108$). This crossing does not show type-I error inflation because the treatment effect may be real. The null simulation below addresses inflation.

Figure 6.2 shows the log e-value and $-\log_{10}(p)$. Smaller p-values appear higher in the lower panel, and the first classical crossing is marked. The classical looks are spaced over ordered event times rather than calendar days, which explains the uneven horizontal spacing.

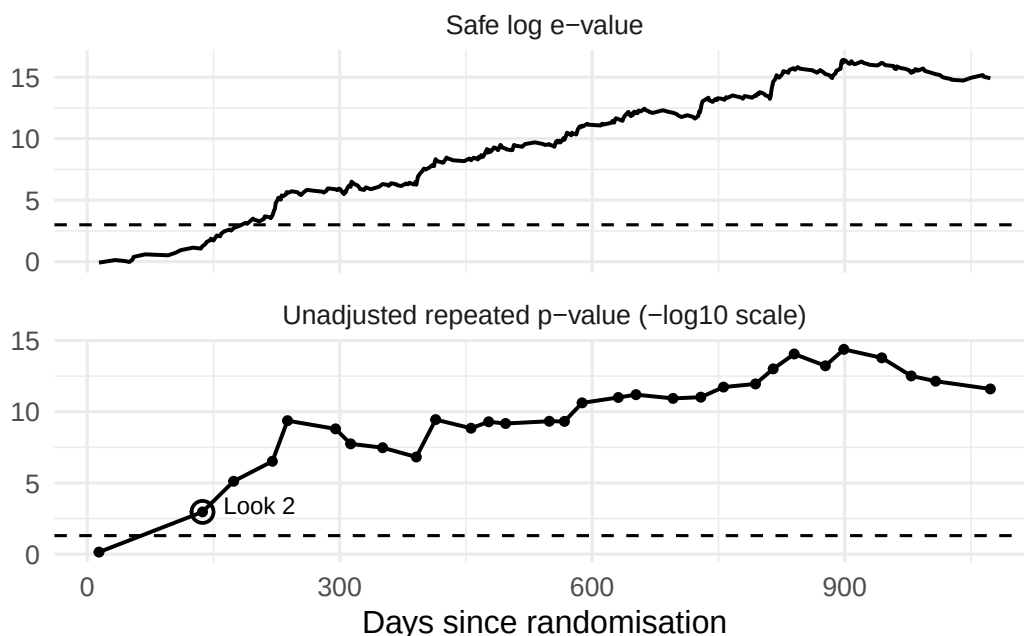


Figure 6.2: Monitoring paths for the ACTG 175 survival endpoint. The upper panel shows the safe log e-value and the lower panel shows the unadjusted repeated p-values on a negative log10 scale. The first classical crossing of the nominal threshold is marked.

To compare the two monitoring rules under the null, treatment labels are permuted within

the recorded `strat` groups. This preserves the treatment counts within each stratum while removing the treatment–outcome association. For each permutation, the safe rule rejects if the e-process ever reaches $1/\alpha$, and the unadjusted rule rejects if any interim p-value is at most α .

Across 250 permutations, the safe rule rejected in 12 cases (0.048, Monte Carlo standard error 0.0135), whereas the unadjusted repeated-logrank procedure rejected in 102 cases (0.408, standard error 0.0311). The safe estimate is consistent with the nominal level $\alpha = 0.05$. The much higher rejection rate of the repeated-logrank procedure illustrates the type-I error inflation caused by testing at multiple interim looks without adjustment. These estimates are conditional on the observed event and censoring pattern, the chosen monitoring schedule, and the within-`strat` permutation scheme. The calculation is a conditional permutation diagnostic rather than a reconstruction of the trial’s historical randomisation or monitoring protocol.

6.5 Safe t-tests for CD4 and CD8 change

ACTG 175 records CD4 and CD8 counts at baseline and in a 20 ± 5 week assessment window. We analyse the change from baseline in each count, comparing `Other therapy` with `ZDV only`.

For CD4, a larger increase indicates a better immune response. The one-sided alternative is therefore that the mean CD4 change is greater under `Other therapy`. The design uses 0.3 as the minimum relevant standardised effect, while the estimated difference is reported on the original cell-count scale.

CD8 change is analysed in the same direction, although its clinical interpretation is less direct than that of CD4 change. It is treated as a secondary outcome.

Both analyses use an independent two-sample Gaussian model with a common variance. Participants with a missing week-20 measurement are omitted separately for each endpoint, so the two analyses may contain different patients.

```
make_t_endpoint <- function(data, variable) {
  endpoint_data <- data.frame(
    treatment_binary = data$treatment_binary,
    value = data[[variable]]
  )

  endpoint_data <- endpoint_data[
    !is.na(endpoint_data$value) & !is.na(endpoint_data$treatment_binary),
    ,
    drop = FALSE
  ]

  x <- endpoint_data$value[endpoint_data$treatment_binary == "Other therapy"]
  y <- endpoint_data$value[endpoint_data$treatment_binary == "ZDV only"]

  list(
    x = x,
    y = y,
    n_total = length(x) + length(y),
    mean_difference = mean(x) - mean(y)
  )
}
```

```

    )
  }

  cd4_t_data <- make_t_endpoint(actg, "cd4_change_20")
  cd8_t_data <- make_t_endpoint(actg, "cd8_change_20")

  cd4_mean_diff_fmt <- fmt_number(cd4_t_data$mean_difference)
  cd8_mean_diff_fmt <- fmt_number(cd8_t_data$mean_difference)
  cd4_n_fmt <- format(cd4_t_data$n_total, big.mark = ",")
  cd8_n_fmt <- format(cd8_t_data$n_total, big.mark = ",")

```

After excluding missing measurements, the CD4 analysis includes 2,139 patients and the estimated mean difference, calculated as **Other therapy** minus **ZDV only**, is 50.41 cells per cubic millimetre. The CD8 analysis includes 2,139 patients and gives a mean difference of 10.35 cells per cubic millimetre. Positive differences favour **Other therapy** under the chosen one-sided alternatives.

The same safe t-test design is used for both endpoints, with level 0.05, target power 0.8, and minimum relevant standardised effect 0.3. The planning calculation assumes equal group sizes. The tests use the observed group sizes.

```

design_t <- safestats::designSafeT(
  deltaMin = delta_min_t,
  alpha = alpha_single,
  beta = beta_design,
  alternative = "greater",
  testType = "twoSample",
  ratio = 1,
  seed = 2026,
  pb = FALSE
)

safe_cd4 <- safestats::safeTTest(
  x = cd4_t_data$x,
  y = cd4_t_data$y,
  designObj = design_t,
  varEqual = TRUE,
  ciValue = 1 - alpha_single
)

safe_cd8 <- safestats::safeTTest(
  x = cd8_t_data$x,
  y = cd8_t_data$y,
  designObj = design_t,
  varEqual = TRUE,
  ciValue = 1 - alpha_single
)

```

For each endpoint, the safe test returns an e-value and an anytime-valid interval. The classical comparison uses a one-sided pooled t-test and a fixed-sample two-sided interval. The week 20 measurements have no reliable accrual order, so only the final safe interval is reported.

Table 6.3 compares the safe e-values with $1/\alpha$ and the classical p-values with α . The interval columns report the difference in mean cell-count change.

Table 6.3: Safe and classical t-test summaries for 20-week CD4 and CD8 change. The safe e-value threshold is $1/\alpha$.

Endpoint	Analysed n	Safe e-value	Safe anytime-valid interval	Classical p-value	Classical interval
CD4 change at 20 weeks	2,139	5.90e+13	[31.47, 69.34]	<0.001	[38.6, 62.22]
CD8 change at 20 weeks	2,139	7.08e-07	[-40.72, 61.43]	0.262	[-21.5, 42.21]

For CD4 change, the safe e-value is 5.90e+13 and the classical one-sided p-value is <0.001. For CD8 change, the corresponding values are 7.08e-07 and 0.262.

6.6 Event-free status at 96 weeks

The survival outcome can also be summarised at a fixed time. At 96 weeks, the question is whether a participant remained free of the composite event. The analysis includes only participants whose status at that time is known, as defined in Section 6.2.

This endpoint is analysed with a classical two-proportions test rather than the safe two-proportions test introduced in Section 3.4. ACTG 175 supplies one retrospectively constructed fixed-horizon table among participants with known 96-week status, not a pre-specified stream of complete blocks. We therefore do not retrofit a safe block design after observing the follow-up and censoring structure.

The one-sided comparison tests whether the event-free probability is larger under **Other therapy** than under **ZDV only**. A two-sided fixed-sample confidence interval is reported for the risk difference

$$P_{\text{Other therapy}} - P_{\text{ZDV only}}.$$

Both calculations use the uncorrected normal approximation.

```
bin_a <- actg_bin |>
  dplyr::filter(treatment_binary == "ZDV only")

bin_b <- actg_bin |>
  dplyr::filter(treatment_binary == "Other therapy")

na_bin <- nrow(bin_a)
nb_bin <- nrow(bin_b)

ya_bin <- as.integer(sum(bin_a$event_free_t0))
yb_bin <- as.integer(sum(bin_b$event_free_t0))

event_a_bin <- na_bin - ya_bin
event_b_bin <- nb_bin - yb_bin

pa_hat <- ya_bin / na_bin
pb_hat <- yb_bin / nb_bin
risk_diff_hat <- pb_hat - pa_hat

classical_bin_p <- stats::prop.test(
```

```

x = c(yb_bin, ya_bin),
n = c(nb_bin, na_bin),
alternative = "greater",
correct = FALSE
)

classical_bin_ci <- stats::prop.test(
  x = c(yb_bin, ya_bin),
  n = c(nb_bin, na_bin),
  alternative = "two.sided",
  conf.level = 1 - alpha_single,
  correct = FALSE
)

```

Table 6.4: Event-free status at 96 weeks by treatment group.

Treatment	Event-free	Event by 96 weeks	Total	Event-free proportion
ZDV only	352	125	477	0.7379
Other therapy	1282	197	1479	0.8668

Table 6.4 shows the event-free counts and proportions at 96 weeks.

The event-free proportions are 0.7379 under **ZDV only** and 0.8668 under **Other therapy**, giving an estimated risk difference of 0.1289. The classical one-sided p-value is <0.001 , read against $\alpha = 0.05$. The two-sided confidence interval for the risk difference is $[0.08576, 0.172]$.

The binary endpoint is not included in the e-value family analysed in Section 6.7.

6.7 Arm-specific comparisons with eBH

The earlier analyses pool all three active treatments into **Other therapy**. The eBH analysis instead treats the active arms separately and controls the false discovery rate across the resulting claims.

The family was defined in advance for this chapter as the nine combinations of three active arms (ZDV + ddI, ZDV + ddC, and ddI) and three endpoints (survival, CD4 change, and CD8 change). Each active arm is compared with ZDV. The fixed-horizon binary endpoint is excluded because it uses the pooled **Other therapy** comparison rather than an arm-specific contrast. The CD8 comparisons are included, but their directional interpretation is weaker, as noted in Section 6.5.

The nine e-values are dependent because the comparisons share the ZDV group and the endpoints are measured on the same patients. Since the PRDS condition is not established here, we use the arbitrary-dependence guarantee of eBH from Proposition 5.1. The comparison with BH and BY is discussed in Section 5.1 and Section 5.6.

For each active arm, the safe logrank and safe t-tests are applied to the comparison with ZDV using the same effect directions as in Section 6.3 and Section 6.5. A hazard ratio below one favours the active arm for survival, while larger CD4 and CD8 changes favour

the active arm. The same design alternatives are reused across cells to keep the e-value constructions comparable. The target-power calculations for the pooled comparisons do not apply to the smaller arm-specific contrasts. The resulting nine e-values are then entered into the eBH procedure.

```

active_arms <- c("ZDV + ddI", "ZDV + ddC", "ddI")
endpoint_levels <- c(
  "Composite survival",
  "CD4 change",
  "CD8 change"
)

e_logrank_arm <- function(active) {
  sub <- actg[actg$arm %in% c("ZDV", active), , drop = FALSE]
  sub$arm_pair <- factor(as.character(sub$arm), levels = c("ZDV", active))

  fit <- safestats::safeLogrankTest(
    survival::Surv(days, cens) ~ arm_pair,
    data = sub,
    designObj = design_lr,
    exact = TRUE
  )

  unname(fit$eValue)
}

e_ttest_arm <- function(active, outcome) {
  x <- actg[[outcome]][actg$arm == active]
  y <- actg[[outcome]][actg$arm == "ZDV"]

  x <- x[!is.na(x)]
  y <- y[!is.na(y)]

  fit <- safestats::safeTTest(
    x = x,
    y = y,
    designObj = design_t,
    varEqual = TRUE
  )

  unname(fit$eValue)
}

e_for_cell <- function(active, endpoint) {
  switch(
    endpoint,
    "Composite survival" = e_logrank_arm(active),
    "CD4 change"         = e_ttest_arm(active, "cd4_change_20"),
    "CD8 change"         = e_ttest_arm(active, "cd8_change_20")
  )
}

family <- do.call(rbind, lapply(active_arms, function(active) {
  data.frame(

```

```

    arm = active,
    endpoint = endpoint_levels,
    e_value = vapply(
      endpoint_levels,
      function(endpoint) e_for_cell(active, endpoint),
      numeric(1)
    ),
    stringsAsFactors = FALSE
  )
}))

```

The eBH procedure is applied at the FDR level 0.1 fixed in Section 6.1. Writing the ordered e-values as $e_{(1)} \geq \dots \geq e_{(K)}$, it selects the largest k satisfying

$$e_{(k)} \geq \frac{K}{\alpha k}.$$

```

ebh <- function(e, alpha) {
  K <- length(e)
  ord <- order(e, decreasing = TRUE)
  crit <- K / (alpha * seq_len(K))
  passes <- e[ord] >= crit
  k_star <- max(c(0L, which(passes)))

  reject <- logical(K)
  reject[ord[seq_len(k_star)]] <- TRUE

  list(
    reject = reject,
    k_star = k_star
  )
}

K_family <- nrow(family)
ebh_fit <- ebh(family$e_value, alpha_fdr)

family$pseudo_p <- pmin(1, 1 / family$e_value)
family$reject <- ebh_fit$reject

```

Table 6.5 reports all nine results, and Figure 6.3 compares the ranked e-values with the eBH boundary.

Table 6.5: Arm-specific e-values and eBH decisions for the nine-hypothesis treatment-arm by endpoint family. Pseudo-p-values are $\min(1, 1/e_i)$.

Endpoint	Arm	Safe e-value	Pseudo-p	Decision
Composite survival	ZDV + ddI	6.74e+04	1.48e-05	Reject
Composite survival	ZDV + ddC	2.46e+04	4.06e-05	Reject
Composite survival	ddI	4689	2.13e-04	Reject
CD4 change	ZDV + ddI	6.46e+13	1.55e-14	Reject
CD4 change	ZDV + ddC	1.84e+06	5.43e-07	Reject
CD4 change	ddI	5.22e+08	1.91e-09	Reject
CD8 change	ZDV + ddI	0.001324	1	Do not reject
CD8 change	ZDV + ddC	1.94e-08	1	Do not reject
CD8 change	ddI	0.01633	1	Do not reject

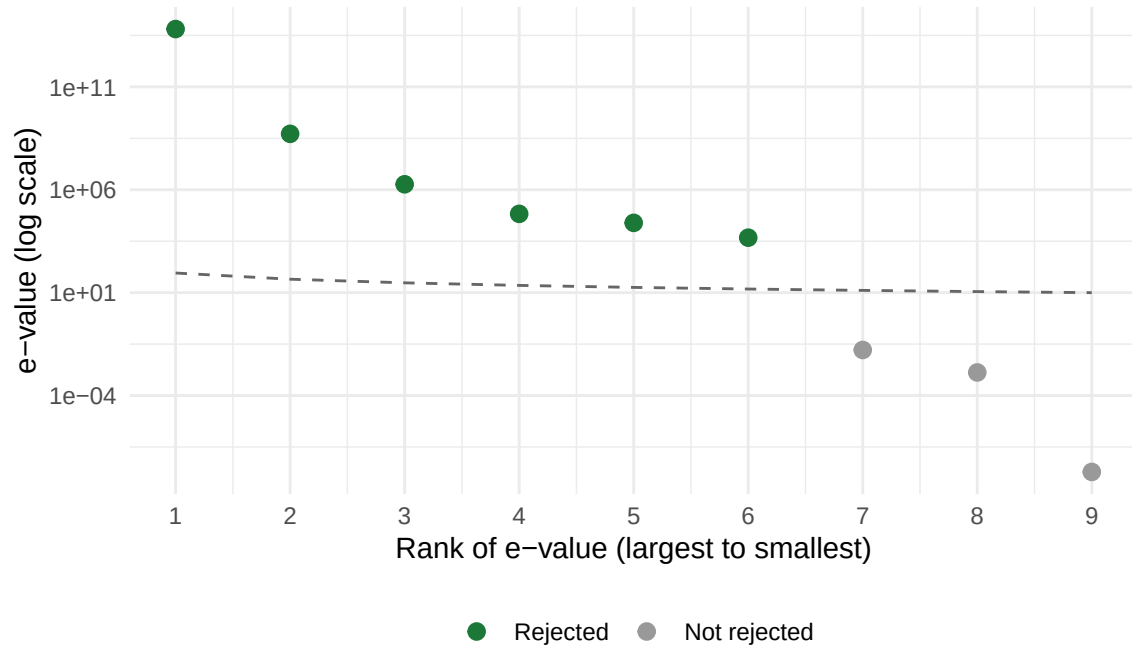


Figure 6.3: Ranked e-values and the eBH boundary for the nine-hypothesis family. Points up to the last rank satisfying the eBH criterion are rejected.

At an FDR level of 0.1, eBH rejects 6 of the 9 hypotheses. The rejected hypotheses are Composite survival for ZDV + ddI, CD4 change for ZDV + ddI, Composite survival for ZDV + ddC, CD4 change for ZDV + ddC, Composite survival for ddI, and CD4 change for ddI. This is not a reanalysis of ACTG 175's primary efficacy claims. The nine-hypothesis family was defined for this chapter and does not reproduce the trial's original analysis plan. Any causal interpretation is also subject to the sample-selection and missingness limitations discussed in Section 6.2.

The arm-specific family answers a different question from the pooled comparisons earlier in the chapter. Pooling asks whether the active treatments are beneficial on average relative to ZDV monotherapy, whereas the eBH analysis examines which individual arm–endpoint hypotheses are rejected after multiplicity adjustment. The two analyses therefore need

not agree cell by cell.

The e-values are compared directly with the eBH thresholds. The pseudo-p-values $p_i^* = \min(1, 1/e_i)$ in Table 6.5 are used only for the equivalent BH representation of eBH. They are not p-values from the underlying safe tests. Because the nine e-values arise from overlapping comparisons within the same trial, they are combined using eBH rather than multiplied. Multiplication is reserved for the separate-study construction in Section 6.8.

6.8 Optional continuation with the BCG vaccine studies

ACTG 175 is a single trial, whereas optional continuation concerns evidence accumulated across separate studies. The BCG data provide controlled studies with distinct participants and completed two-by-two tables, allowing the product construction to be examined across studies.

The data come from the published meta-analysis of Colditz et al. (1994) and are available as `metadat::dat.bcg` (Viechtbauer et al. 2026). The studies are shown in publication-year order, with author used to order ties. This is only an approximate chronology because publication year may not match the date at which complete results became available.

Each study is summarised by the tuberculosis cases and group sizes among vaccinated participants and controls, together with the reported allocation method. In Table 6.6, an entry such as 4/123 means four tuberculosis cases among 123 participants. The risk ratio is the observed tuberculosis risk in the vaccinated group divided by that in the control group. Values below one favour vaccination, one indicates equal observed risks, and values above one favour control.

For the e-value calculation, each completed two-by-two table is conditioned on its row and column totals. The number of tuberculosis cases among vaccinated participants then follows Fisher’s noncentral hypergeometric distribution, indexed by the conditional odds ratio ω . The null $\omega = 1$ gives the central hypergeometric distribution. The working alternative $\omega = 0.5$ was fixed in Section 6.1 before the study-level e-values were calculated. It corresponds to lower tuberculosis odds under vaccination.

The study-level e-value is the probability of the observed cell count under $\omega = 0.5$ divided by its probability under $\omega = 1$. Under the conditional null, this likelihood ratio has expectation one. Values above one favour the working alternative relative to the null, while values below one favour the null relative to that alternative. The probabilities are computed with `BiasedUrn` (Fog 2024). The displayed risk ratio is descriptive and is distinct from the conditional odds ratio used in the e-value calculation.

For studies with random allocation, conditioning on the margins also has a randomisation interpretation. For alternate or systematic allocation, validity relies on the conditional model rather than randomisation alone.

Assuming independence across studies, the product of the study-level e-variables is a test supermartingale by Proposition 4.2 and hence an e-process for the intersection null of no effect in each study by Proposition 4.1. Extending the sequence requires each new e-variable to remain conditionally valid given the preceding evidence and inclusion history.

The analysis is retrospective. Result-dependent publication or study inclusion could violate conditional validity and create the accumulation bias discussed in the introduction

(ter Schure and Grünwald 2019; Pawel and Held 2024). The historical sequence therefore does not itself establish that optional-continuation conditions were satisfied.

The e-values in Section 6.7 arise from overlapping comparisons within one trial and are combined with eBH. The BCG e-values arise from separate studies and are multiplied under the model, independence, and selection assumptions above. After all 13 studies, the product is $5.25\text{e}+23$, compared with the threshold $1/\alpha = 20$.

Figure 6.4 traces the running product in approximate publication-year order. Its path depends on the chosen within-year order, although the final product does not. Under the assumptions above, the product may be evaluated after any study without fixing the total number of studies in advance.

The sharp decrease at the final study is caused by TPT Madras (1980). Its estimated risk ratio is 1.012, indicating nearly equal tuberculosis rates in the vaccinated and control groups. That result is much more compatible with the null $\omega = 1$ than with the working alternative $\omega = 0.5$, so its study-level e-value is only $3.60\text{e}-27$. Multiplying by this value reduces the running product from $1.46\text{e}+50$ to $5.25\text{e}+23$. A running product of e-values need not increase. Evidence that conflicts with the working alternative produces an e-value below one and reduces the accumulated evidence.

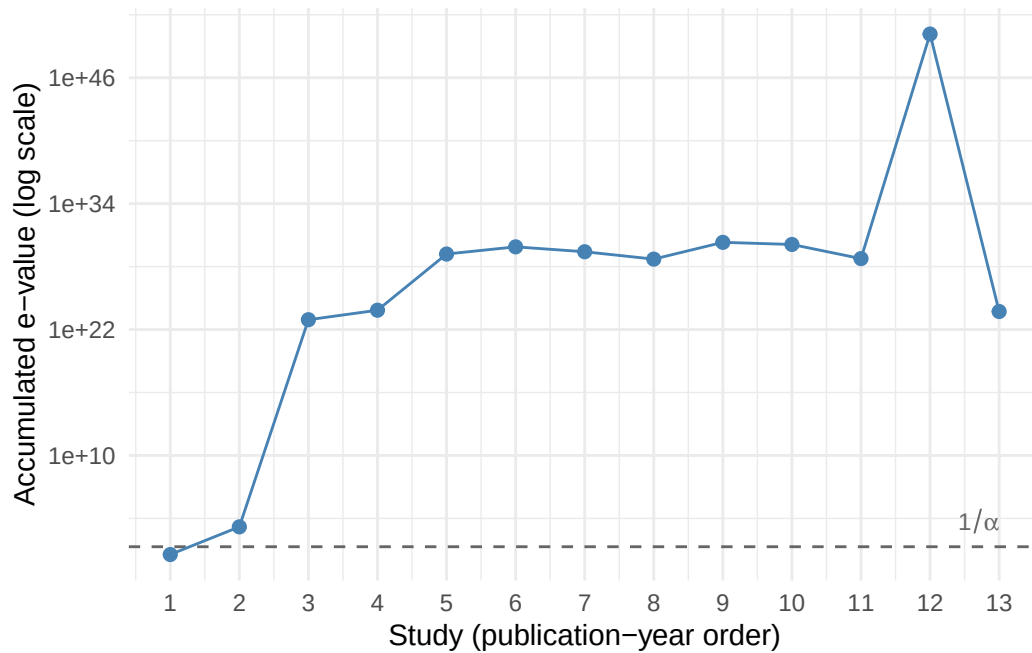


Figure 6.4: Running product of the BCG study e-values in approximate publication-year order. The dashed line marks $1/\alpha$. The anytime-valid interpretation is conditional on the modelling, independence, and study-selection assumptions stated in the text.

Table 6.6: BCG study results in approximate publication-year order. TB/ n gives the number of tuberculosis cases divided by the group size. The product column is the cumulative product of the study-level e-values.

Study	Allocation	Vaccinated (TB/ n)	Control (TB/ n)	Risk ratio	e-value	Product
Aronson (1948)	random	4/123	11/139	0.4109	3.609	3.609
Ferguson & Simes (1949)	random	6/306	29/303	0.2049	435.8	1573
Stein & Aronson (1953)	alternate	180/1541	372/1451	0.4556	5.50e+19	8.65e+22
Rosenthal et al (1960)	random	3/231	11/220	0.2597	8.047	6.96e+23
Rosenthal et al (1961)	systematic	17/1716	65/1665	0.2538	2.26e+05	1.58e+29
Coetzee & Berjak (1968)	random	29/7499	45/7277	0.6254	4.851	7.64e+29
Comstock & Webster (1969)	systematic	5/2498	3/2341	1.5620	0.3407	2.60e+29
Frimodt-Moller et al (1973)	alternate	33/5069	47/5808	0.8045	0.1979	5.15e+28
Vandiviere et al (1973)	random	8/2545	10/629	0.1977	39.74	2.05e+30
Comstock et al (1974)	systematic	186/50634	141/27338	0.7122	0.629	1.29e+30
Comstock et al (1976)	systematic	27/16913	29/17854	0.9828	0.04492	5.79e+28
Hart & Sutherland (1977)	random	62/13598	248/12867	0.2366	2.52e+21	1.46e+50
TPT Madras (1980)	random	505/88391	499/88391	1.0120	3.60e-27	5.25e+23

The BCG analysis demonstrates the product construction but is not a new meta-analysis of vaccine efficacy. The null specifies no effect in each study, the studies may be heterogeneous, several did not use random allocation, and the historical publication and inclusion process is not modelled.

6.9 Power and stopping behaviour of the safe t-test

A simulation estimates the type-I error, power, and stopping sample size of the safe two-sample t-test used in Section 6.5.

The simulation follows the Gaussian model used by the test. Both groups have variance one. Their means are equal under the null, while under the alternative the treatment mean is higher by $\delta = 0.3$. The safe test is evaluated at ten interim looks between 5 observations per group and its planned sample size. Sampling stops at the first look for which the e-value reaches $1/\alpha$. A conventional fixed-sample two-sample t-test is planned using the same effect size, significance level, and target power. The ten-look schedule is coarser than the monitoring scheme used to determine the safe test's planned sample size, so its simulated power need not equal the design target of 0.8.

The safe design plans for 190 observations per group, compared with 139 for the fixed-sample test.

Across 500 simulations under the null, the estimated type-I error is 0.018 with Monte Carlo standard error 0.005946. Under the alternative, the estimated power is 0.712 with standard error 0.02025, and the mean stopping sample size is 131.8 observations per group. The type-I error estimate is a finite Monte Carlo check. The anytime-valid guarantee itself follows from the e-process property and Ville's inequality.

Table 6.7 summarises the operating-characteristic estimates, and Figure 6.5 shows the stopping-sample-size distribution under the design alternative.

Table 6.7: Safe sequential and fixed-sample t-test characteristics at the same effect size, significance level, and target power. Safe-test error and power values are Monte Carlo estimates; fixed-sample values are design targets.

Procedure	Type-I error	Power	Planned n / group	Mean n / group under alternative
Safe t-test	0.018	0.712	190	131.8
Fixed-sample t-test	0.05	0.8	139	139

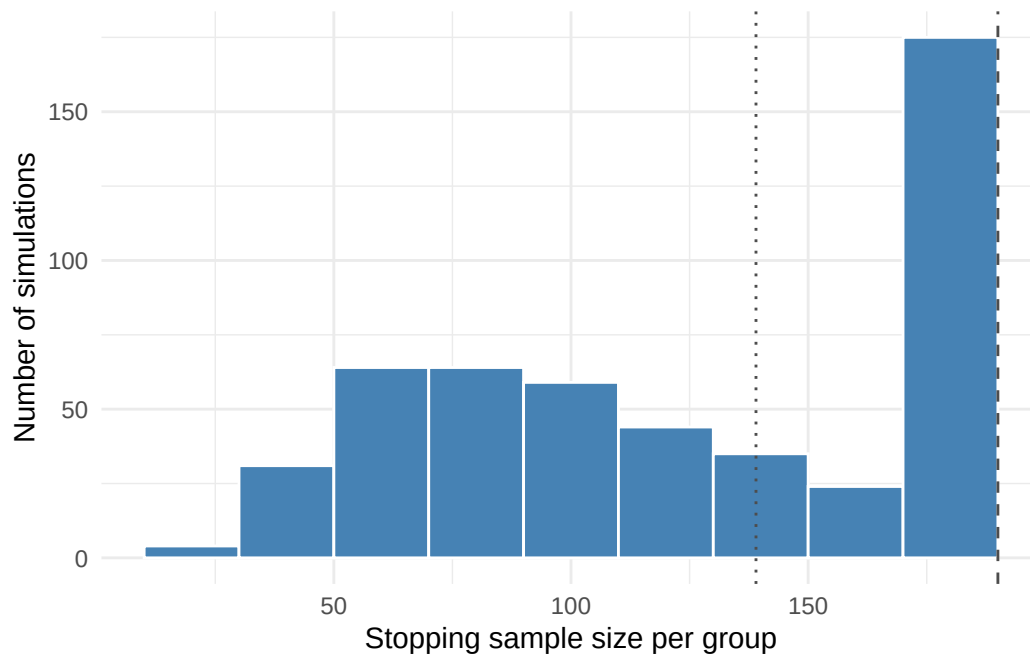


Figure 6.5: Stopping sample size per group for the safe t-test under the design alternative. The dashed line marks its planned sample size and the dotted line the fixed-sample requirement.

Under the design alternative, early threshold crossings reduce the mean sample size, while simulations without a crossing use the full planned sample. These results apply to the Gaussian model, effect size, and monitoring schedule specified above. They do not imply that the safe test is uniformly more powerful or more efficient than a fixed-sample test.

Chapter 7

Discussion and Conclusion

This thesis presents an applied introduction to e-values for readers familiar with standard statistical testing. It covers the expectation bound and betting interpretation in Chapter 2; safe z, t, two-proportions, and logrank tests in Chapter 3; e-processes and confidence sequences in Chapter 4; e-value FDR methods in Chapter 5; and the applications in Chapter 6. Its contribution is expository and comparative: it explains the existing framework from first principles, shows how several procedures are implemented in `safestats` (Turner et al. 2022), compares the resulting e-value procedures with their classical counterparts on shared data and designs, and makes the assumptions behind the applications explicit.

7.1 Main conclusions

At a single analysis time, Markov’s inequality turns an e-variable into a level- α test. The main practical advantage appears sequentially: conditionally valid factors multiply into a test supermartingale, and Ville’s inequality bounds the probability that the process ever crosses $1/\alpha$. The same threshold may therefore be monitored continuously without fixing a stopping time in advance. Confidence sequences provide the corresponding time-uniform guarantee for estimation (Ramdas et al. 2023; Ramdas and Wang 2025; Howard et al. 2021), as developed in Section 4.5.

The shared decision rule does not make safe-test construction automatic. The safe z-test uses a Gaussian likelihood ratio; the safe t-test must handle unknown variance; and binary and survival outcomes require their own constructions. Growth-rate optimality guides the choice among valid e-variables, but the model, nuisance parameters, and working alternative remain problem-specific (Grünwald et al. 2024). The sequential error argument becomes generic only after that problem-specific construction has been established.

The way e-values are combined also depends on the inferential setting. Arithmetic averages preserve e-validity under arbitrary dependence when several e-variables concern the same null, as discussed in Section 2.7. Products are valid when the factors are independent under the null they jointly test or when they form a sequence of conditionally valid contributions to a common null. If the factors correspond to different study-specific nulls, their product targets the intersection null that all of those nulls hold. Products are not a general method for combining dependent evidence about different hypotheses. For multiple testing, eBH provides false discovery rate control under arbitrary dependence, while the e-Partitioning formulation of the e-Closure Principle extends the guarantee to a collection of admissible

rejection sets (Wang and Ramdas 2022; Xu et al. 2025). In both settings, flexibility is obtained by proving a uniform guarantee before the data-dependent choice is made: over the evidence path in sequential testing and over the admissible collection in simultaneous FDR control.

E-values should also be kept separate from effect estimation. A large e-value provides evidence against a null, but it does not state how large or scientifically important an effect is. Effect estimates and suitable intervals remain necessary. Nor is a general e-value simply the reciprocal of a p-value. Calibration between the two scales is possible, but it loses information, and the resulting values should not be treated as interchangeable. The relevant question is which procedure matches the way the data were collected and analysed, not which scale produces the more striking number.

7.2 Limitations and practical implications

The word “safe” refers to a mathematical guarantee under a stated null model. It does not make an analysis assumption-free. The model must be appropriate, the e-variable must be valid for that model, and any sequential contribution must satisfy the required conditional expectation bound. Randomisation, censoring, missing outcomes, dependence, and study selection can all determine whether these conditions are credible. A formally valid e-value can still answer the wrong scientific question if the population, endpoint, or model has been chosen poorly. Safety is therefore conditional rather than unconditional (Pawel and Held 2024).

The case study made these restrictions concrete. The ACTG 175 analyses in Section 6.2 used the reduced participant sample distributed with the R package rather than the full randomised cohort. The endpoint corresponds to the percentage-based composite endpoint reported for ACTG 175, but the available sample and the thesis-specific analyses do not reproduce the full published analysis. The monitoring analysis in Section 6.4 replayed the observed event sequence and illustrated the difference between a safe e-process and repeated unadjusted logrank tests, but it did not reconstruct the trial’s historical monitoring. The arm-specific family in Section 6.7 was defined for this thesis rather than taken from the original trial protocol. Their scientific interpretation is therefore limited to the available sample and the thesis-specific analysis choices.

The BCG analysis in Section 6.8 uses the historical trials to illustrate the product construction. Multiplying the study-level e-values required a conditional model and an independence assumption across studies. A prospective extension would additionally require each newly included study to remain conditionally valid given the preceding evidence and inclusion history. Historical publication and study selection were not modelled. When a study is more compatible with the null than with the working alternative, its e-value is below one and the running product falls.

The choice of working alternative affects efficiency throughout the framework. Likelihood-ratio e-values and many safe tests are tuned to a particular effect or distribution. When the construction is fixed in advance, misspecifying the alternative does not by itself invalidate the null guarantee, but it can reduce e-power, slow evidence accumulation, or make expected log growth negative even when the null is false. Mixtures and plug-in strategies reduce reliance on a single fixed alternative, at the cost of additional design choices. Validity and efficiency should therefore be assessed separately.

The flexibility of the anytime-valid procedures studied here can carry an efficiency cost. The universal Markov threshold $1/\alpha$ can be conservative, fixed-sample power may be lower than for a classical test designed for one analysis time, and confidence sequences are wider than fixed-sample confidence intervals. The fixed-sample comparison in Section 2.11 shows this cost directly. Sequential monitoring may offset part of it by allowing early stopping when evidence accumulates quickly. The simulation in Section 6.9 shows how the balance depends on the true effect, the chosen alternative, and the monitoring schedule. These results illustrate particular designs rather than establishing a general power advantage. The type-I error guarantees come from the e-process theory, not from finite simulation results.

E-values are most attractive when the analysis genuinely requires repeated monitoring, optional continuation, dependence-robust multiple testing, or post-hoc selection from a protected collection. For one fixed analysis with no such flexibility, a classical procedure may be more efficient.

7.3 Future work

The case study points to several practical extensions: covariate-adjusted or stratified survival analyses, regression-based e-processes, and principled treatment of missing outcomes. Such methods must preserve e-validity without sacrificing too much efficiency, and their guarantees must remain explicit about the null model under which they hold.

Further practical guidance is needed on choosing the working alternative and monitoring schedule and on interpreting planned versus expected sample sizes. These decisions can materially affect performance while leaving null validity unchanged. Comparisons based on realistic monitoring plans and several plausible alternatives would be more informative than fixed-sample power alone.

Software support is still limited. The `safestats` package implements several important procedures (Turner et al. 2022), but e-value software is less broad and standardised than software for classical inference. More consistent interfaces and direct support for reporting effect estimates alongside e-values would make the methods easier to use correctly. The exhaustive eBH+ implementation in Section 5.4 is intended only for small teaching examples; polynomial algorithms are available for important cases, but scalable software remains an important practical requirement (Xu et al. 2025).

7.4 Conclusion

E-values are best viewed as a complement to fixed-horizon inference, not as a general replacement. They are most useful when monitoring, continuation, or selection is data-dependent, provided that the null model, filtration, and combination rule justify the construction. The working alternative primarily determines efficiency.

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